

with(LinearAlgebra)

[&x, Add, Adjoint, BackwardSubstitute, BandMatrix, Basis, BezoutMatrix, BidiagonalForm, BilinearForm, (1)

CARE, CharacteristicMatrix, CharacteristicPolynomial, Column, ColumnDimension, ColumnOperation, ColumnSpace, CompanionMatrix, CompressedSparseForm, ConditionNumber, ConstantMatrix, ConstantVector, Copy, CreatePermutation, CrossProduct, DARE, DeleteColumn, DeleteRow, Determinant, Diagonal, DiagonalMatrix, Dimension, Dimensions, DotProduct, EigenConditionNumbers, Eigenvalues, Eigenvectors, Equal, ForwardSubstitute, FrobeniusForm, FromCompressedSparseForm, FromSplitForm, GaussianElimination, GenerateEquations, GenerateMatrix, Generic, GetResultDataType, GetResultShape, GivensRotationMatrix, GramSchmidt, HankelMatrix, HermiteForm, HermitianTranspose, HessenbergForm, HilbertMatrix, HouseholderMatrix, IdentityMatrix, IntersectionBasis, IsDefinite, IsOrthogonal, IsSimilar, IsUnitary, JordanBlockMatrix, JordanForm, KroneckerProduct, LA_Main, LUDecomposition, LeastSquares, LinearSolve, LyapunovSolve, Map, Map2, MatrixAdd, MatrixExponential, MatrixFunction, MatrixInverse, MatrixMatrixMultiply, MatrixNorm, MatrixPower, MatrixScalarMultiply, MatrixVectorMultiply, MinimalPolynomial, Minor, Modular, Multiply, NoUserValue, Norm, Normalize, NullSpace, OuterProductMatrix, Permanent, Pivot, PopovForm, ProjectionMatrix, QRDecomposition, RandomMatrix, RandomVector, Rank, RationalCanonicalForm, ReducedRowEchelonForm, Row, RowDimension, RowOperation, RowSpace, ScalarMatrix, ScalarMultiply, ScalarVector, SchurForm, SingularValues, SmithForm, SplitForm, StronglyConnectedBlocks, SubMatrix, SubVector, SumBasis, SylvesterMatrix, SylvesterSolve, ToeplitzMatrix, Trace, Transpose, TridiagonalForm, UnitVector, VandermondeMatrix, VectorAdd, VectorAngle, VectorMatrixMultiply, VectorNorm, VectorScalarMultiply, ZeroMatrix, ZeroVector, Zip]

▼ Vokabular med eksempler eller som minimum chat/wiki definitioner

▼ Undervisning fra Troels

▼ *Ting vi rent faktisk har fået undervisning i*

▼ *Matrix Notation*

$$\bar{A} = \begin{matrix} & \text{columns } n & \\ \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & a_{ij} & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} & \begin{matrix} | \\ \text{rows } m \\ | \end{matrix} & \begin{matrix} \text{scalar elements} \\ \downarrow \\ [a_{ij}] \end{matrix} \end{matrix}$$

$$\text{size}(\bar{A}) = m \times n$$

$\bar{A}$ by column vector

Square matrices
will be particularly
interesting $m=n$

$$\bar{A} = [\bar{b}_1, \bar{b}_2, \dots, \bar{b}_n], \quad \bar{b}_i = \begin{bmatrix} b_{1i} \\ \vdots \\ b_{mi} \end{bmatrix}$$

Matrix notation

Matrix algebra: $\bar{y} = \bar{A} \bar{x}$ "similar to numbers"

Matrix equality: $\bar{A} = \bar{B} \Leftrightarrow [a_{ij}] = [b_{ij}]$
 $\text{size}(\bar{A}) = \text{size}(\bar{B})$

Matrix transpose:

$$\bar{A}^T = [a_{ij}]^T = [a_{ji}]$$

$\bar{A}$ by row vector

$$\bar{A} = [\bar{a}_1^T, \bar{a}_2^T, \dots, \bar{a}_m^T]^T, \quad \text{since } \bar{a}_i = (\bar{a}_i^T)^T$$

$$\bar{a}_i = [a_{i1}, \dots, a_{in}]$$

Special matrices (square)

Diagonal

$$\begin{bmatrix} a & & & \\ & b & & \\ & & c & \\ & & & \ddots \\ & & & & z \end{bmatrix}$$

Triangular

lower/upper

$$\begin{bmatrix} a & & & \\ & b & & \\ & & c & \\ & & & \ddots \\ & & & & z \end{bmatrix}$$

Scalar

$$\begin{bmatrix} k & & & \\ & k & & \\ & & \ddots & \\ & & & k \end{bmatrix} = k \cdot \mathbf{I}$$

Identity

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix} = \mathbf{I}$$

Null (in various forms) $\mathbf{0}$?

Identitets Matrix

$$I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Symmetrisk Matrix

En symmetrisk matrice er en matrice, hvor den transponerede matrice er magen til. Husk at den transponerede matrice er en matrice hvor rækker og kolonner skifter retning, man kan se det som at matricen roterer omkring den diagonale akse. Yderligere skal en matrice have lige mange rækker og kolonner for at kunne være symmetrisk. Et eksempel kunne være:

$$A = \begin{bmatrix} -74 & -32 & -2 & -72 \\ -32 & -76 & -93 & 27 \\ -2 & -93 & 57 & -77 \\ -72 & 27 & -77 & -98 \end{bmatrix}$$

Så følgende er sandt:

$$A = A^T$$

Echelonform

Echelonform er der hvor vi vil skabe et trappe fra venstre mod højre. Vi opnår det vha gaussisk elimination.

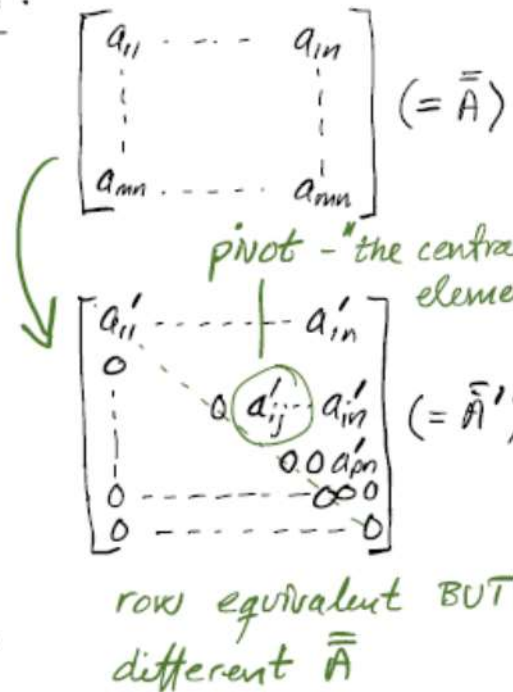
Echelon form

A triangular matrix, such that:

- all 0's are below, left
- first $a'_{ij} \neq 0$ in each row is to the right of the $a'(\neq 0)$ in the row above AND has only 0's below it

! first a'_{ij} 's are not necessarily on one diagonal

! there can be rows with all 0's



Reduceret echelonform er når alle de forreste pivot points samtidig er 1 og flest mulige andre værdier er 0

$$\text{echelon} := \begin{bmatrix} -45 & -85 & 89 & 9 & 39 \\ 68 & 19 & 92 & 81 & -35 \\ 58 & 25 & -2 & 35 & 26 \\ -43 & 17 & -46 & 80 & -74 \\ -85 & 81 & -29 & 20 & 13 \end{bmatrix} :$$

Som vi kan på på echelonform ved:

GaussianElimination(echelon) =

$$\begin{bmatrix} -45 & -85 & 89 & 9 & 39 \\ 0 & -\frac{985}{9} & \frac{10192}{45} & \frac{473}{5} & \frac{359}{15} \\ 0 & 0 & -\frac{306688}{4925} & -\frac{130448}{4925} & \frac{284547}{4925} \\ 0 & 0 & 0 & \frac{2407141}{19168} & -\frac{6986535}{306688} \\ \vdots & \vdots & \vdots & \vdots & 11097123357 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

$$\text{ReducedRowEchelonForm}(\text{echelon}) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Et bedre eksempel vil være:

$$A := \begin{bmatrix} 13 & -9 & 5 & 2 \\ 2 & 7 & -9 & -8 \\ 5 & 4 & 7 & -9 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 2 \\ 2 & -2 & -8 \\ 5 & 4 & -2 \end{bmatrix}$$

som på echelon bliver:

$$\text{GaussianElimination}(A) = \begin{bmatrix} 4 & 5 & 2 \\ 0 & -\frac{9}{2} & -9 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

og på reduceret echelon bliver:

$$\text{ReducedRowEchelonForm}(A) = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Her er det nemlig ikke handy dandy perfekt som i forrige eksempel.

Reduced row echelon form [\[edit \]](#)

A matrix is in **reduced row echelon form** (also called **row canonical form**) if it satisfies the following conditions:

- It is in row echelon form.
- The leading entry in each nonzero row is 1 (called a leading one).
- Each column containing a leading 1 has zeros in all its other entries.

If the first two conditions are verified, the last condition is equivalent to:

- Each column containing a leading 1 has zeros in all entries above the leading 1.

While a matrix may have several echelon forms, its reduced echelon form is unique.

Given a matrix in reduced row echelon form, if one permutes the columns in order to have the leading 1 on the diagonal, the resulting matrix is of the form

$$\begin{pmatrix} I & X \\ 0 & 0 \end{pmatrix},$$

where I is the [identity matrix](#) of dimension j equal to the rank of the entire matrix, X is a matrix with j rows and $n - j$ columns, and 0 represents $n - j$ [matrices](#) of appropriate size. Since a permutation of columns is not a row operation, the resulting matrix is not necessarily in reduced row echelon form.

In the Gaussian elimination method, this corresponds to a permutation of the unknowns in the original linear system. The resulting system is in row echelon form, in which the first j coefficients are unconstrained, and the remaining $n - j$ are determined by the first j coefficients.

▼ Spor for en matrix

Sporet for en matrice er alle værdier på diagonalen:

In [linear algebra](#), the **trace** of a [square matrix](#) $\mathbf{A}$, denoted $\text{tr}(\mathbf{A})$,^[1] is defined to be the sum of elements on the main diagonal (the line of elements from the upper left to the lower right) of $\mathbf{A}$. The trace is only defined for a square matrix ($n \times n$).

It can be proven that the trace of a matrix is the sum of its [eigenvalues](#) (counted with multiplicities). It can also be proven that the trace of the product of two matrices $\mathbf{A}$ and $\mathbf{B}$ of appropriate sizes. This implies that [similar matrices](#) have the same trace. As a consequence, the trace of a matrix is invariant under a [change of basis](#). The trace of a [linear operator](#) mapping a finite-dimensional [vector space](#) into itself, since all matrices describing such an operator have the same trace.

The trace is related to the derivative of the [determinant](#) (see [Jacobi's formula](#)).

Definition [\[edit\]](#)

The **trace** of an $n \times n$ [square matrix](#) $\mathbf{A}$ is defined as^{[1][2][3]:34}

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}$$

where a_{ii} denotes the entry on the i th row and i th column of $\mathbf{A}$. The entries of $\mathbf{A}$ can be [real numbers](#), [complex numbers](#), or elements of a [field](#) F . The trace is not defined for non-square matrices.

Example [\[edit\]](#)

Let $\mathbf{A}$ be a matrix, with

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 3 \\ 11 & 5 & 2 \\ 6 & 12 & -5 \end{pmatrix}$$

Then

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^3 a_{ii} = a_{11} + a_{22} + a_{33} = 1 + 5 + (-5) = 1$$

Properties [\[edit\]](#)

Basic properties [\[edit\]](#)

The trace is a [linear mapping](#). That is,^{[1][2]}

$$\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$$

$$\text{tr}(c\mathbf{A}) = c \text{tr}(\mathbf{A})$$

for all square matrices $\mathbf{A}$ and $\mathbf{B}$, and all [scalars](#) c .^{[3]:34}

A matrix and its **transpose** have the same trace: ^{[1][2][3]:34}

$$\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{A}^T).$$

This follows immediately from the fact that transposing a square matrix does not affect elements along the

Geometrisk Multiplicitet

Den geometriske multiplicitet er givet ved antallet af vektorer som beskriver nulrummet af matricen. Altså er nullitet=geometrisk multiplicitet

Geometrisk Multiplicitet

Geometrisk multiplicitet af en egen værdi er antallet af lineært uafhængige egenvektorer for den pågældende egen værdi. Vi skal finde egenvektorerne for $\lambda = 9$:

$$Q - 9I = \begin{pmatrix} 13 - 9 & 5 & 2 \\ 2 & 7 - 9 & -8 \\ 5 & 4 & 7 - 9 \end{pmatrix} = \begin{pmatrix} 4 & 5 & 2 \\ 2 & -2 & -8 \\ 5 & 4 & -2 \end{pmatrix}$$

Vi skal løse:

$$\begin{pmatrix} 4 & 5 & 2 \\ 2 & -2 & -8 \\ 5 & 4 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Efter rækkeoperationer ser vi, at vi har et ledigt frihedsgrad, dvs. vi har én lineært uafhængig egenvektor, hvilket betyder, at den geometriske multiplicitet er 1.

Algebraisk Multiplicitet

Den algebraiske multiplicitet er fundet ved at kigge på roden af lambda. Vist i eksemplet her:

Nu har vi altså et resultat, som matcher det maple giver os. Det betyder også at vi nu kan benytte maple til at løse den karakteriske ligning for os, så vi kan få spektret givet:

`Determinant(X)` Error, (in LinearAlgebra:-Determinant) invalid input: LinearAlgebra:-Determinant expects its 1st argument, A, to be of

type Matrix() but received X $\xrightarrow{\text{simplify}}$ Error, invalid input: simplify uses a 1st argument, s, which is missing $\xrightarrow{\text{solve for lambda}}$ $[[\lambda = \lambda]]$

For at finde den algebraiske multiplicitet kigger vi på leddet mellem simplify og solve, og ser at der står $-(\lambda - 9)^3$.

Det betyder reelt at determinanten kan beskrives ved $-(\lambda - 9)^3 = 0$, hvilket fortæller os at den algebraiske multiplicitet er 3, da lambda opstår som 3 separate løsninger, samtidig med at den i den endelige ligning har en rod på 3.

Fra https://01005.compute.dtu.dk/enotes/13_-_Egenvaerdier_og_eigenvektorer.pdf

||| Sætning 13.39 Egenskaber for multipliciteter

Givet en reel $(n \times n)$ -matrix A .

1. A har højst n forskellige reelle egenverdier, og også summen af de reelle egenverdiers algebraiske multipliciteter er højst n .
2. A har højst n forskellige komplekse egenverdier, og summen af de komplekse egenverdiers algebraiske multipliciteter er lig med n .
3. Er λ en reel eller kompleks egenverdi for A , gælder der:

$$1 \leq \text{gm}(\lambda) \leq \text{am}(\lambda) \leq n \quad (13-58)$$

Det vil sige, at den geometriske multiplicitet af en egenverdi mindst vil være 1, at den vil være mindre end eller lig med den algebraiske multiplicitet af egenverdien, som igen vil være mindre eller lig antallet af søjler og rækker i A .

▼ Egenverdi og vektor

Det er nok mig der misser et eller andet, men jeg kan virkelig ikke se den store forskel mellem egenbaser og værdier, og så nullitet og nulvektorer. Kig i kursusgang 5 hvis der skal bruges mere information

Først opskriver vi matricen som beskriver en spejling langs y-aksen:

$$A := \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} :$$

Denne matrice beskriver at vi går -1 langs x-aksen og ikke rykker på y-aksen, for derefter at bibeholde en normal y vektor. Tænk på 3blue1brown's videoer om dette.

For at kunne finde egenverdier og egenvektorer, laves den karakteristiske matrix X :

$$X := \begin{bmatrix} -1 - \lambda & 0 \\ 0 & 1 - \lambda \end{bmatrix} :$$

Nu kan den karakteristiske ligning findes ved at finde determinanten:

$$((-1 - \lambda) \cdot (1 - \lambda)) - 0 \cdot 0 = (-1 - \lambda)(1 - \lambda) \stackrel{\text{simplify}}{=} \lambda^2 - 1$$

Denne ligning kan vi så sætte lig 0:

$$\lambda^2 - 1 = 0 \xrightarrow{\text{solve}} \{\lambda = 1\}, \{\lambda = -1\}$$

Og vi kan hurtigt se at lambda enten skal være 1 eller -1, selv uden at bruge solve.

For lige at tilføje lidt ekstra, kan vi sige at spektret for egenverdierne i denne opgave er

$\{\lambda = 1\}, \{\lambda = -1\}$ og at den algebraiske multiplicitet er 2, da lambda står i ^2.

Egenvektorne findes ved at sætte egenverdierne ind og derefter finde nulrummet af matricen på reduceret echelonform:

$$\begin{bmatrix} -1 - (-1) & 0 \\ 0 & 1 - (-1) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ \vdots & \vdots \end{bmatrix}$$

Der byttes rundt på række 1 og 2:

$$\begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} :$$

Og der reduceres nu til reduceret echelonform:

$$\begin{bmatrix} \frac{0}{2} & \frac{2}{2} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \vdots & \vdots \end{bmatrix}$$

Nulrummet findes nu ved at tilføre variabler:

$$\begin{bmatrix} 0X_1 & 1X_2 \\ 0X_1 & 0X_2 \end{bmatrix} = \begin{bmatrix} 0 = 1X_2 \\ 0 = 0 \end{bmatrix} :$$

Det kan derefter omskrives til vektoren:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} X_2 :$$

som så er den ene egenvektor og den første vektor i nulrummet.

Den anden vektor er givet ved samme fremgangsmetode:

$$\begin{bmatrix} -1 - 1 & 0 \\ 0 & 1 - 1 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ \vdots & \vdots \end{bmatrix}$$

Række 1 divideres nu med -2:

$$\begin{bmatrix} -\frac{2}{-2} & \frac{0}{-2} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \vdots & \vdots \end{bmatrix}$$

Og nu står matricen på reduceret echelonform

Nulrummet findes nu ved at tilføre variabler:

$$\begin{bmatrix} 1X_1 & 0X_2 \\ 0X_1 & 0X_2 \end{bmatrix} = \begin{bmatrix} 1X_1 = 0 \\ 0 = 0 \end{bmatrix} :$$

og giver vektoren:

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} X_1 :$$

Da der er to beskrivende vektorer af nulrummet, er den geometriske multiplicitet 2 i alt, eller 1 per valgt egenvektor.

▼ **Bekræftende billeder:**

YOUR INPUT

Find the null space of $\begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$.

SOLUTION

The reduced row echelon form of the matrix is $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ (for steps, see [rref calculator](#)).

To find the null space, solve the matrix equation $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

If we take $x_1 = t$, then $x_2 = 0$.

Thus, $\vec{x} = \begin{bmatrix} t \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} t$.

This is the null space.

The nullity of a matrix is the dimension of the basis for the null space.

Thus, the nullity of the matrix is 1.

ANSWER

The basis for the null space is $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ **A**.

The nullity of the matrix is 1 **A**.

YOUR INPUT

Find the null space of $\begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}$.

SOLUTION

The reduced row echelon form of the matrix is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ (for steps, see [ref calculator](#)).

To find the null space, solve the matrix equation $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

If we take $x_2 = t$, then $x_1 = 0$.

Thus, $\vec{x} = \begin{bmatrix} 0 \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} t$.

This is the null space.

The nullity of a matrix is the dimension of the basis for the null space.

Thus, the nullity of the matrix is 1.

ANSWER

The basis for the null space is $\left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ **A**.

The nullity of the matrix is 1 **A**.

YOUR INPUT

Find the eigenvalues and eigenvectors of $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$.

SOLUTION

Start from forming a new matrix by subtracting λ from the diagonal entries of the given matrix:

$$\begin{bmatrix} -\lambda - 1 & 0 \\ 0 & 1 - \lambda \end{bmatrix}.$$

The determinant of the obtained matrix is $(1 - \lambda)(-\lambda - 1)$ (for steps, see [determinant calculator](#)).

Solve the equation $(1 - \lambda)(-\lambda - 1) = 0$.

The roots are $\lambda_1 = -1$, $\lambda_2 = 1$ (for steps, see [equation solver](#)).

These are the eigenvalues.

Next, find the eigenvectors.

- $\lambda = -1$

$$\begin{bmatrix} -\lambda - 1 & 0 \\ 0 & 1 - \lambda \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

The null space of this matrix is $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ (for steps, see [null space calculator](#)).

This is the eigenvector.

- $\lambda = 1$

$$\begin{bmatrix} -\lambda - 1 & 0 \\ 0 & 1 - \lambda \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}$$

The null space of this matrix is $\left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ (for steps, see [null space calculator](#)).

This is the eigenvector.

ANSWER

Eigenvalue: -1 , multiplicity: 1 , eigenvector: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Eigenvalue: 1 , multiplicity: 1 , eigenvector: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Egenvektorer

Nu finder vi egenvektorerne for hver af egenværdierne.

1. Egenværdi $\lambda = -1$:

Vi skal løse $(A - (-1)I)\mathbf{v} = 0$:

$$\left(\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} - (-1) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) \mathbf{v} = 0 \implies \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Denne matrix giver ligningen $2v_2 = 0$, hvilket betyder, at $v_2 = 0$.

Egenvektoren kan dermed skrives som:

$$\mathbf{v} = \begin{pmatrix} v_1 \\ 0 \end{pmatrix} \quad \text{for ethvert } v_1 \neq 0$$

En mulig egenvektor er:

$$\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

2. Egenværdi $\lambda = 1$:

Vi skal løse $(A - I)\mathbf{v} = 0$:

$$\left(\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) \mathbf{v} = 0 \implies \begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Denne matrix giver ligningen $-2v_1 = 0$, hvilket betyder, at $v_1 = 0$.

Egenvektoren kan dermed skrives som:

$$\mathbf{v} = \begin{pmatrix} 0 \\ v_2 \end{pmatrix} \quad \text{for ethvert } v_2 \neq 0$$

En mulig egenvektor er:

$$\mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

3x3 matrice

Først skal den karakteriske matrice (X) dannes ud fra A:

$$A := \begin{bmatrix} 13 & 5 & 2 \\ 2 & 7 & -8 \\ 5 & 4 & 7 \end{bmatrix} :$$

$$X := \begin{bmatrix} 13 - \lambda & 5 & 2 \\ 2 & 7 - \lambda & -8 \\ 5 & 4 & 7 - \lambda \end{bmatrix} :$$

Da vi nu har X, skal determinanten findes, for at vi kan få den karakteriske ligning:

$$(13 - \lambda) \cdot (-1)^{1+1} \cdot ((7 - \lambda) \cdot (7 - \lambda) - (-8) \cdot 4) + 5 \cdot (-1)^{1+2} \cdot (2 \cdot (7 - \lambda) - (-8) \cdot 5) + 2 \cdot (-1)^{1+3} \cdot (2 \cdot 4 - (7 - \lambda) \cdot 5)$$

$$(13 - \lambda) ((7 - \lambda)^2 + 32) - 324 + 20\lambda \quad (1.1.1.8.2.1)$$

simplify

$$-(\lambda - 9)^3 \quad (1.1.1.8.2.2)$$

Men i stedet for bare at få den simplificeret, vil vi gerne først lave det om til en kvadratisk ligning:

$$(13 - \lambda) ((7 - \lambda)^2 + 32) - 324 + 20\lambda \rightarrow$$

$$(13 - \lambda) (7^2 - 14\lambda + \lambda^2 + 32) - 324 + 20\lambda \rightarrow$$

$$(13 - \lambda) \cdot (49 - 14\lambda + \lambda^2 + 32) - 324 + 20\lambda \rightarrow$$

$$(13 - \lambda) \cdot (81 - 14\lambda + \lambda^2) - 324 + 20\lambda \rightarrow$$

$$13 \cdot 81 - 13 \cdot 14\lambda + 13 \cdot \lambda^2 - \lambda \cdot 81 - \lambda \cdot (-14\lambda) - \lambda \cdot \lambda^2 - 324 + 20\lambda \rightarrow$$

$$1053 - 182\lambda + 13\lambda^2 - 81\lambda - (-14\lambda^2) - \lambda^3 - 324 + 20\lambda \rightarrow$$

$$1053 - 182\lambda + 13\lambda^2 - 81\lambda - (-14\lambda^2) - \lambda^3 - 324 + 20\lambda =$$

$$-\lambda^3 + 27\lambda^2 - 243\lambda + 729$$

Nu har vi altså et resultat, som matcher det maple giver os. Det betyder også at vi nu kan benytte maple til at løse ligningen for os, så vi kan få spektret givet:

$$\text{Determinant}(X) = -\lambda^3 + 27\lambda^2 - 243\lambda + 729 \xrightarrow{\text{simplify}} -(\lambda - 9)^3 \xrightarrow{\text{solve for lambda}} [[\lambda = 9], [\lambda = 9], [\lambda = 9]]$$

For at finde den algebraiske multiplicitet kigger vi på leddet mellem simplify og solve, og ser at der står $-(\lambda - 9)^3$.

Det betyder reelt at determinanten kan beskrives ved $-(\lambda - 9)^3 = 0$, hvilket fortæller os at den algebraiske multiplicitet er 3, da lambda opstår som 3 separate løsninger, samtidig med at den i den endelige ligning har en rod på 3.

For at finde den geometriske rod, indsættes 9 på lambdas plads, og vi får derefter en matrice som vha nulrummet kan give os egenvektoren.

For at finde nulrummet skal vi have omskrevet matricen til reduceret echelonform:

$$A := \begin{bmatrix} 13 & -9 & 5 & 2 \\ 2 & 7 & -9 & -8 \\ 5 & 4 & 7 & -9 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 2 \\ 2 & -2 & -8 \\ 5 & 4 & -2 \end{bmatrix}$$

For at få denne på reduceret echelonform divideres række 1 med 4 i første omgang:

$$\begin{bmatrix} \frac{4}{4} & \frac{5}{4} & \frac{2}{4} \\ 2 & -2 & -8 \\ 5 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 1 & \frac{5}{4} & \frac{1}{2} \\ 2 & -2 & -8 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Derefter trækkes række 1 gange 2 fra række 2:

$$\begin{bmatrix} 1 & \frac{5}{4} & \frac{1}{2} \\ 2 - 1 \cdot 2 & -2 - \frac{5}{4} \cdot 2 & -8 - \frac{1}{2} \cdot 2 \\ 5 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 1 & \frac{5}{4} & \frac{1}{2} \\ 0 & -\frac{9}{2} & -9 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Nu ganges række 1 med 5 og trækkes fra række 3:

$$\begin{bmatrix} 1 & \frac{5}{4} & \frac{1}{2} \\ 0 & -\frac{9}{2} & -9 \\ 5 - 1 \cdot 5 & 4 - \frac{5}{4} \cdot 5 & -2 - \frac{1}{2} \cdot 5 \end{bmatrix} = \begin{bmatrix} 1 & \frac{5}{4} & \frac{1}{2} \\ 0 & -\frac{9}{2} & -9 \\ 0 & -\frac{9}{4} & -\frac{9}{2} \end{bmatrix}$$

Så ganges række to med $-\frac{2}{9}$ for at få et 1 tal i midten:

$$\begin{bmatrix} 1 & \frac{5}{4} & \frac{1}{2} \\ 0 \cdot \left(-\frac{2}{9}\right) & -\frac{9}{2} \cdot \left(-\frac{2}{9}\right) & -9 \cdot \left(-\frac{2}{9}\right) \\ 0 & -\frac{9}{4} & -\frac{9}{2} \end{bmatrix} = \begin{bmatrix} 1 & \frac{5}{4} & \frac{1}{2} \\ 0 & 1 & 2 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Derefter kan række 2 gange $\frac{5}{4}$ trækkes fra række 1:

$$\begin{bmatrix} 1 - \frac{0 \cdot 5}{4} & \frac{5}{4} - \frac{1 \cdot 5}{4} & \frac{1}{2} - \frac{2 \cdot 5}{4} \\ 0 & 1 & 2 \\ 0 & -\frac{9}{4} & -\frac{9}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Til sidst kan række 2 ganges med $\frac{9}{4}$ og ligges til række 3:

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ 0 + \frac{0 \cdot 9}{4} & -\frac{9}{4} + \frac{1 \cdot 9}{4} & -\frac{9}{2} + \left(\frac{2 \cdot 9}{4}\right) \end{bmatrix} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Hvilket har givet os matricen på reduceret echelonform.

For så at finde nulrummet sættes variabler ind i matricen og den bliver sat lig med en 0-vektor:

$$\begin{bmatrix} 1 X_1 & 0 X_2 & -2 X_3 \\ 0 X_1 & 1 X_2 & 2 X_3 \\ 0 X_1 & 0 X_2 & 0 X_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} :$$

Ud fra dette kan vi opstille ligningerne:

$$\begin{bmatrix} 1 X_1 - 2 X_3 = 0 \\ 1 X_2 + 2 X_3 = 0 \\ 0 = 0 \end{bmatrix} = \begin{bmatrix} 1 X_1 = 2 X_3 \\ 1 X_2 = -2 X_3 \\ 0 = 0 \end{bmatrix} :$$

Nu har vi så en 1 ubeskrevet variabel, nemlig X_3 .

Derfor kan vi opsætte følgende vektor:

$$\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} X_3 :$$

Lige præcis hvor "1" kommer fra har jeg ikke rigtig fanget, men tror det er fordi vi skal have en variabel mindst 1 gang på z-positionen, og samtlige regneudgaver jeg har brugt siger det samme:

$$NullSpace(A) = \left\{ \begin{bmatrix} 2 \\ -2 \\ \vdots \end{bmatrix} \right\}$$

Billeder af fremgangsmetoden:

YOUR INPUT

Find the eigenvalues and eigenvectors of $\begin{bmatrix} 13 & 5 & 2 \\ 2 & 7 & -8 \\ 5 & 4 & 7 \end{bmatrix}$.

SOLUTION

Start from forming a new matrix by subtracting λ from the diagonal entries of the given matrix

$$\begin{bmatrix} 13 - \lambda & 5 & 2 \\ 2 & 7 - \lambda & -8 \\ 5 & 4 & 7 - \lambda \end{bmatrix}.$$

The determinant of the obtained matrix is $-(\lambda - 9)^3$ (for steps, see [determinant calculator](#)).

Solve the equation $-(\lambda - 9)^3 = 0$.

The roots are $\lambda_1 = 9$, $\lambda_2 = 9$, $\lambda_3 = 9$ (for steps, see [equation solver](#)).

These are the eigenvalues.

Next, find the eigenvectors.

$$\lambda = 9$$

$$\begin{bmatrix} 13 - \lambda & 5 & 2 \\ 2 & 7 - \lambda & -8 \\ 5 & 4 & 7 - \lambda \end{bmatrix} = \begin{bmatrix} 4 & 5 & 2 \\ 2 & -2 & -8 \\ 5 & 4 & -2 \end{bmatrix}$$

The null space of this matrix is $\left\{ \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \right\}$ (for steps, see [null space calculator](#)).

This is the eigenvector.

ANSWER

Eigenvalue: 9 λ , multiplicity: 3 λ , eigenvector: $\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \lambda$.

Determinant udregnes ved:

YOUR INPUT

Calculate $\begin{vmatrix} 13 - \lambda & 5 & 2 \\ 2 & 7 - \lambda & -8 \\ 5 & 4 & 7 - \lambda \end{vmatrix}.$

SOLUTION

Subtract column 1 multiplied by $\frac{7}{2} - \frac{\lambda}{2}$ from column 2: $C_2 = C_2 - \left(\frac{7}{2} - \frac{\lambda}{2}\right) C_1.$

$$\begin{vmatrix} 13 - \lambda & 5 & 2 \\ 2 & 7 - \lambda & -8 \\ 5 & 4 & 7 - \lambda \end{vmatrix} = \begin{vmatrix} 13 - \lambda & -\frac{(\lambda-13)(\lambda-7)}{2} + 5 & 2 \\ 2 & 0 & -8 \\ 5 & \frac{5\lambda}{2} - \frac{27}{2} & 7 - \lambda \end{vmatrix}$$

Add column 1 multiplied by 4 to column 3: $C_3 = C_3 + 4C_1.$

$$\begin{vmatrix} 13 - \lambda & -\frac{(\lambda-13)(\lambda-7)}{2} + 5 & 2 \\ 2 & 0 & -8 \\ 5 & \frac{5\lambda}{2} - \frac{27}{2} & 7 - \lambda \end{vmatrix} = \begin{vmatrix} 13 - \lambda & -\frac{(\lambda-13)(\lambda-7)}{2} + 5 & 54 - 4\lambda \\ 2 & 0 & 0 \\ 5 & \frac{5\lambda}{2} - \frac{27}{2} & 27 - \lambda \end{vmatrix}$$

Expand along row 2:

$$\begin{vmatrix} 13 - \lambda & -\frac{(\lambda-13)(\lambda-7)}{2} + 5 & 54 - 4\lambda \\ 2 & 0 & 0 \\ 5 & \frac{5\lambda}{2} - \frac{27}{2} & 27 - \lambda \end{vmatrix} = \\ (2)(-1)^{2+1} \begin{vmatrix} -\frac{(\lambda-13)(\lambda-7)}{2} + 5 & 54 - 4\lambda \\ \frac{5\lambda}{2} - \frac{27}{2} & 27 - \lambda \end{vmatrix} + (0)(-1)^{2+2} \begin{vmatrix} 13 - \lambda & 54 - 4\lambda \\ 5 & 27 - \lambda \end{vmatrix} + \\ (0)(-1)^{2+3} \begin{vmatrix} 13 - \lambda & -\frac{(\lambda-13)(\lambda-7)}{2} + 5 \\ 5 & \frac{5\lambda}{2} - \frac{27}{2} \end{vmatrix} = -2 \begin{vmatrix} -\frac{(\lambda-13)(\lambda-7)}{2} + 5 & 54 - 4\lambda \\ \frac{5\lambda}{2} - \frac{27}{2} & 27 - \lambda \end{vmatrix}$$

The determinant of a 2x2 matrix is $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$

$$\begin{vmatrix} -\frac{(\lambda-13)(\lambda-7)}{2} + 5 & 54 - 4\lambda \\ \frac{5\lambda}{2} - \frac{27}{2} & 27 - \lambda \end{vmatrix} = \left(-\frac{(\lambda-13)(\lambda-7)}{2} + 5\right) \cdot (27 - \lambda) - (54 - 4\lambda) \left(\frac{5\lambda}{2} - \frac{27}{2}\right) \\ = \frac{\lambda^3}{2} - \frac{27\lambda^2}{2} + \frac{243\lambda}{2} - \frac{729}{2}$$

$$\text{Finally, } (-2) \cdot \left(\frac{\lambda^3}{2} - \frac{27\lambda^2}{2} + \frac{243\lambda}{2} - \frac{729}{2} \right) = -(\lambda - 9)^3.$$

ANSWER

$$\begin{vmatrix} 13 - \lambda & 5 & 2 \\ 2 & 7 - \lambda & -8 \\ 5 & 4 & 7 - \lambda \end{vmatrix} = -(\lambda - 9)^3 \approx -729 (0.111111111111111\lambda -$$

Nulrum udregnes ved:

YOUR INPUT

Find the null space of $\begin{bmatrix} 4 & 5 & 2 \\ 2 & -2 & -8 \\ 5 & 4 & -2 \end{bmatrix}$.

SOLUTION

The reduced row echelon form of the matrix is $\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ (for steps, see [ref calculator](#)).

To find the null space, solve the matrix equation $\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

If we take $x_3 = t$, then $x_1 = 2t$, $x_2 = -2t$.

$$\text{Thus, } \vec{x} = \begin{bmatrix} 2t \\ -2t \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} t.$$

This is the null space.

The nullity of a matrix is the dimension of the basis for the null space.

Thus, the nullity of the matrix is 1.

ANSWER

The basis for the null space is $\left\{ \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \right\}$ **A**.

The nullity of the matrix is 1 **A**.



▼ **Eigenbase**

Eigenbasen er alle vektorer som beskriver matricen:

I have the matrix

$$A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}$$

I've calculated the Eigenvalues and Eigenvectors as follows with help in a previous question:

$\lambda = 2$ or 8

$$E_{\lambda=2} = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = a \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \mid a, b \in \mathbb{R} \right\}$$

$$E_{\lambda=8} = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \mid a \in \mathbb{R} \right\}$$

Is the Eigenbasis simply:

$$E_{\lambda=2} = \text{span} \left(\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right)$$

$$E_{\lambda=8} = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right)$$

linear-algebra

matrices

eigenvalues-eigenvectors

matrix-equations

proof-explanation

Share Cite Follow

edited Aug 25, 2015 at 16:57



Michael Hardy

1

asked Aug 25, 2015 at 16:52



Gabi

157



1



1

▲ 8

Add a comment

Answers

Sorted by: Highest score (default)

An eigenbasis is a basis in which every vector is an eigenvector.

In your case,

1 2 3 4 5 6 7 8 9 10

$$\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

is an eigenbasis for your matrix A .

▼ Kofaktor

Kofaktor er basically at man finder determinanten af en matrice som er 1 række og 1 søjle mindre end den samlede matrice, og så indsætter værdien af den determinant på C_{IJ} pladsen

In our case, the cofactor matrix would be $\begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$.

YOUR INPUT

Find the cofactor matrix of $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

SOLUTION

The cofactor matrix consists of all cofactors of the given matrix, which are calculated according to the formula $C_{ij} = (-1)^{i+j} M_{ij}$, where M_{ij} is the [minor](#), i.e. the determinant of the submatrix formed by deleting row i and column j from the given matrix.

Calculate all cofactors:

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} = -3 \text{ (for steps, see [determinant calculator](#)).$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} = 6 \text{ (for steps, see [determinant calculator](#)).$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = -3 \text{ (for steps, see [determinant calculator](#)).$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix} = 6 \text{ (for steps, see [determinant calculator](#)).$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} = -12 \text{ (for steps, see [determinant calculator](#)).$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = 6 \text{ (for steps, see [determinant calculator](#)).$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} = -3 \text{ (for steps, see [determinant calculator](#)).$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} = 6 \text{ (for steps, see [determinant calculator](#)).$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} = -3 \text{ (for steps, see [determinant calculator](#)).$$

Thus, the cofactor matrix is $\begin{bmatrix} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{bmatrix}$.

ANSWER

The cofactor matrix is $\begin{bmatrix} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{bmatrix}$.

▼ Determinant

$$A := \begin{bmatrix} 1.0 & 0.2 & 1.6 \\ 3.0 & 0.6 & 1.2 \\ 2.0 & 0.8 & 0.4 \end{bmatrix} = \begin{bmatrix} 1.0 & 0.2 & 1.6 \\ 3.0 & 0.6 & 1.2 \\ 2.0 & 0.8 & 0.4 \end{bmatrix}$$

For at finde determinanten med co-factor expansion, skal der først vælges en række og eller en søjle. I dette eksempel benyttes række 1:

Grundformen for denne matrice vil se ud på følgende måde:

$$A = \begin{bmatrix} A_1 & A_2 & A_3 \\ A_4 & A_5 & A_6 \\ A_7 & A_8 & A_9 \end{bmatrix} = A_1 \cdot (-1)^{1+1} \cdot \begin{vmatrix} A_5 & A_6 \\ A_8 & A_9 \end{vmatrix} + A_2 \cdot (-1)^{1+2} \cdot \begin{vmatrix} A_4 & A_6 \\ A_7 & A_9 \end{vmatrix} + A_3 \cdot (-1)^{1+3} \cdot \begin{vmatrix} A_4 & A_5 \\ A_7 & A_8 \end{vmatrix}$$

Når man så når 2x2 matricerne er det nemmeste at finde prikproduktet af disse matricer og indsætte i stedet for matricerne:

$$A = \begin{bmatrix} A_1 & A_2 & A_3 \\ A_4 & A_5 & A_6 \\ A_7 & A_8 & A_9 \end{bmatrix} = A_1 \cdot (-1)^{1+1} \cdot (A_5 \cdot A_9 - A_8 \cdot A_6) + A_2 \cdot (-1)^{1+2} \cdot (A_4 \cdot A_9 - A_7 \cdot A_6) + A_3 \cdot (-1)^{1+3} \cdot (A_4 \cdot A_8 - A_7 \cdot A_5)$$

Derefter får man determinanten af matricen. Med værdier indsat får man derfor:

$$A := \begin{bmatrix} 1.0 & 0.2 & 1.6 \\ 3.0 & 0.6 & 1.2 \\ 2.0 & 0.8 & 0.4 \end{bmatrix} = 1.0 \cdot (-1)^{1+1} \cdot \begin{vmatrix} 0.6 & 1.2 \\ 0.8 & 0.4 \end{vmatrix} + 0.2 \cdot (-1)^{1+2} \cdot \begin{vmatrix} 3.0 & 1.2 \\ 2.0 & 0.4 \end{vmatrix} + 1.6 \cdot (-1)^{1+3} \cdot \begin{vmatrix} 3.0 & 0.6 \\ 2.0 & 0.8 \end{vmatrix} :$$

Og nu kan prikprodukterne af 2x2 matricerne findes:

$$A := \begin{bmatrix} 1.0 & 0.2 & 1.6 \\ 3.0 & 0.6 & 1.2 \\ 2.0 & 0.8 & 0.4 \end{bmatrix} = 1.0 \cdot (-1)^{1+1} \cdot (0.6 \cdot 0.4 - 0.8 \cdot 1.2) + 0.2 \cdot (-1)^{1+2} \cdot (3.0 \cdot 0.4 - 2.0 \cdot 1.2) + 1.6 \cdot (-1)^{1+3} \cdot (3.0 \cdot 0.8 - 2.0 \cdot 0.6)$$

$$A := \begin{bmatrix} 1.0 & 0.2 & 1.6 \\ 3.0 & 0.6 & 1.2 \\ \vdots & \vdots & \vdots \end{bmatrix} = 1.440 \quad (1.1.1.11.1)$$

Altså er determinanten 1.44.

▼ Gaussisk Elimination

Gauss elimination er udført ved at tage

$$\begin{bmatrix} A_1 & A_2 & A_3 & A_4 \\ A_5 & A_6 & A_7 & A_8 \\ A_9 & A_{10} & A_{11} & A_{12} \end{bmatrix} :$$

Først omskriver vi ligningssystemet til en matrice:

$$\begin{bmatrix} -1 & 3 & -2 & 7 \\ 3 & 0 & 3 & -3 \\ 2 & 1 & 2 & -1 \end{bmatrix}$$

Nu flipper vi række 2 og 3, da vi allerede nu kan se vi så har et 0 som står rigtigt:

$$\begin{bmatrix} -1 & 3 & -2 & 7 \\ 2 & 1 & 2 & -1 \\ 3 & 0 & 3 & -3 \end{bmatrix}$$

Da vi nu har omskrevet til en nemmere formel, kan vi påbegynde gauss-elimination:

For at ændre 2 tallet i række 2 søjle 1 til et 0 som det skal være, stiller vi en ligning op som hedder:

$$A_1 \cdot X + A_5 = 0$$

Denne ligning fortæller at tallet i den øverste række skal ganges med x og lægges sammen med den position vi vil ændre til 0, og så skal resultatet være 0.

I denne opgave bliver det altså:

$$-1 \cdot 2 + 2 = 0$$

Nu skal de 2 så ganges på resten af tallene i rækken, og ligges til i rækken nedenunder, så

matricen bliver til:

$$\begin{bmatrix} A_1 & A_2 & A_3 & A_4 \\ A_5 + A_1 \cdot X & A_6 + A_2 \cdot X & A_7 + A_3 \cdot X & A_8 + A_4 \cdot X \\ A_9 & A_{10} & A_{11} & A_{12} \end{bmatrix}$$

Som i opgaven bliver til:

$$\begin{bmatrix} -1 & 3 & -2 & 7 \\ 2 + (-1) \cdot 2 & 1 + 3 \cdot 2 & 2 + (-2) \cdot 2 & -1 + 7 \cdot 2 \\ 3 & 0 & 3 & -3 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 3 & -2 & 7 \\ 0 & 7 & -2 & 13 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.12.1)$$

Næste skridt er at A_9 skal ændre til et 0 også, hvilket vi kan opnå ved at sige:

$$A_1 \cdot X + A_9 = 0$$

Som i dette tilfælde bliver:

$$-1 \cdot 3 + 3 = 0$$

Det gentages så for hele rækken:

$$\begin{bmatrix} -1 & 3 & -2 & 7 \\ 0 & 7 & -2 & 13 \\ 3 + (-1) \cdot 3 & 0 + 3 \cdot 3 & 3 + (-2) \cdot 3 & (-3) + 7 \cdot 3 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 3 & -2 & 7 \\ 0 & 7 & -2 & 13 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.12.2)$$

Nu skal man så indse at hele den nederste række kan divideres med 3, så det gør vi:

$$\begin{bmatrix} -1 & 3 & -2 & 7 \\ 0 & 7 & -2 & 13 \\ 0 & 3 & -1 & 6 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 3 & -2 & 7 \\ 0 & 7 & -2 & 13 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.12.3)$$

Så for at skabe noget der kan gå op i 7-tallet ovenfor, ganges den nederste række nu med 7:

$$\begin{bmatrix} -1 & 3 & -2 & 7 \\ 0 & 7 & -2 & 13 \\ 0 & 21 & -7 & 42 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 3 & -2 & 7 \\ 0 & 7 & -2 & 13 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.12.4)$$

For at kunne ændre 21 til 0, ganger vi 7-tallet ovenover med -3, for at eliminere 21. Altså gør vi:

$$\begin{bmatrix} A_1 & A_2 & A_3 & A_4 \\ A_5 & A_6 & A_7 & A_8 \\ A_9 + A_5 \cdot (-3) & A_{10} + A_6 \cdot (-3) & A_{11} + A_7 \cdot (-3) & A_{12} + A_8 \cdot (-3) \end{bmatrix}$$

Hvor -3-tallet egentlig er x, for at formen passer fra tidligere eksempler.

I denne opgave bliver resultatet:

$$\begin{bmatrix} -1 & 3 & -2 & 7 \\ 0 & 7 & -2 & 13 \\ 0 & 21 + 7 \cdot (-3) & -7 + (-2) \cdot (-3) & 42 + 13 \cdot (-3) \end{bmatrix} \quad \begin{bmatrix} -1 & 3 & -2 & 7 \\ 0 & 7 & -2 & 13 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.12.5)$$

Da vi nu har 3 ligninger (husk at fjerde kolonne er egentlig resultatet af de 3 foregående, som egentlig er en ligning) kan vi regne ud hvad der XYZ er:

$$\begin{bmatrix} -1x + 3y - 2z = 7 \\ 0x + 7y - 2z = 13 \\ 0x + 0y - 1z = 3 \end{bmatrix} :$$

Ud fra disse ligninger kan vi uddrage at:

$$\begin{bmatrix} x = 2 \\ z = -3 \\ y = 1 \end{bmatrix} \rightarrow \begin{bmatrix} x = 2 \\ y = 1 \\ z = -3 \end{bmatrix}$$

Det er vigtigt at huske på z og y er byttet rundt, da vi swappede dem i step 1.

▼ Cramers Formel

Ved at benytte Cramers formel er fremgangsmetoden noget anderledes, da vi nu skal finde determinanten for A_x og for hele matricen også. Først gør vi det for hele matricen:

$$\overline{A} = \begin{bmatrix} -1 & 3 & -2 \\ 3 & 0 & 3 \\ 2 & 1 & 2 \end{bmatrix} \quad \left(\begin{bmatrix} 1.0 & 0.2 & 1.6 \\ 3.0 & 0.6 & 1.2 \\ \vdots & \vdots & \vdots \end{bmatrix} = 1.440 \right) = \begin{bmatrix} -1 & 3 & -2 \\ 3 & 0 & 3 \\ \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.13.1)$$

Ligesom i kursusgang 1, tager vi udgangspunkt i at finde determinanten via den øverste række:

$$\begin{bmatrix} -1 & 3 & -2 \\ 3 & 0 & 3 \\ 2 & 1 & 2 \end{bmatrix} = (-1) \cdot (-1)^{1+1} \cdot (0 \cdot 2 - 1 \cdot 3) + 3 \cdot (-1)^{1+2} \cdot (3 \cdot 2 - 2 \cdot 3) + (-2) \cdot (-1)^{1+3}$$

$$+3 \cdot (3 \cdot 1 - 2 \cdot 0)$$

$$\begin{bmatrix} -1 & 3 & -2 \\ 3 & 0 & 3 \\ \vdots & \vdots & \vdots \end{bmatrix} = -3 \quad (1.1.13.2)$$

Altså er determinanten af A = -3

Nu skal vi så finde determinanten af A_x skal man indsætte resultaterne fra ligningen, altså

$$\begin{array}{rrrrr} -x & + & 3y & - & 2z & = & 7 \\ 3x & & & + & 3z & = & -3 \\ 2x & + & y & + & 2z & = & -1 \end{array} \quad \text{i dette tilfælde, på hhv. 1.}$$

2. og 3. kolonne i matricen (i dette tilfælde er det også x-y-z i kolonnerne). Den determinant skal så divideres med determinanten for den originale matrice. Altså:

$$\begin{bmatrix} A_1 & A_2 & A_3 \\ A_4 & A_5 & A_6 \\ A_7 & A_8 & A_9 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} \rightarrow \begin{bmatrix} A_x & A_2 & A_3 \\ A_y & A_5 & A_6 \\ A_z & A_8 & A_9 \end{bmatrix} \rightarrow \begin{bmatrix} A_1 & A_x & A_3 \\ A_4 & A_y & A_6 \\ A_7 & A_z & A_9 \end{bmatrix} \rightarrow \begin{bmatrix} A_1 & A_2 & A_x \\ A_4 & A_5 & A_y \\ A_7 & A_8 & A_z \end{bmatrix}$$

Hvilket i denne opgave bliver til:

$$\begin{bmatrix} 7 & 3 & (-2) \\ (-3) & 0 & 3 \\ (-1) & 1 & 2 \end{bmatrix} = 7 \cdot (-1)^{1+1} \cdot (0 \cdot 2 - 1 \cdot 3) + (-3) \cdot (-1)^{1+2} \cdot (3 \cdot 2 - (1 \cdot (-2))) + (-1) \cdot (-1)^{1+3} \cdot (3 \cdot 3 - 0 \cdot (-2))$$

$$\begin{bmatrix} 7 & 3 & -2 \\ -3 & 0 & 3 \\ \vdots & \vdots & \vdots \end{bmatrix} = -6 \quad (1.1.13.3)$$

Det er vigtigt at bemærke, at i stedet for at benytte den øverste række som vi plejer, benytter vi nu den første søjle i stedet. Resultatet vi har fået nu er resultat for x, så nu kan den findes for hhv. y og z:

$$\begin{bmatrix} -1 & 7 & -2 \\ 3 & -3 & 3 \\ 2 & -1 & 2 \end{bmatrix} = (-1) \cdot (-1)^{1+1} \cdot ((-3) \cdot 2 - (-1) \cdot 3) + 3 \cdot (-1)^{1+2} \cdot (7 \cdot 2 - (-1) \cdot (-2)) + 2 \cdot (-1)^{1+3} \cdot (7 \cdot 3 - (-3) \cdot (-2))$$

$$\begin{bmatrix} -1 & 7 & -2 \\ 3 & -3 & 3 \\ \vdots & \vdots & \vdots \end{bmatrix} = -3 \quad (1.1.13.4)$$

$$\begin{bmatrix} -1 & 3 & 7 \\ 3 & 0 & -3 \\ 2 & 1 & -1 \end{bmatrix} = (-1) \cdot (-1)^{1+1} \cdot (0 \cdot (-1) - 1 \cdot (-3)) + 3 \cdot (-1)^{1+2} \cdot (3 \cdot (-1) - 1 \cdot 7)$$

$$+ 2 \cdot (-1)^{1+3} \cdot (3 \cdot (-3) - 0 \cdot 7)$$

$$\begin{bmatrix} -1 & 3 & 7 \\ 3 & 0 & -3 \\ \vdots & \vdots & \vdots \end{bmatrix} = 9$$

(1.1.1.13.5)

Disse determinanter kan nu indsættes i en ny matrice, som i virkeligheden er en søjlevektor, som hedder:

$$\begin{bmatrix} \frac{-6}{-3} \\ \frac{-3}{-3} \\ \frac{9}{-3} \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 1 \\ \vdots \end{bmatrix}$$

(1.1.1.13.6)

Hvilket også matcher resultatet fra første del af opgaven, hvor gauss eliminerings bliver benyttet.

▼ Rækkerum

Taget fra besvarelse af opgave 1.5 kursusgang 2

For at finde rækkerummet af en matrice, skal vi først finde ud af hvor mange rækker der er lineært uafhængige. Det er her vi allerede fundet ved at reducere matricen til echelonform.

Nu kan vi så finde rækkerummet ved at indse rangen på 3, er det samme som rækkerummets størrelse. Basen for rækkerummet er bestået af vektorer skabt ud fra de resterende rækker. Altså er basen for rækkerummet i dette tilfælde:

$$\begin{bmatrix} 1 \\ 3 \\ -5 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \\ 2 \\ -7 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ -4 \\ 20 \end{bmatrix} :$$

Og så tror jeg man scorer bonuspoint ved lige at nævne at fra den originale matrice er det rækker 1, 2 og 4 som skaber en base for rækkerummet. Altså er basen for det originale rækkerum:

$$\begin{bmatrix} -2 \\ -5 \\ 8 \\ 0 \\ -17 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ -5 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 7 \\ -13 \\ 5 \\ -3 \end{bmatrix} :$$

Det passer umiddelbart også med at Troels henter til række 1, 2 og 4 i løsningsarket.

▼ Søjlerum

Taget fra besvarelse af opgave 1.5 kursusgang 2

Søjlerummet finder vi ved igen først at få matrixen på echelonform. Når den er givet, kigger vi på pivoterne af vores matrice og finder:

$$\begin{bmatrix} 1 & 3 & -5 & 1 & 5 \\ 0 & 1 & -2 & 2 & -7 \\ 0 & 0 & 0 & -4 & 20 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} :$$

Ud fra dette kan vi se at vores pivoter er til stede i søjle 1, 2 og 4, hvilket i denne opgave også passer med den originale række 1,2 og 4 (HVILKET ER HELT TILFÆLDIGT - tror jeg).

Ud fra disse pivoter er basen for søjlerummet givet ved:

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -4 \\ 0 \end{bmatrix} :$$

Og i den originale matrice vil basen være:

$$\begin{bmatrix} -2 \\ 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 3 \\ 11 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 7 \\ 5 \end{bmatrix} :$$

Hvilket også passer med den meget minimale information som troels giver i sit svarark.

▼ Transponeret Matrix

Ved en transponeret matrice forstås det at vi flipper matricen langs diagonalen, så rækker bliver til søjler og søjler bliver til søjler:

$$Eks := \begin{bmatrix} 26 & 95 & -78 \\ -3 & 63 & -4 \\ -62 & 10 & 5 \\ -83 & -61 & -91 \\ 9 & -26 & -44 \\ 88 & -20 & -38 \end{bmatrix} :$$

$$\text{Transpose}(Eks) = \begin{bmatrix} 26 & -3 & -62 & -83 & 9 & 88 \\ 95 & 63 & 10 & -61 & -26 & -20 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Bemærk at der før var 3 søjler og 6 rækker, og nu 6 søjler og 3 rækker.

YOUR INPUT

Calculate $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}^T$.

SOLUTION

The rows become the columns, and the columns become the rows.

Thus, the transpose of the matrix is $\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$.

ANSWER

The transpose is $\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$ **A**.

▼ Diagonal Matrix

En diagonal matrice har kun værdier langs diagonalen og skal vær nXn stor.

$$\begin{bmatrix} -44 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 \\ 0 & 0 & -59 & 0 \\ 0 & 0 & 0 & 99 \end{bmatrix} :$$

Derudover er der noget advanced math til at lave en matrice diagonal, men da vi ikke har haft det

i kurset, vises det bare her med et screenshot.

YOUR INPUT

Diagonalize $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.

SOLUTION

First, find the eigenvalues and eigenvectors (for steps, see [eigenvalues and eigenvectors calculation](#)).

Eigenvalue: 6, eigenvector: $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

Eigenvalue: 3, eigenvector: $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

Eigenvalue: -2 , eigenvector: $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.

Form the matrix P , whose column i is eigenvector no. i : $P = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$.

Form the diagonal matrix D whose element at row i , column i is eigenvalue no. i : $D = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{bmatrix}$.

The matrices P and D are such that the initial matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix} = PDP^{-1}$.

ANSWER

$$\begin{bmatrix} 1 & 1 & -1 \end{bmatrix}$$

$$P = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \text{ A}$$

$$D = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \text{ A}$$

▼ Rang af en Matrix

Opgave 1.5

Find rangen af A , B og C .

$$A = \begin{Bmatrix} -1 & 3 \\ 3 & -9 \end{Bmatrix} \quad B = \begin{Bmatrix} 30 & -70 & 50 \\ -36 & 84 & -60 \end{Bmatrix} \quad C$$

Check både række og søjle og sammenlign resultaterne.

For matricen C , identificer de elementære transformationsmatricer der bringer C til echelonform, evt. implementer i Matlab eller lignende.

$$A := \begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix}$$

$$A := \begin{bmatrix} -1 & 3 \\ \vdots & \vdots \end{bmatrix} \quad (1.1.1.18.1)$$

$$B := \begin{bmatrix} 30 & -70 & 50 \\ -36 & 84 & -60 \end{bmatrix}$$

$$B := \begin{bmatrix} 30 & -70 & 50 \\ \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.18.2)$$

$$C := \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$$

$$C := \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.18.3)$$

Manuelle løsninger:

For at løse opgaverne benyttes gaussisk elimination (gør sig gældende for ABC)

$$A = \begin{bmatrix} -1 & 3 \\ 3 + (-1) \cdot 3 & -9 + 3 \cdot 3 \end{bmatrix} \quad \begin{bmatrix} -1 & 3 \\ \vdots & \vdots \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ \vdots & \vdots \end{bmatrix} \quad (1.1.1.18.4)$$

$$B = \begin{bmatrix} 30 & -70 & 50 \\ -36 + 30 \cdot 1.2 & 84 + (-70) \cdot 1.2 & -60 + 50 \cdot 1.2 \end{bmatrix} \quad \begin{bmatrix} 30 & -70 & 50 \\ \vdots & \vdots & \vdots \end{bmatrix} = \begin{bmatrix} 30 & -70 & 50 \\ \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.18.5)$$

$$C = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 + 3 \cdot 2 & 42 + 0 \cdot 2 & 24 + 2 \cdot 2 & 54 + 2 \cdot 2 \\ 21 & -21 & 0 & -15 \end{bmatrix} \quad \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.18.6)$$

$$\begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 21 & -21 & 0 & -15 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 21 + 3 \cdot (-7) & -21 + 0 \cdot (-7) & 0 + 2 \cdot (-7) & -15 + 2 \cdot (-7) \end{bmatrix} \quad \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.18.7)$$

$$\begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & -21 & -14 & -29 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & -21 + 42 \cdot 0.5 & -14 + 28 \cdot 0.5 & -29 + 58 \cdot 0.5 \end{bmatrix} \quad \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.18.8)$$

Da vi så kan finde rangen af hver matrice ved at tælle antallet af rækker med andet end 0'er i, bliver resultatet:

$$A = 1$$

$$B = 1$$

$$C = 2$$

MAPLE-LØSNINGER:

$$\text{Rank}(A)$$

$$1$$

(1.1.1.18.9)

$$\text{Rank}(B)$$

$$1$$

(1.1.1.18.10)

$$\text{Rank}(C)$$

$$2$$

(1.1.1.18.11)

$$\text{GaussianElimination}(A)$$

$$\begin{bmatrix} -1 & 3 \\ \vdots & \vdots \end{bmatrix}$$

(1.1.1.18.12)

$$\text{GaussianElimination}(B)$$

$$\begin{bmatrix} 30 & -70 & 50 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

(1.1.1.18.13)

$$\text{GaussianElimination}(C)$$

$$\begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

(1.1.1.18.14)

▼ Nullitet

HUSK AT I OPGAVER MED LIGNINGER ER DET KUN NULRUMMET AF KOEFFECIENTMATRICEN MAN SKAL BRUGE, IKKE AUGMENTED MATRICE

For at finde nulrummet skal vi igen have matricen på echelonform:

$$A := \begin{bmatrix} 1 & 3 & -5 & 1 & 5 \\ 0 & 1 & -2 & 2 & -7 \\ 0 & 0 & 0 & -4 & 20 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} :$$

Men ikke nok med den skal på echelonform, så skal vi have den på reduceret echelonform, så alle pivotpunkter får værdien 1:

For at opnå dette ganger vi række 2 med 3 tre og trækker det fra række 1, så "3" tallet i række 1 søjle 2 bliver 0:

$$\begin{bmatrix} 1 & 3 - 1 \cdot 3 & -5 - (-2) \cdot 3 & 1 - 2 \cdot 3 & 5 - (-7) \cdot 3 \\ 0 & 1 & & -2 & 2 & -7 \\ 0 & 0 & 0 & -4 & 20 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & -5 & 26 \\ 0 & 1 & -2 & 2 & -7 \\ 0 & 0 & 0 & -4 & 20 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Nu kan vi så dividere række 3 med -4 for at ændre det forreste -4 til et 1-tal:

$$\begin{bmatrix} 1 & 0 & 1 & -5 & 26 \\ 0 & 1 & -2 & 2 & -7 \\ \frac{0}{-4} & \frac{0}{-4} & \frac{0}{-4} & -\frac{4}{-4} & \frac{20}{-4} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & -5 & 26 \\ 0 & 1 & -2 & 2 & -7 \\ 0 & 0 & 0 & 1 & -5 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Derefter kan vi gange række 3 med -5 og trække det fra række 1:

$$\begin{bmatrix} 1 - 0 \cdot (-5) & 0 - 0 \cdot (-5) & 1 - 0 \cdot (-5) & -5 - 1 \cdot (-5) & 26 - (-5) \cdot (-5) \\ & 0 & 1 & -2 & 2 & -7 \\ & 0 & 0 & 0 & 1 & -5 \\ & 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 2 & -7 \\ 0 & 0 & 0 & 1 & -5 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Til sidst skal vi gange række 3 med 2 og trække det fra række 2 for at få:

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 + 0 \cdot 2 & 1 + 0 \cdot 2 & -2 + 0 \cdot 2 & 2 - 1 \cdot 2 & -7 - (-5) \cdot 2 \\ 0 & 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -5 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Og vi bekræfter lige resultatet med maple:

$$\text{ReducedRowEchelonForm}(A) = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -5 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Nu kigger vi så på antal søjler med alt andet end rene nuller, hvilket er omsonst at skrive i dette tilfælde, da sådanne søjler ikke findes i denne matrice.

Nu navngives hver søjle:

$$\begin{bmatrix} 1X_1 & 0X_2 & 1X_3 & 0X_4 & 1X_5 \\ 0X_1 & 1X_2 & -2X_3 & 0X_4 & 3X_5 \\ 0X_1 & 0X_2 & 0X_3 & 1X_4 & -5X_5 \\ 0X_1 & 0X_2 & 0X_3 & 0X_4 & 0X_5 \end{bmatrix} :$$

Derefter kan hver række sættes lig med 0, og blive til ligninger:

$$\begin{bmatrix} X_1 + 0X_2 + X_3 + 0X_4 + 1X_5 = 0 \\ 0X_1 + X_2 - 2X_3 + 0X_4 + 3X_5 = 0 \\ 0X_1 + 0X_2 + 0X_3 + X_4 - 5X_5 = 0 \\ 0X_1 + 0X_2 + 0X_3 + 0X_4 + 0X_5 = 0 \end{bmatrix} \rightarrow \begin{bmatrix} X_1 + X_3 + X_5 = 0 \\ X_2 - 2X_3 + 3X_5 = 0 \\ X_4 - 5X_5 = 0 \\ 0 = 0 \end{bmatrix}$$

Nu har vi to variabler tilbage, som vi ikke kender til, nemlig X_3 og X_5 .

Vi kan derfor opstiller følgende ligninger:

$$X_1 = -X_3 - X_5 :$$

$$X_2 = 2X_3 - 3X_5 :$$

$$X_4 = 5X_5 :$$

Med disse ligninger kan vi opstille en vektor med alle vores variabler som hedder:

$$\begin{bmatrix} -X_3 - X_5 \\ 2X_3 - 3X_5 \\ X_3 \\ 5X_5 \\ X_5 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} X_3 + \begin{bmatrix} -1 \\ -3 \\ 0 \\ 5 \\ 0 \end{bmatrix} X_5 :$$

Altså kan nulrummet beskrives med det ovenstående og det bekræftes også af maple:

$$NullSpace(A) = \left\{ \begin{bmatrix} -1 \\ -3 \\ 0 \\ 5 \\ \vdots \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 0 \\ 0 \\ \vdots \end{bmatrix} \right\}$$

Da nulrummet kan beskrives med 2 vektorer, er nulliteten 2.

Yderligere kan vi finde dimensionen af nulrummet ved at trække rangen af matricen fra antallet af søjler: $5 - 3 = 2$

▼ **Billedeksempler:**

$$A \vec{x} = \vec{v}$$

$$\begin{array}{l} 2x + 5y + 3z = -3 \\ 4x + 0y + 8z = 0 \\ 1x + 3y + 0z = 2 \end{array} \rightarrow \begin{array}{c} A \\ \left[\begin{array}{ccc} 2 & 5 & 3 \\ 4 & 0 & 8 \\ 1 & 3 & 0 \end{array} \right] \end{array} \begin{array}{c} \vec{x} \\ \left[\begin{array}{c} x \\ y \\ z \end{array} \right] \end{array}$$

SOLUTION

The reduced row echelon form of the matrix is $\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ (for step 4, see [this calculator](#)).

To find the null space, solve the matrix equation $\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$.

If we take $x_3 = t$, $x_5 = s$, then $x_1 = -s - t$, $x_2 = -3s + 2t$, $x_4 = 5s$.

$$\text{Thus, } \vec{x} = \begin{bmatrix} -s - t \\ -3s + 2t \\ t \\ 5s \\ s \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} t + \begin{bmatrix} -1 \\ -3 \\ 0 \\ 5 \\ 1 \end{bmatrix} s.$$

This is the null space.

The nullity of a matrix is the dimension of the basis for the null space.

Thus, the nullity of the matrix is 2.

ANSWER

$$\left(\begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 0 \\ 5 \\ 1 \end{bmatrix} \right)$$

The basis for the null space is $\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 5 \\ 1 \end{bmatrix} \right\}$. **A**

The nullity of the matrix is **2 A**.

Example

Find an explicit description of $\text{Nul } A$ where

$$A = \begin{bmatrix} 3 & 6 & 6 & 3 & 9 \\ 6 & 12 & 13 & 0 & 3 \end{bmatrix}$$

Solution: Row reduce augmented matrix corresponding

$$\begin{bmatrix} 3 & 6 & 6 & 3 & 9 & 0 \\ 6 & 12 & 13 & 0 & 3 & 0 \end{bmatrix} \sim \dots \sim \begin{bmatrix} 1 & 2 & 0 & 13 & 3 & 0 \\ 0 & 0 & 1 & -6 & -3 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2x_2 - 13x_4 - 33x_5 \\ x_2 \\ 6x_4 + 15x_5 \\ x_4 \\ x_5 \end{bmatrix}$$

Navigation icons: back, forward, search, etc.

Jiwen He, University of Houston

Math 2331, Linear Algebra

Null Spaces, Column Spaces, & Linear Transformations

Null Space

Column Space

Null A

Null Space: Example (cont.)

$$= x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -13 \\ 0 \\ 6 \\ 1 \end{bmatrix} + x_5 \begin{bmatrix} -33 \\ 0 \\ 15 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \check{0} \\ 0 \end{bmatrix} \quad \begin{bmatrix} \hat{0} \\ 0 \end{bmatrix} \quad \begin{bmatrix} \check{1} \\ 1 \end{bmatrix}$$

Then

$$\text{Nul } A = \text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$$

Observations:

1. Spanning set of $\text{Nul } A$, found using the method in the example, is automatically linearly independent:

$$c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -13 \\ 0 \\ 6 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} -33 \\ 0 \\ 15 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$\Rightarrow$

$$c_1 = \text{-----} \quad c_2 = \text{-----} \quad c_3 = \text{-----}$$

2. If $\text{Nul } A \neq \{\mathbf{0}\}$, the the number of vectors in the spanning set for $\text{Nul } A$ equals the number of free variables in $A\mathbf{x} = \mathbf{0}$

▼ Homogent Ligningssystem

1.2 Homogent system $Ax = 0$

Homogent system $Ax = 0$

- Ligningssystemet $Ax = b$ kaldes homogent, hvis $b = 0$ (nulvektoren!).
- Et homogent system har altid mindst én løsning, nemlig $x = 0$ (nulvektoren): Den trivielle løsning.
- Ved reduktion af totalmatricen for $Ax = 0$ til echelonform optræder der kun to tilfælde:

$$\begin{bmatrix} \# & * & * & 0 \\ 0 & \# & * & 0 \\ 0 & 0 & \# & 0 \end{bmatrix} \quad \begin{bmatrix} \# & * & * & 0 \\ 0 & 0 & \# & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- I den første situation er præcis én løsning, nemlig den trivielle.

1

- I den anden situation er der uendeligt mange løsninger. I eksemplet er x_2 fri og x_1 og x_3 basale variable.

▼ Inhomogent Ligningssystem

1.3 Inhomogent system $Ax = b$

Inhomogent system $Ax = b$

- *Theorem 6 Struktursætningen.* Den fuldstændige løsning til det inhomogene system $Ax = b$ kan skrives som summen af en partikulær løsning og den fuldstændige løsning til den homogene ligning $Ax = 0$.
- Forudsat selvfølgelig, at systemet $Ax = b$ overhovedet har en løsning!.
- Bevis: Første del: Forskellen mellem to løsninger til den inhomogene ligning løser den homogene:

$$Ax = b \wedge Ay = b \implies Ax - Ay = 0 \implies A(x - y) = 0$$

- Anden del: Summen af en løsning til den inhomogene og en løsning til den homogene løser den inhomogene:

$$Ax = b \wedge Az = 0 \implies Ax + Az = b \implies A(x + z) = b$$

▼ Vektorrum

What do you understand by the term vector space?

Vector space can be imagined as a 3D space with a set of objects known as vectors, that can perform certain properties.

Answer: A vector space is a set of vectors that can be added together and multiplied by a scalar.

Go through the definition and various axioms related to vector spaces.

Explanation:

Definition:

A field is a set F of numbers with the property that if $a, b \in F$, then $a + b$, $a - b$, ab , and a / b are also in F (assuming, of course, that b not equal to 0 in the expression a / b)

Example:

We're familiar with how to add, subtract, multiply, and divide the following sets of numbers

$$N = \{0, 1, 2, 3, \dots\}$$

$$Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

$$Q = \{a/b \mid a, b \in Z, b \text{ not equal to } 0\}$$

$$R = \text{all real numbers}$$

$$C = \{a + bi \mid a, b \in R\}$$

However, not all of these sets of numbers are fields of numbers. For example, 3 and 5 are in N , but $3 - 5$ are not. Also, 3 and 5 are in Z , but $3 / 5$ are not. This shows that N and Z are not fields of numbers. However, Q , R , and C are all fields of numbers

Axioms of Vector Spaces:

Vector can be defined using various **axioms**. Let x, y and z be the elements of vector space V and a & b be the elements of the field F .

- 1) Closed under Addition: For every element x, y in V , $x + y$ is also in V
- 2) Closed under **Scalar Multiplication**: For every element x in V and scalar a in F , ax is in V
- 3) Commutativity of Addition: For every element x, y in V , $x + y = y + x$
- 4) Existence of Multiplicative Identity: There exists an element in F notated as 1 so that for all x in V , $1x = x$.
- 5) Existence of Additive Inverse: For every element x in V , there exists another element in V such that $x + (-x) = 0$
- 6) Existence of Additive Identity
- 7) Associativity of **Scalar Multiplication**
- 8) Distribution of scalars to element
- 9) Distribution of elements to scalar
- 10) Associativity of Scalar Multiplication

Thus, a vector space is a set of vectors that can be added together and multiplied by a scalar.

Et vektorrum (eller lineært rum) er en samling af objekter kaldet vektorer, som kan tilføjes sammen og skaleres ved hjælp af skalarer (tallet fra en given felt, typisk de reelle tal $\mathbb{R}$ eller komplekse tal $\mathbb{C}$). Et vektorrum skal opfylde visse aksiomer, der beskriver hvordan vektoraddition og skalarmultiplikation fungerer.

Definition af et vektorrum

Et vektorrum V over et felt F (typisk $\mathbb{R}$ eller $\mathbb{C}$) er en mængde udstyret med to operationer:

1. **Vektoraddition:** En binær operation, der tager to vektorer $\mathbf{u}, \mathbf{v} \in V$ og returnerer en tredje vektor $\mathbf{u} + \mathbf{v} \in V$.
2. **Skalarmultiplikation:** En operation, der tager en skalar $a \in F$ og en vektor $\mathbf{v} \in V$ og returnerer en ny vektor $a\mathbf{v} \in V$.

Disse operationer skal opfylde følgende aksiomer for alle $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ og skalarer $a, b \in F$:

Aksiomer for vektoraddition

1. **Kommutativitet:** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
2. **Associativitet:** $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
3. **Nulvektor:** Der eksisterer en vektor $\mathbf{0} \in V$ sådan at $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for alle $\mathbf{v} \in V$.
4. **Additiv invers:** For hver vektor $\mathbf{v} \in V$ eksisterer en vektor $-\mathbf{v} \in V$ sådan at $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.

Aksiomer for skalarmultiplikation

1. **Distributivitet mht. vektorer:** $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
2. **Distributivitet mht. skalarer:** $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$
3. **Associativitet:** $a(b\mathbf{v}) = (ab)\mathbf{v}$
4. **Enhedsvektor:** $1\mathbf{v} = \mathbf{v}$ (hvor 1 er multiplikativ identitet i F)

Eksempler på vektorrum

1. **Reelle koordinatrum:** Mængden af n -tupler af reelle tal, $\mathbb{R}^n$, med de sædvanlige addition og skalarmultiplikation.
2. **Komplekse koordinatrum:** Mængden af n -tupler af komplekse tal, $\mathbb{C}^n$.

3. **Funktionenrum:** Mængden af alle kontinuerte funktioner fra et interval $[a, b]$ til $\mathbb{R}$.
4. **Polynomier:** Mængden af polynomier af grad højst n , P_n .

Basis og dimension

- **Basis:** En mængde af vektorer i et vektorrum V kaldes en basis, hvis vektorene er lineært uafhængige og spænder hele rummet. Enhver vektor i V kan skrives som en linearkombination af basisvektorene.
- **Dimension:** Dimensionen af et vektorrum er antallet af vektorer i en basis for rummet.

For eksempel, i $\mathbb{R}^3$, er $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, hvor $\mathbf{e}_1 = (1, 0, 0)$, $\mathbf{e}_2 = (0, 1, 0)$, og $\mathbf{e}_3 = (0, 0, 1)$ en basis, og dimensionen af $\mathbb{R}^3$ er 3.

▼ Base for et Vektorrum

Basis og dimension

- **Basis:** En mængde af vektorer i et vektorrum V kaldes en basis, hvis vektorene er lineært uafhængige og spænder hele rummet. Enhver vektor i V kan skrives som en linearkombination af basisvektorene.
- **Dimension:** Dimensionen af et vektorrum er antallet af vektorer i en basis for rummet.

For eksempel, i $\mathbb{R}^3$, er $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, hvor $\mathbf{e}_1 = (1, 0, 0)$, $\mathbf{e}_2 = (0, 1, 0)$, og $\mathbf{e}_3 = (0, 0, 1)$ en basis, og dimensionen af $\mathbb{R}^3$ er 3.

▼ Spænd af Vektorer

Et spænd af vektorer beskriver hvor stort et område vektorene kan nå. Det vil altså sige, alle de koordinater vektorene kan nå fra origo. Hvis vektorne beskrevet af opgavet eksempelvis kun er 2 dimensionelle, er spændet af vektorer begrænset til 2 dimension - men ikke nødvendigvis til XY planet eller andre velkendte plan.

In **mathematics**, the **linear span** (also called the **linear hull**^[1] or just **span**) of a **set** S of **vectors** (from a **vector space**), denoted $\text{span}(S)$,^[2] is defined as the set of all **linear combinations** of the vectors in S .^[3] For example, two **linearly independent vectors** span a **plane**. The linear span can be characterized either as the **intersection** of all **linear subspaces** that contain S , or as the smallest subspace containing S . The linear span of a set of vectors is therefore a vector space itself. Spans can be generalized to **matroids** and **modules**.

To express that a vector space V is a linear span of a subset S , one commonly uses the following phrases—either: S spans V , S is a **spanning set** of V , V is spanned/generated by S , or S is a **generator** or generator set of V .

▼ Lineær afhængighed

Opgave 1.6

Check lineær uafhængighed for vektorerne:

- a) $a_1 = [-3, 6, -1, 1, -7]$, $a_2 = [1, -2, 2, 3, -1]$ og $a_3 = [2, -4, 5, 8, -4]$
 b) $a_1 = [-3, 1, 2]$, $a_2 = [-7, -1, -4]$, $a_3 = [6, -2, -4]$, $a_4 = [1, 3, 8]$

Sammenlign resultatet for de to tilfælde.

▼ A)

Først skal vi have omskrevet vektorerne til en matrice:

$$A := \begin{bmatrix} -3 & 6 & -1 & 1 & -7 & 0 \\ 1 & -2 & 2 & 3 & -1 & 0 \\ 2 & -4 & 5 & 8 & -4 & 0 \end{bmatrix} :$$

Det er vigtigt at bemærke, at de 3 nuller til sidst er tilføjet, da hver vektor skal give 0, for at finde ud af om de er lineært uafhængige, så det der egentlig er sket er:

$$\begin{bmatrix} \widehat{A_1} = 0 \\ \widehat{A_2} = 0 \\ \widehat{A_3} = 0 \end{bmatrix} :$$

Man kan finde ud af om de er lineært uafhængige ved enten at finde determinanten, eller ved at bruge den generelle formel. Det er den generelle formel som vil blive benyttet her, da tricket med determinanten kun kan benyttes på kvadratiske matricer.

Den generelle formel siger at:

Den kan så reduceres yderligere, for at få samme resultat som maple og matlab (uden vi lige kan lure hvorfor). Det gøres ved at gange række 2 med -2 på række 1:

$$\begin{bmatrix} 1 & -2 & 2 + 1 \cdot (-2) & 3 + 2 \cdot (-2) & -1 + (-2) \cdot (-2) & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.25.1.5)$$

Da vi kan se der er nuller i bunden, kan vi repræsentere den sidste vektor med de to først, og altså er de lineært afhængige

▼ *Chr's fuckup*

Først skal vi have omskrevet vektorerne til en matrice:

$$A := \begin{bmatrix} -3 & 6 & -1 & 1 & -7 & 0 \\ 1 & -2 & 2 & 3 & -1 & 0 \\ 2 & -4 & 5 & 8 & -4 & 0 \end{bmatrix} :$$

Det er vigtigt at bemærke, at de 3 nuller til sidst er tilføjet, da hver vektor skal give 0, for at finde ud af om de er lineært uafhængige, så det der egentlig er sket er:

$$\begin{bmatrix} \widehat{A_1} = 0 \\ \widehat{A_2} = 0 \\ \widehat{A_3} = 0 \end{bmatrix} :$$

Man kan finde ud af om de er lineært uafhængige ved enten at finde determinanten, eller ved at bruge den generelle formel. Det er den generelle formel som vil blive benyttet her, da tricket med determinanten kun kan benyttes på kvadratiske matricer.

Den generelle formel siger at:

$$C_1 \cdot \widehat{A_1} + C_2 \cdot \widehat{A_2} + C_3 \cdot \widehat{A_3} = 0$$

For at finde C værdierne, opstilles en matrice med alle vektorer lig 0, og vha gaussisk elimination vil vi reducere udtrykket til echelonform. Først flippes række 1 og 2:

$$\begin{bmatrix} -3 & 6 & -1 & 1 & -7 & 0 \\ 1 & -2 & 2 & 3 & -1 & 0 \\ 2 & -4 & 5 & 8 & -4 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ -3 & 6 & -1 & 1 & -7 & 0 \\ 2 & -4 & 5 & 8 & -4 & 0 \end{bmatrix} :$$

Dernæst indser vi at vi ved at gange med 3 kan skabe nogle nuller:

$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ -3 + 1 \cdot 3 & 6 + (-2) \cdot 3 & -1 + 2 \cdot 3 & 1 + 3 \cdot 3 & -7 + (-1) \cdot 3 & 0 + 0 \cdot 3 \\ 2 & -4 & 5 & 8 & -4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 5 & 10 & -10 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.25.1.1.1)$$

Nu kan vi så gange række 1 på række 3 sammen med -2 to og få:

$$\begin{aligned} & [[1, -2, 2, 3, -1, 0], \\ & \quad [0, 0, 5, 10, -10, 0], \\ & \quad [2 + 1 \cdot (-2), -4 + (-2) \cdot (-2), 5 + 2 \cdot (-2), 8 + 3 \cdot (-2), -4 + (-1) \cdot (-2), 0 + 0 \cdot (-2)]] \end{aligned}$$

$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 5 & 10 & -10 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.25.1.1.2)$$

Som giver os mulighed for at flippe række 2 og 3:

$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 5 & 10 & -10 & 0 \end{bmatrix}$$

Nu kan vi så gange række 2 på række 3 sammen med -5 og få en nul-række:

$$\begin{bmatrix} 1 & -2 & & 2 & & 3 & & -1 & 0 \\ 0 & 0 & & 1 & & 2 & & -2 & 0 \\ 0 & 0 & 5 + 1 \cdot (-5) & 10 + 2 \cdot (-5) & -10 + (-2) \cdot (-5) & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.25.1.1.3)$$

Nu er matricen så reduceret til sin echelonform. Herfra kigger vi på den søjle længst mod højre, som ikke bare er 0:

$$\begin{bmatrix} -1 = 0 \\ -2 = 0 \\ 0 = 0 \end{bmatrix} :$$

Nu skal vi så mindes hvilken række som hører til hvilken C værdi, i dette tilfælde er det:

$$\begin{bmatrix} C_2 \\ C_3 \\ C_1 \end{bmatrix} :$$

Så skal vi nu se det som om at C_2 og C_3 kan findes vha en C_1 værdi, som vi sætter til 1 for nemheds skyld. Altså må 1 C_2 være lig 1 C_1 og 1 C_3 være lig 2 C_2:

$$\begin{bmatrix} C_2 \rightarrow -1 = C_1 \\ C_3 \rightarrow -2 = 2 C_1 \\ C_1 \rightarrow C_1 = C_1 \end{bmatrix} :$$

Resultatet kan så dobbelttjekkes ved at indsætte værdierne i grundformlen:

$$C_1 \cdot \widehat{A}_1 + C_2 \cdot \widehat{A}_2 + C_3 \cdot \widehat{A}_3 = 0 \rightarrow 1 \cdot \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \end{bmatrix} + 1 \cdot \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \end{bmatrix} \\ + 2 \cdot \begin{bmatrix} 2 & -4 & 5 & 8 & -4 \end{bmatrix} \\ 1 \cdot \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \end{bmatrix} + 1 \cdot \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 2 & -4 & 5 & 8 & -4 \end{bmatrix} \\ \begin{bmatrix} \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad (1.1.1.25.1.1.4)$$

$$1 \cdot \begin{bmatrix} -3 \\ 6 \\ -1 \\ 1 \\ -7 \end{bmatrix} + 1 \cdot \begin{bmatrix} 1 \\ -2 \\ 2 \\ 3 \\ -1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 2 \\ -4 \\ 5 \\ 8 \\ -4 \end{bmatrix} \\ \begin{bmatrix} 2 \\ -4 \\ 11 \\ 20 \\ \vdots \end{bmatrix} \quad (1.1.1.25.1.1.5)$$

▼ B)

$$A := \begin{bmatrix} -3 & 1 & 2 \\ -7 & -1 & -4 \\ 6 & -2 & -4 \\ 1 & 3 & 8 \\ -1 & 2 & 5 \end{bmatrix} = \begin{bmatrix} -3 & 1 & 2 \\ -7 & -1 & -4 \\ 6 & -2 & -4 \\ 1 & 3 & 8 \\ -1 & 2 & 5 \end{bmatrix}$$

Ligesom før vil vi forsøge at reducere matricen til sin echelonform. Først flipper vi række 1 med række 4:

$$\begin{bmatrix} -3 & 1 & 2 \\ -7 & -1 & -4 \\ 6 & -2 & -4 \\ 1 & 3 & 8 \\ -1 & 2 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 8 \\ -7 & -1 & -4 \\ 6 & -2 & -4 \\ -3 & 1 & 2 \\ -1 & 2 & 5 \end{bmatrix}$$

Herfra får vi rettet første søjle til, så det er rene nuller:

$$\begin{bmatrix} 1 & 3 & 8 \\ -7 + 1 \cdot 7 & -1 + 3 \cdot 7 & -4 + 8 \cdot 7 \\ 6 + 1 \cdot (-6) & -2 + 3 \cdot (-6) & -4 + 8 \cdot (-6) \\ -3 + 1 \cdot 3 & 1 + 3 \cdot 3 & 2 + 8 \cdot 3 \\ -1 + 1 \cdot 1 & 2 + 3 \cdot 1 & 5 + 8 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 8 \\ 0 & 20 & 52 \\ 0 & -20 & -52 \\ 0 & 10 & 26 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Herfra kan vi så gange række 2 og 3 sammen med 1:

$$\begin{bmatrix} 1 & 3 & 8 \\ 0 & 20 + (-20) \cdot 1 & 52 + (-52) \cdot 1 \\ 0 & -20 & -52 \\ 0 & 10 & 26 \\ 0 & 5 & 13 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 8 \\ 0 & 0 & 0 \\ 0 & -20 & -52 \\ 0 & 10 & 26 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Nu kan række 4 ganges med 2, og ganges på række 3:

$$\begin{bmatrix} 1 & 3 & 8 \\ 0 & 0 & 0 \\ 0 & -20 + 10 \cdot 2 & -52 + 26 \cdot 2 \\ 0 & 10 & 26 \\ 0 & 5 & 13 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 10 & 26 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Ligeledes kan række 5 ganges med 2 og ganges på række 4:

$$\begin{bmatrix} 1 & 3 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 10 + 5 \cdot (-2) & 26 + 13 \cdot (-2) \\ 0 & 5 & 13 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Nu kan række 2 så flippes med række 5, og vi har matricen på sin echelonform:

$$\begin{bmatrix} 1 & 3 & 8 \\ 0 & 5 & 13 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Den kan omskrives til sin reducerede echelonform ved at dividere række 2 med 5:

$$\begin{bmatrix} 1 & 3 & 8 \\ 0 & \frac{5}{5} & \frac{13}{5} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 8 \\ 0 & 1 & \frac{13}{5} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Herefter gange række 2 på række 1 sammen med -3:

$$\begin{bmatrix} 1 & 3 + 1 \cdot (-3) & 8 + \frac{13}{5} \cdot (-3) \\ 0 & 1 & \frac{13}{5} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & \frac{1}{5} \\ 0 & 1 & \frac{13}{5} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

(1.1.1.25.2.1)

Og resultatet kan blive verificeret af maple:

ReducedRowEchelonForm(A)

$$\begin{bmatrix} 1 & 0 & \frac{1}{5} \\ 0 & 1 & \frac{13}{5} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

(1.1.1.25.2.2)

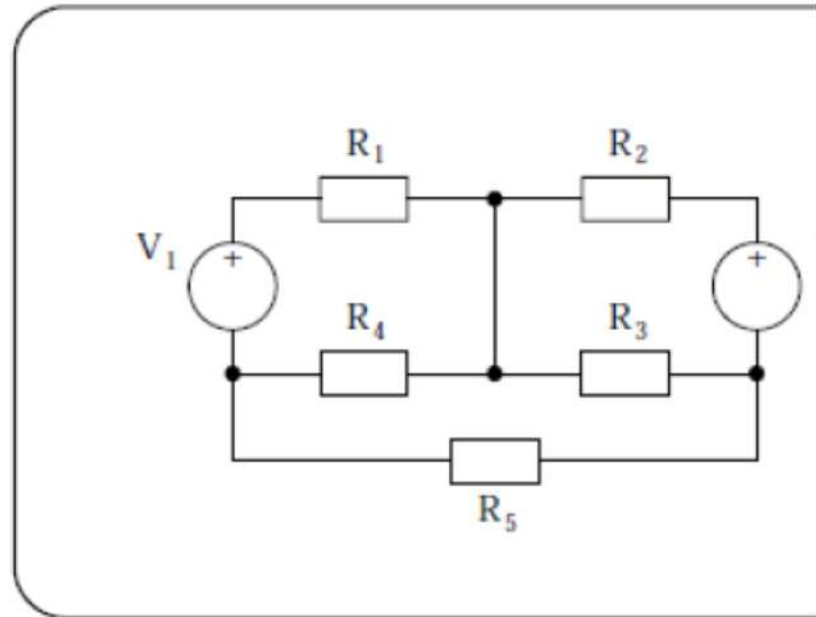
Ud fra dette kan vi konkludere at matricen er lineært afhængig.

▼ Lineær Algebra i et Kredsløb

Opgave 1.3

Beregn strømmen igennem R_3 . Plusretningen er tilhøjre på
mgerne vha. Kirchoffs maskeligninger og løs ligningssystem
Generatorer og modstande har følgende værdier:

$$\begin{array}{llll} V_1 = 1 \text{ V} & R_1 = 1 \text{ } \Omega & R_3 = 3 \text{ } \Omega & R_5 = 5 \text{ } \Omega \\ V_2 = 2 \text{ V} & R_2 = 2 \text{ } \Omega & R_4 = 4 \text{ } \Omega & \end{array}$$



Vi starter lige med at stille variabler op:

$$V_1 := 1 :$$

$$V_2 := 2 :$$

$$R_1 := 1 :$$

$$R_2 := 2 :$$

$$R_3 := 3 :$$

$$R_4 := 4 :$$

$$R_5 := 5 :$$

▼ Forkerte fremgangsmetoder/ihvertfald nogen hvor jeg fucker op

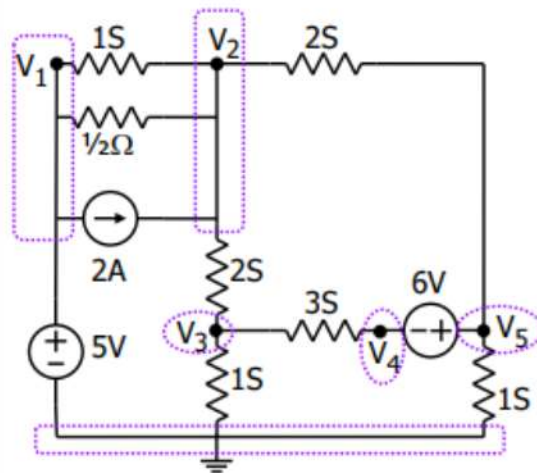
Nu kan vi opstille maskeligninger. For at stille dem korrekt op bruger jeg slide 12 og 13 fra Ole Kiels slides i KDS kursusgang 2:



Systematisk opstilling af ligninger

1. Identificer alle knudepunkterne (N inkl. stel)
 2. Udnævn et knudepunkt til stel
 3. Opstil ligninger for alle (N-1) knudepunkter (Knodepunktsmetoden eller KCL)
 4. Opstil ligninger ordnet
 5. Opstil og løs matrix
- => Altid resultat

Strøm



$$V_1:$$

$$V_2: (1 + 2 + 2 + 2)S \cdot V_2 - (1 + 2)S \cdot V_1$$

$$V_3: (1 + 2 + 3)S \cdot V_3$$

$$V_4: (3 + 1 + 2)S \cdot V_4 - 3$$

$$V_5:$$

Enheder udeladt:

$$\begin{array}{rrrr} V_1 & -0 \cdot V_2 & -0 \cdot V_3 & -0 \cdot V_4 & -0 \cdot V_5 \\ -3 \cdot V_1 & +7 \cdot V_2 & -2 \cdot V_3 & -0 \cdot V_4 & -2 \cdot V_5 \\ -0 \cdot V_1 & -2 \cdot V_2 & +6 \cdot V_3 & -3 \cdot V_4 & -0 \cdot V_5 \\ -0 \cdot V_1 & -2 \cdot V_2 & -3 \cdot V_3 & +6 \cdot V_4 & -0 \cdot V_5 \\ -0 \cdot V_1 & -0 \cdot V_2 & -0 \cdot V_3 & -V_4 & +V_5 \end{array}$$

ESD1: Kredslobsteori #3

2022



Systematisk opstilling af ligninger

$$\begin{array}{rrrrrr} V_1 & -0 \cdot V_2 & -0 \cdot V_3 & -0 \cdot V_4 & -0 \cdot V_5 & = & 5 \\ -3 \cdot V_1 & +7 \cdot V_2 & -2 \cdot V_3 & -0 \cdot V_4 & -2 \cdot V_5 & = & 2 \\ -0 \cdot V_1 & -2 \cdot V_2 & +6 \cdot V_3 & -3 \cdot V_4 & -0 \cdot V_5 & = & 0 \\ -0 \cdot V_1 & -2 \cdot V_2 & -3 \cdot V_3 & +6 \cdot V_4 & -0 \cdot V_5 & = & -18 \\ -0 \cdot V_1 & -0 \cdot V_2 & -0 \cdot V_3 & -V_4 & +V_5 & = & 6 \end{array}$$

Ligningerne kan beskrives med matricer:
(som I lærer om senere)

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -3 & 7 & -2 & 0 & -2 \\ 0 & -2 & 6 & -3 & 0 \\ 0 & -2 & -3 & 6 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 0 \\ -18 \\ 6 \end{bmatrix}$$

$$\underline{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -3 & 7 & -2 & 0 & -2 \\ 0 & -2 & 6 & -3 & 0 \\ 0 & -2 & -3 & 6 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}$$

Konduktansmatrix

$$\underline{V} = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{bmatrix}$$

Spændingsvektor

$$\underline{I} = \begin{bmatrix} 5 \\ 2 \\ 0 \\ -18 \\ 6 \end{bmatrix}$$

Strømvektor

$$\underline{G} \cdot \underline{V} = \underline{I}$$

Matlab: (";" = ny række)

G = [1 0 0 0 0; -3 7 -2 0 -2; 0 -2 6 -3 0; 0 -2 -3 6 0; 0 0 0 -1 1];

I = [5; 2; 0; -18; 6];

V = G^-1 * I % (eller V = G \ I, ikke V = I / G)

V =

5.0000

3.9231

0.6154

-1.3846

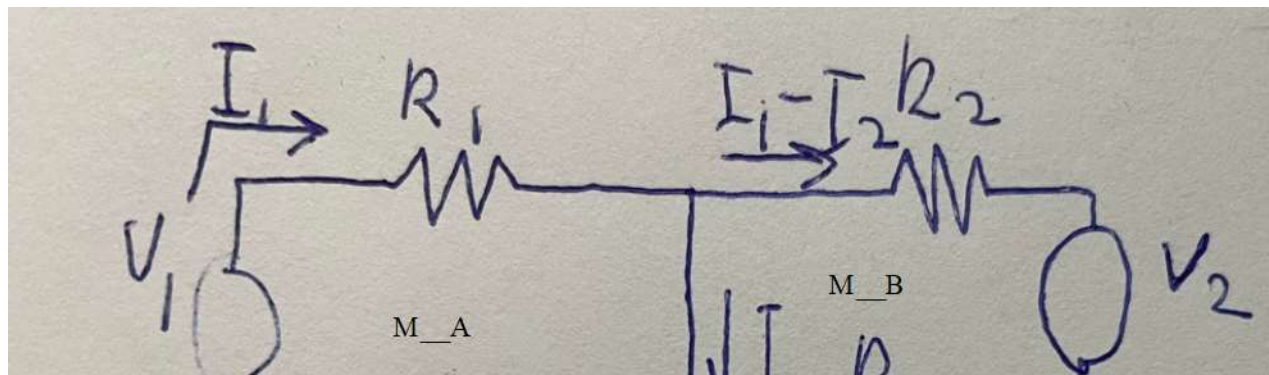
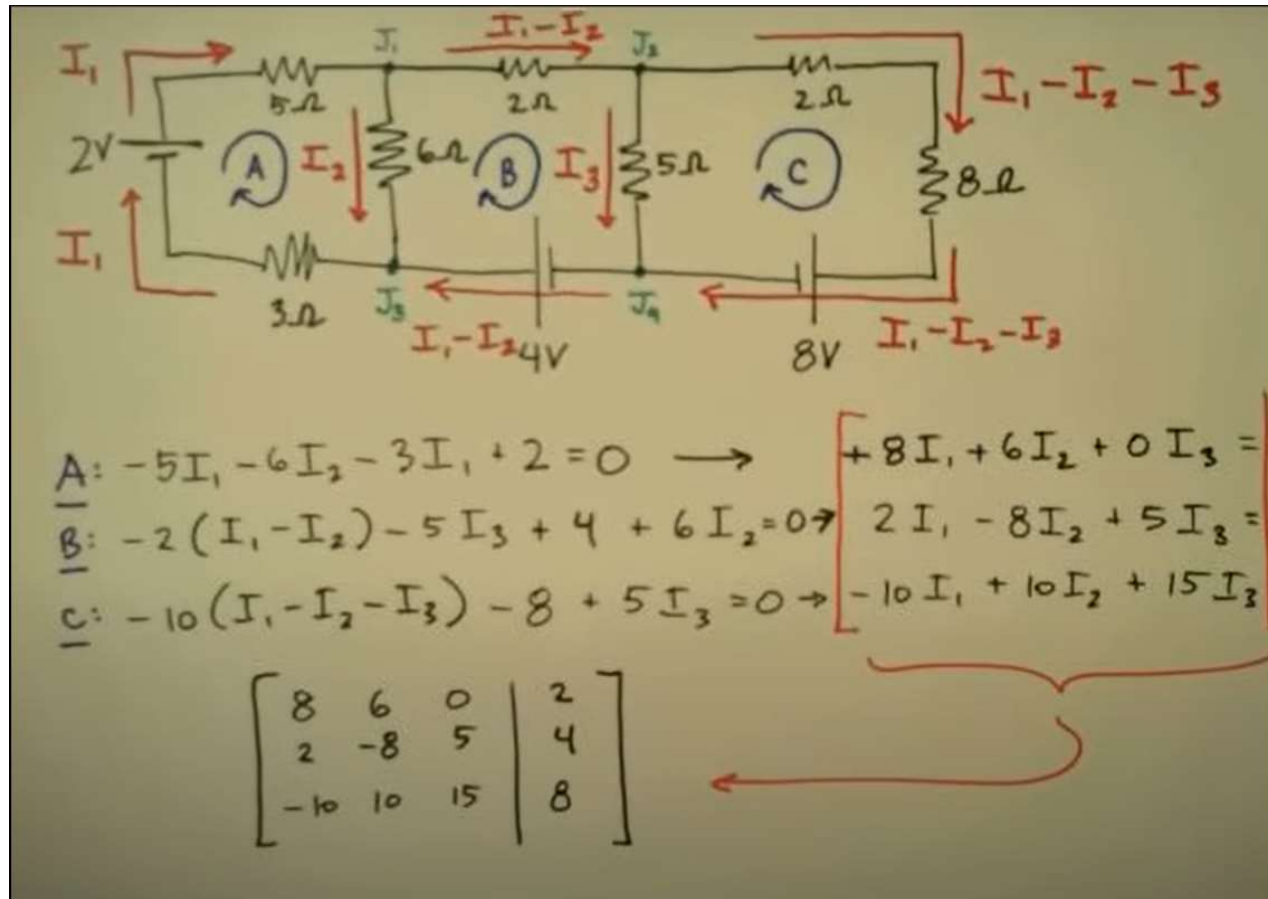
4.6154

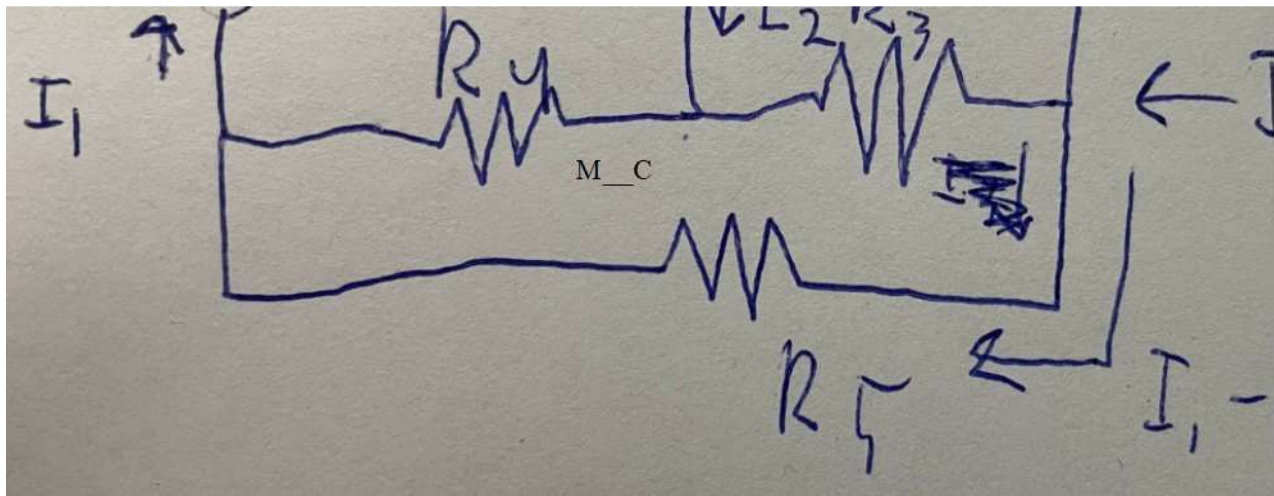
D.v.s:

- Der findes en metode til at løse ligningerne
- Baggrunden lærer I om senere

Eller, det troede jeg at jeg ville. I stedet benytter jeg følgende video: https://www.youtube.com/watch?v=_xomX-d8XU4

Fremgangsmetoden er at vi laver masker og derefter bestemmer en retning. Jeg læser opgaven som om strømmen løber fra venstre mod højre. Screenshottet herunder er fra videoen og billedet er min jank version:





$$M_A := -R_1 \cdot I_1 - R_4 \cdot I_2 - 0 \cdot I_3 + V_1 = 0 :$$

$$M_B := -R_2 \cdot (I_1 - I_2) + V_2 - R_3 \cdot I_3 = 0 :$$

$$M_C := -R_5 \cdot (I_1 - I_2 - I_3) = 0 :$$

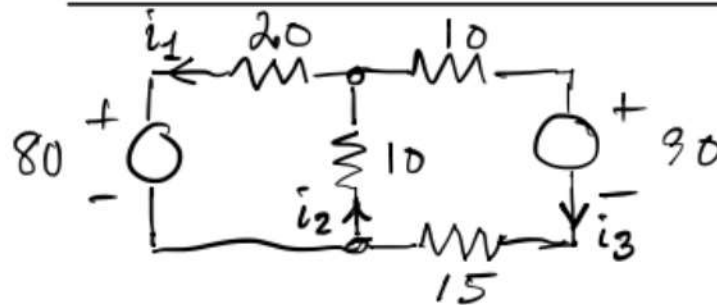
Da vi nu har et sæt ligninger magen til dem fra videoen, kan vi opstille en matrix ud fra dem:

$$A := \begin{bmatrix} -R_1 & -R_2 & 0 & V_1 \\ -R_2 & R_2 & R_3 & V_2 \\ -R_5 & R_5 & R_5 & 0 \end{bmatrix} = \begin{bmatrix} -1 & -2 & 0 & 1 \\ -2 & 2 & 3 & 2 \\ -5 & 5 & 5 & 0 \end{bmatrix}$$

Hvilket for some reason er forkert, så

nu prøver jeg troels' fremgangsmetode med først at opstille KCL ligninger og derefter KVL ligninger:

Ex.: Solution by Gauss elimination



$$\begin{aligned} i_1 - i_2 + i_3 &= 0 \\ -i_1 + i_2 - i_3 &= 0 \\ 10i_2 + 20i_3 &= 90 \\ 20i_1 + 10i_2 &= 80 \end{aligned}$$

$$\bar{X} = \bar{Z}$$

$$\bar{A} \approx \begin{bmatrix} 1 & -1 & 1 & | & 0 \\ -1 & 1 & -1 & | & 0 \\ 0 & 10 & 25 & | & 90 \\ 20 & 10 & 0 & | & 80 \end{bmatrix} \xrightarrow{\text{Gauss}} \begin{bmatrix} 1 & -1 & 1 & | & 0 \\ 0 & 10 & 25 & | & 90 \\ 0 & 0 & -95 & | & 90 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

using partial pivoting (swapping)

$$I_1 - I_2 + I_3 = 0:$$

$$-I_1 + I_2 - I_3 = 0:$$

$$R_1 \cdot I_1 + R_4 \cdot I_1 = V_1:$$

$$R_2 \cdot I_2 + R_3 \cdot I_2 = V_2:$$

$$R_4 \cdot I_1 + R_3 \cdot I_2 + R_5 \cdot I_3 = 0:$$

Som på matrixform bliver til:

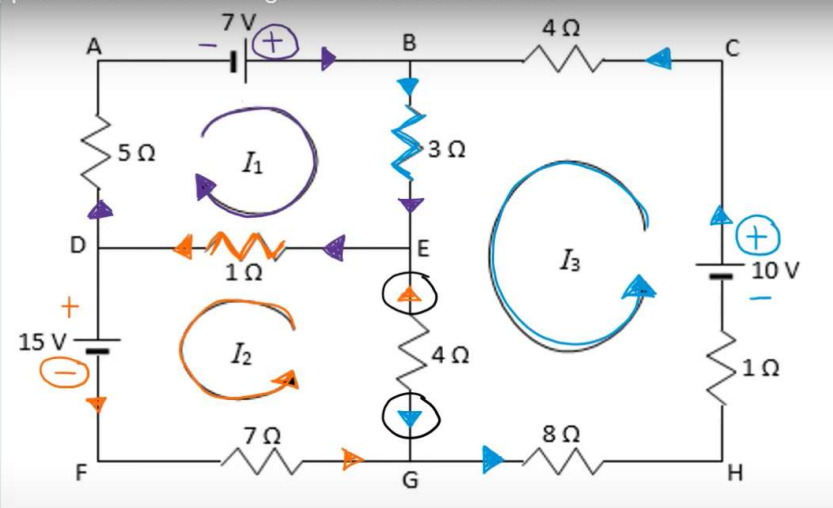
$$A := \begin{bmatrix} 1 & -1 & 1 & 0 \\ -1 & 1 & -1 & 0 \\ R_1 + R_4 & 0 & 0 & V_2 \\ 0 & R_2 + R_3 & 0 & V_2 \\ R_4 & R_3 & R_5 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 & 0 \\ -1 & 1 & -1 & 0 \\ 5 & 0 & 0 & 2 \\ 0 & 5 & 0 & 2 \\ 4 & 3 & 5 & 0 \end{bmatrix}$$

$$\text{ReducedRowEchelonForm}(A) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Sidste forsøg bruger jeg <https://www.youtube.com/watch?v=LVDX7T-jX3I> som guideline: DET HER ER GULDGRUBEN!!!!!!!!!!!!!!!!!!!!!!

Applications in Linear Algebra: Electrical Networks



$$\begin{bmatrix} \text{Resistance} \\ \text{(in Ohms } \Omega) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} \text{Current Flow} \\ \text{(in amps)} \end{bmatrix}$$

Loop 1: $(3+1+5)I_1 + 1I_2 + 3I_3 = 7$

Loop 2: $1I_1 + (7+4+1)I_2 - 4I_3 = -15$

Loop 3: $3I_1 - 4I_2 + (4+3+4+8+1)I_3 = 10$

$$\begin{bmatrix} 9 & 1 & 3 \\ 1 & 12 & -4 \\ 3 & -4 & 20 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 7 \\ -15 \\ 10 \end{bmatrix}$$

Først opstilles maskeligninger:

$$M_A := (R_1 + R_4) \cdot I_1 - R_4 \cdot I_3 = V_1 :$$

$$M_B := (R_2 + R_3) \cdot I_2 - R_3 \cdot I_3 = -V_2 :$$

$$M_C := -R_4 \cdot I_1 - R_3 \cdot I_2 + (R_4 + R_3 + R_5) \cdot I_3 = 0 :$$

Det kan vi skrive over på matrixform ved:

$$A := \begin{bmatrix} R_1 + R_4 & 0 & -R_4 & V_1 \\ 0 & R_2 + R_3 & -R_3 & -V_2 \\ -R_4 & -R_3 & R_4 + R_3 + R_5 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 0 & -4 & 1 \\ 0 & 5 & -3 & -2 \\ -4 & -3 & 12 & 0 \end{bmatrix}$$

$$\text{ReducedRowEchelonForm}(A) = \begin{bmatrix} 1 & 0 & 0 & \frac{27}{175} \\ 0 & 1 & 0 & -\frac{76}{175} \\ 0 & 0 & 1 & -\frac{2}{35} \end{bmatrix} \xrightarrow{\text{at 5 digits}} \begin{bmatrix} 1. & 0. & 0. & 0.15429 \\ 0. & 1. & 0. & -0.43429 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

FUCKING FINALLY!!!!

Det der er gjort er at jeg har opstillet ligninger for hver maske. Med det skal forstås at jeg har kørt hver maske rundt med uret, og så sat eksempelvis R_{-1} og R_{-4} til at være positive og derefter har jeg tjekket om nogen af modstandende også er påvirket af en anden maske, og det er R_{-4} eksempelvis, da den er påvirket af maske M_C , og da strømmen går i modsat retning igennem den modstand i maske M_C (relativt til maske M_A), så skal fortegnet ændres. Ligeledes skal V_{-2} modsatrettes da strømmen løber modsat ind i den spændingsgenerator end den gør i V_{-1} .

HUSK AT SE VIDEOEN EN GANG TIL INDEN EKSAMEN

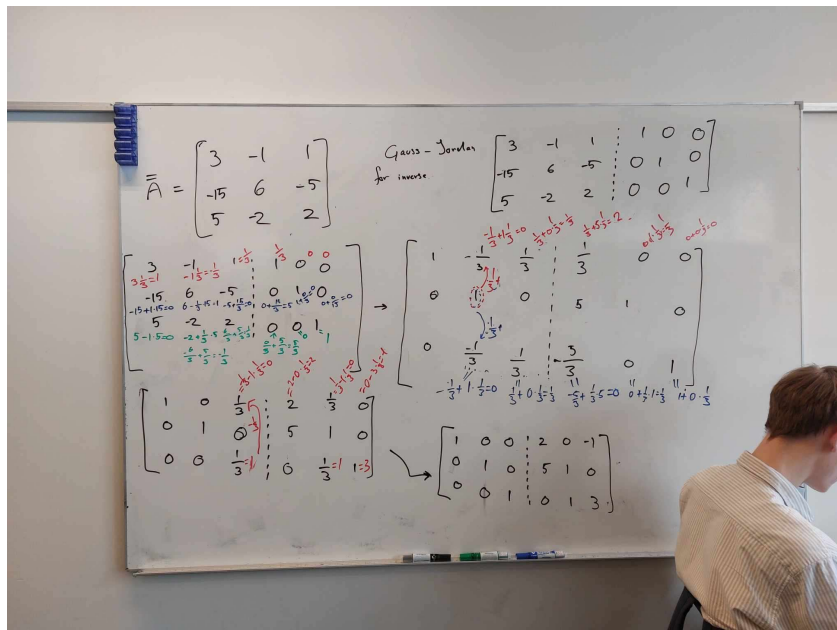
Gauss-Jordan - Mangler forståelse

Opgave 1.7

Beregn vha. Gauss-Jordan-metoden den inverse matrix af A og B , de ikke eksisterer, så forklar hvorfor.

$$A = \begin{Bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{Bmatrix} \quad B = \begin{Bmatrix} 2 & 8 & 0 & 0 \\ 9 & -4 & 0 & 0 \\ 0 & 0 & 7 & 1 \\ 0 & 0 & 6 & -2 \end{Bmatrix}$$

See Kreyszig for detaljer.



Fremgangsmetoden:

<https://www.intmath.com/matrices-determinants/inverse-matrix-gauss-jordan-elimination.php>

▼ A)

YOUR INPUT

Perform the Gauss-Jordan elimination (reduce completely) on $\left[\begin{array}{cc|c} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{array} \right]$.

SOLUTION

Divide row 1 by 3: $R_1 = \frac{R_1}{3}$.

$$\left[\begin{array}{cc|c} 1 & -\frac{1}{3} & \frac{1}{3} \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{array} \right]$$

Add row 1 multiplied by 15 to row 2: $R_2 = R_2 + 15R_1$.

$$\left[\begin{array}{cc|c} 1 & -\frac{1}{3} & \frac{1}{3} \\ 0 & 1 & 0 \\ 5 & -2 & 2 \end{array} \right]$$

Subtract row 1 multiplied by 5 from row 3: $R_3 = R_3 - 5R_1$.

$$\left[\begin{array}{cc|c} 1 & -\frac{1}{3} & \frac{1}{3} \\ 0 & 1 & 0 \\ 0 & -\frac{1}{3} & \frac{1}{3} \end{array} \right]$$

Add row 2 multiplied by $\frac{1}{3}$ to row 1: $R_1 = R_1 + \frac{R_2}{3}$.

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{1}{3} \\ 0 & 1 & 0 \\ 0 & -\frac{1}{3} & \frac{1}{3} \end{array} \right]$$

Add row 2 multiplied by $\frac{1}{3}$ to row 3: $R_3 = R_3 + \frac{R_2}{3}$.

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{1}{3} \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{3} \end{array} \right]$$

ANSWER

ANSWER

The reduced matrix is $\left[\begin{array}{cc|c} 1 & 0 & \frac{1}{3} \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{3} \end{array} \right] \approx \left[\begin{array}{cc|c} 1 & 0 & 0.3333333333333333 \\ 0 & 1 & 0 \\ 0 & 0 & 0.3333333333333333 \end{array} \right]$. **A**

▼ **B)**

YOUR INPUT

Perform the Gauss-Jordan elimination (reduce completely) on

$$\left[\begin{array}{ccc|c} 2 & 8 & 0 & 0 \\ 9 & -4 & 0 & 0 \\ 0 & 0 & 7 & 1 \\ 0 & 0 & 6 & -2 \end{array} \right]$$

SOLUTION

Divide row 1 by 2: $R_1 = \frac{R_1}{2}$.

$$\left[\begin{array}{ccc|c} 1 & 4 & 0 & 0 \\ 9 & -4 & 0 & 0 \\ 0 & 0 & 7 & 1 \\ 0 & 0 & 6 & -2 \end{array} \right]$$

Subtract row 1 multiplied by 9 from row 2: $R_2 = R_2 - 9R_1$.

$$\left[\begin{array}{ccc|c} 1 & 4 & 0 & 0 \\ 0 & -40 & 0 & 0 \\ 0 & 0 & 7 & 1 \\ 0 & 0 & 6 & -2 \end{array} \right]$$

Divide row 2 by -40 : $R_2 = -\frac{R_2}{40}$.

$$\left[\begin{array}{ccc|c} 1 & 4 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 7 & 1 \\ 0 & 0 & 6 & -2 \end{array} \right]$$

Subtract row 2 multiplied by 4 from row 1: $R_1 = R_1 - 4R_2$.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 7 & 1 \\ 0 & 0 & 6 & -2 \end{array} \right]$$

Divide row 3 by 7: $R_3 = \frac{R_3}{7}$.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{7} \\ 0 & 0 & 6 & -2 \end{array} \right]$$

Subtract row 3 multiplied by 6 from row 4: $R_4 = R_4 - 6R_3$.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{7} \\ 0 & 0 & 0 & -\frac{20}{7} \end{array} \right]$$

ANSWER

The reduced matrix is

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{7} \\ 0 & 0 & 0 & -\frac{20}{7} \end{array} \right] \approx$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0.142857142857143 \\ 0 & 0 & 0 & -2.857142857142857 \end{array} \right] \cdot \mathbf{A}$$

▼ *Spektrum af egenverdier*

spektret af en matrice er alle egenverdier som kan bruges i matricen. Altså kan man sige at spektret er alle løsninger for lambda.

$$A := \begin{bmatrix} 6 & 2 & -2 \\ 2 & 5 & 0 \\ -2 & 0 & 7 \end{bmatrix} = \begin{bmatrix} 6 & 2 & -2 \\ 2 & 5 & 0 \\ -2 & 0 & 7 \end{bmatrix}$$

Spektret er fundet ved at finde alle de egenverdier som er til stede i matricen. For at gøre det skal vi danne den karakteristiske matrix X:

$$X := \begin{bmatrix} 6 - \lambda & 2 & -2 \\ 2 & 5 - \lambda & 0 \\ -2 & 0 & 7 - \lambda \end{bmatrix} = \begin{bmatrix} 6 - \lambda & 2 & -2 \\ 2 & 5 - \lambda & 0 \\ -2 & 0 & 7 - \lambda \end{bmatrix}$$

Nu skal determinanten af denne matrice findes:

$$\begin{bmatrix} 6-\lambda & 2 & -2 \\ 2 & 5-\lambda & 0 \\ -2 & 0 & 7-\lambda \end{bmatrix} = (6-\lambda) \cdot (-1)^{1+1} \cdot ((5-\lambda) \cdot (7-\lambda) - 0 \cdot 0) + (2) \cdot (-1)^{1+2} \cdot (2 \cdot (7-\lambda) - (-2) \cdot 0) + (-2) \cdot (-1)^{1+3} \cdot (2 \cdot 0 - (-2) \cdot (5-\lambda))$$

$$\begin{bmatrix} 6-\lambda & 2 & -2 \\ 2 & 5-\lambda & 0 \\ -2 & 0 & 7-\lambda \end{bmatrix} = (6-\lambda)(5-\lambda)(7-\lambda) - 48 + 8\lambda \quad (1.1.1.28.1)$$

$$(6-\lambda)(5-\lambda)(7-\lambda) - 48 + 8\lambda$$

$$(6-\lambda)(5-\lambda)(7-\lambda) - 48 + 8\lambda \quad (1.1.1.28.2)$$

simplify

$$-\lambda^3 + 18\lambda^2 - 99\lambda + 162 \quad (1.1.1.28.3)$$

solve for lambda →

$$[[\lambda=6], [\lambda=3], [\lambda=9]] \quad (1.1.1.28.4)$$

Nu har vi så et tredjegradspolynomie, som vi skal regne ud. Vores gæt er at løsningen er $3 - \lambda$ (som maple også siger herover). Det bevises ved:

$$-\lambda^3 + 18\lambda^2 - 99\lambda + 162$$

Ligningen efter solve for lambda er spektret!!!

▼ Stabilitet ifht egenvektorer

▼ *Fra libretext*

10.4.3: Stability

Eigenvalues can be used to determine whether a fixed point (also known as an equilibrium point) is stable. If a system is initially disturbed around its fixed point yet eventually return to its original location, the fixed point is considered stable. In this concept, imagine a round ball in between two hills. If left alone, the ball will not move. If you disturb the ball by pushing it a little bit up the hill, the ball will roll back to its original position. If you push the ball to the peak of one of the hills. If left undisturbed, the ball will still remain at the peak. If you disturb the ball, it will cause the ball to roll away from the top of the hill. The top of the hill is considered an unstable fixed point.

The eigenvalues of a system linearized around a fixed point can determine the stability of the fixed point. This depends upon the existence of real and imaginary components of the eigenvalues. We will examine each of the possible cases below.

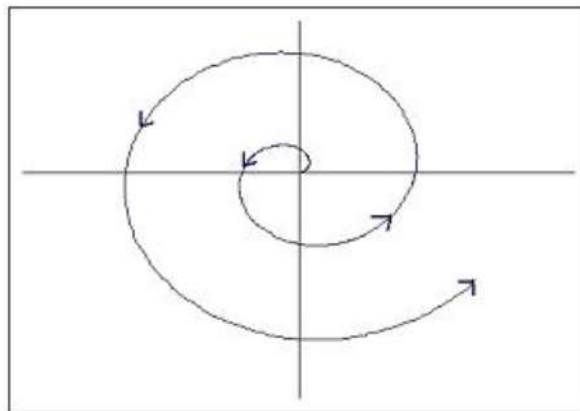
10.4.3.1: Imaginary (or Complex) Eigenvalues

When eigenvalues are of the form $a + bi$, where a and b are real scalars and i is the imaginary unit, the system is stable or unstable depending on the real part a . The cases are when the real part a is positive, negative, and zero. In all cases, when the complex eigenvalues are purely imaginary ($a = 0$), the system is stable.

10.4.3.1.1: Positive Real Part

When the real part is positive, the system is unstable and behaves as an unstable fixed point. The plot of response with time of this situation would look sinusoidal with increasing amplitude.

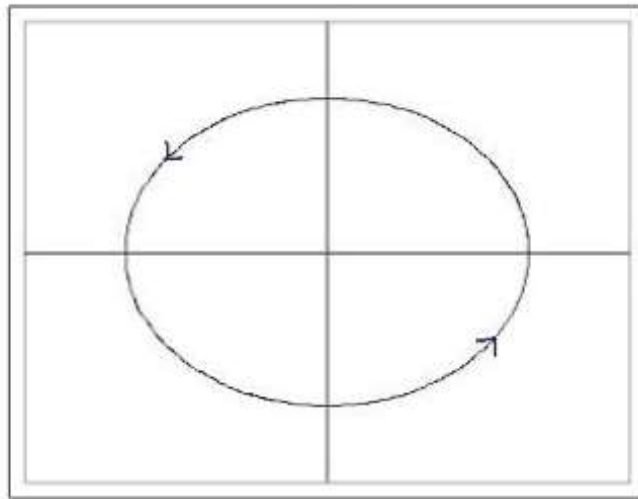
This situation is usually undesirable when attempting to control a process or unit. If a system is disturbed by an external disturbance, the system itself will not go back to steady state.



10.4.3.1.2: Zero Real Part

When the real part is zero, the system behaves as an undamped oscillator. This plot of response with time would look sinusoidal. The figures below should help in

Undamped oscillation is common in many control schemes arising out of competition. Undamped oscillation is considered an unstable process since the system will not go back to steady state

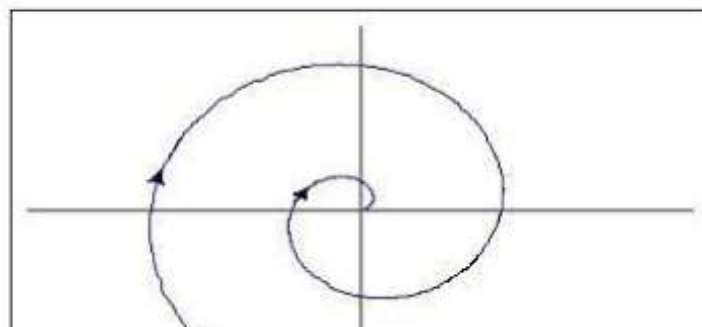


10.4.3.1.3: Negative Real Part

When the real part is negative, then the system is stable and behaves as a damped oscillation. The plot of response with time of this situation would look sinusoidal with

This situation is what is generally desired when attempting to control a process or a system. A disturbance to the system. The oscillation will quickly bring the system back to the steady state. The damping would be needed.

While discussing complex eigenvalues with negative real parts, it is important to note that a negative real part is a necessary but not a sufficient condition of a stable system.





10.4.3.1.4: Complex Part of Eigenvalues

As previously noted, the stability of oscillating systems (i.e. systems with complex eigenvalues) depends on the sign of the complex part of the eigenvalue. The sign of the complex part of the eigenvalue may cause a phase shift of the oscillation.

10.4.3.2: Real Eigenvalues

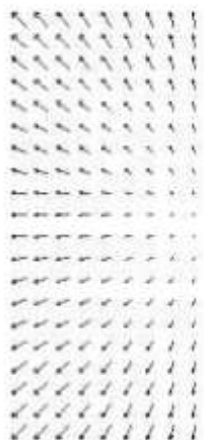
We've seen how to analyze eigenvalues that are complex in form, now we will look at real eigenvalues.

10.4.3.2.1: Zero Eigenvalues

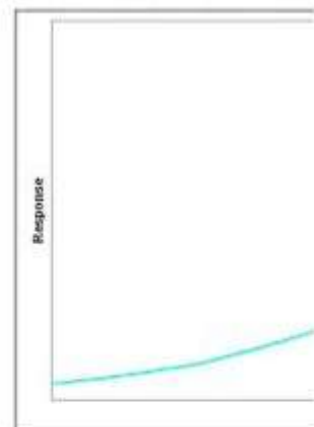
If an eigenvalue has no imaginary part and is equal to zero, the system will be undamped. It will have any non-negative real parts. This is just a trivial case of the complex eigenvalues.

10.4.3.2.2: Positive Eigenvalues

When all eigenvalues are real, positive, and distinct, the system is unstable. On a phase plot, a node pointing out of the same spot (a node) signifies all positive eigenvalues. This is called a node.



Graphically, real and positive eigenvalues will show a typical exponential plot where the response increases over time.



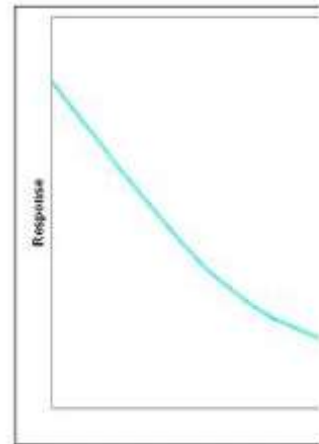
| | | | |

10.4.3.2.3: Negative Eigenvalues

When all eigenvalues are real, negative, and distinct, the system is unstable. Graphically, the system converges to a fixed point. This is called a sink node.



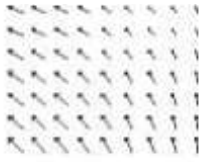
Graphically, real and negative eigenvalues will output an inverse exponential plot



10.4.3.2.4: Positive and Negative Eigenvalues

If the set of eigenvalues for the system has both positive and negative eigenvalues, the series of minimum and maximum points converge at one area in a gradient field. In a surface plot the function looks like a saddle.



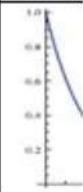
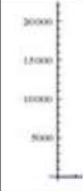
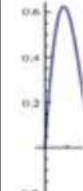
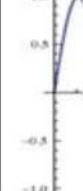


10.4.3.3: Repeated Eigenvalues

If the set of eigenvalues for the system has repeated real eigenvalues, then the eigenvectors are linearly independent, or orthogonal. This is the case of degenerate eigenvalues. In general, the determination of the system's behavior requires further analysis. For the following two possible cases. If the two repeated eigenvalues are positive, then the fixed point is a stable source. If the two repeated eigenvalues are negative, then the fixed point is a stable sink.

10.4.3.4: Summary of Eigenvalue Graphs

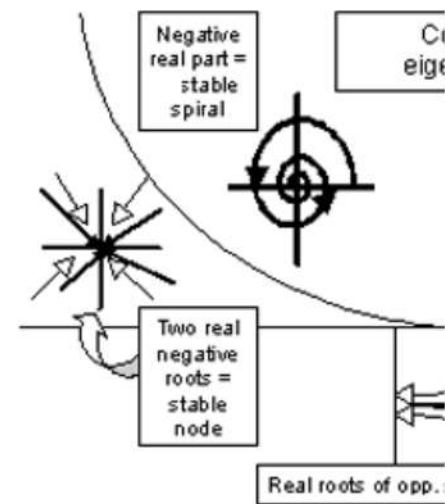
Below is a table summarizing the visual representations of stability that the eigen

Eigenvalues	Graph
All real and negative	
All real and one or more are positive	
All real eigenvalues are negative and there are imaginary parts	
One or more eigenvalues have a positive real part and there are imaginary parts	
Real parts of the eigenvalues are zero and there are imaginary parts	

Note that the graphs from Peter Woolf's lecture from Fall'08 titled Dynamic System

10.4.5: Stability Summary

The following image can work as a quick reference to remind yourself of what ve



The table below gives a complete overview of the stability corresponding to each

Eigenvalue Type	Stability
All Real and +	Unstable
All Real and -	Stable
Mixed + & - Real	Unstable
$+a + bi$	Unstable
$-a + bi$	Stable
$0 + bi$	Unstable
Repeated values	Depends on orthogonality of eigenvectors

▼ Ting vi ikke har fået noget undervisning i efter min overbevisning

▼ Skævsymmetrisk Matrix

En skævsymmetrisk matrix har mange af de samme egenskaber som en symmetrisk, dog med det forbehold at den transponerede matrix er det samme som den negative matrix, altså at:

$$-A = A^T$$

$$A = \begin{bmatrix} 0 & 65 & 24 & -44 \\ -65 & 0 & 76 & 51 \\ -24 & -76 & 0 & -25 \\ 44 & -51 & 25 & 0 \end{bmatrix}$$

HUSK!!!! at en skævsymmetrisk matrix skal have 0'er hele vejen langs diagonalen - tror jeg ihvertfald, selvom løsningen i eksempeksamenssættet ikke opretholder dette

▼ Orthogonal Matrix - mangler forståelse

En orthogonal matrix er en matrix hvor den transponerede matrix ganget med den normale matrix giver en identitetsmatrix som resultat. Formlen lyder:

$$A \cdot A^T = I$$

og kan vises som:

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$A \cdot A^T = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

▼ Chats forklaring:

What is an Orthogonal Matrix?

An **orthogonal matrix** is a square matrix Q whose columns and rows are orthonormal vectors. This means two things:

1. **Orthogonality:** The columns (or rows) of Q are orthogonal to each other. Two vectors are orthogonal if their dot product is zero.
2. **Normalization:** Each column (or row) vector has a unit length (norm equal to 1).

Key Properties

1. **Transpose Equals Inverse:**

$$Q^T = Q^{-1}$$

This means if you multiply an orthogonal matrix by its transpose, you get the identity matrix:

$$Q^T Q = Q Q^T = I$$

where I is the identity matrix.

2. **Preservation of Lengths and Angles:**

Orthogonal matrices preserve the Euclidean length of vectors and the angles between them. If you have a vector $\mathbf{x}$ and you multiply it by an orthogonal matrix Q , the length of $Q\mathbf{x}$ will be the same as the length of $\mathbf{x}$.

3. **Determinant:**

The determinant of an orthogonal matrix is either $+1$ or -1 . This corresponds to whether the transformation described by the matrix preserves orientation (determinant $+1$) or reverses it (determinant -1).

How to Check if a Matrix is Orthogonal

Given a matrix Q , to check if it is orthogonal:

1. Compute Q^T .
2. Multiply Q^T by Q .
3. If $Q^T Q = I$, then Q is orthogonal.

Example

Consider the matrix:

$$Q = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

To verify if Q is orthogonal:

1. Compute the transpose Q^T :

$$Q^T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

2. Multiply Q^T by Q :

$$Q^T Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Since $Q^T Q = I$, Q is indeed an orthogonal matrix.

Applications

1. Rotations and Reflections:

Orthogonal matrices often represent rotations and reflections in space. In two dimensions, any rotation matrix is orthogonal.

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

2. Numerical Stability:

In numerical methods, orthogonal matrices are used because they are stable under floating-point arithmetic. They help maintain numerical accuracy.

3. Simplifying Problems:

Orthogonal transformations can simplify many linear algebra problems, such as diagonalizing symmetric matrices through orthogonal diagonalization.

Summary

Orthogonal matrices are fundamental in linear algebra due to their property of preserving lengths and angles, and their ability to simplify complex linear transformations. By understanding their definition, properties, and applications, you can see how powerful these matrices are in both theoretical and practical contexts.

▼ Hermitisk Matrix - mangler forståelse

In [mathematics](#), a **Hermitian matrix** (or **self-adjoint matrix**) is a [complex square matrix](#) that is equal to its [conjugate transpose](#). The element in the i -th row and j -th column is equal to the [complex conjugate](#) of the element in the j -th row and i -th column.

$$A \text{ is Hermitian} \iff a_{ij} = \overline{a_{ji}}$$

or in matrix form:

$$A \text{ is Hermitian} \iff A = \overline{A^T}.$$

Hermitian matrices can be understood as the complex extension of real [symmetric matrices](#).

If the [conjugate transpose](#) of a matrix A is denoted by A^H , then the Hermitian property can be written compactly as:

$$A \text{ is Hermitian} \iff A = A^H$$

Hermitian matrices are named after [Charles Hermite](#),^[1] who demonstrated in 1855 that matrices of this form always have real [eigenvalues](#). Other, equivalent notations in common use are $A^H = A^\dagger$ = A^* . The notation A^* typically means the [complex conjugate](#) only, and not the [conjugate transpose](#).

$$A = \begin{bmatrix} 0 & \overline{m_{2,1}} & \overline{m_{3,1}} & \overline{m_{4,1}} \\ m_{2,1} & 0 & \overline{m_{3,2}} & \overline{m_{4,2}} \\ m_{3,1} & m_{3,2} & 0 & \overline{m_{4,3}} \\ m_{4,1} & m_{4,2} & m_{4,3} & 0 \end{bmatrix}$$

▼ Skævhermitisk Matrix - mangler forståelse

In [linear algebra](#), a [square matrix](#) with [complex](#) entries is said to be **skew-Hermitian** or **anti-Hermitian** if its conjugate transpose is the negative of the original matrix.^[1] That is, the matrix A is skew-Hermitian if it satisfies the relation

$$A \text{ skew-Hermitian} \iff A^H = -A$$

where A^H denotes the conjugate transpose of the matrix A . In component form, this means that

$$A \text{ skew-Hermitian} \iff a_{ij} = -\overline{a_{ji}}$$

for all indices i and j , where a_{ij} is the element in the i -th row and j -th column of A , and the overline denotes the complex conjugate.

Skew-Hermitian matrices can be understood as the complex versions of real [skew-symmetric matrices](#), or [skew-symmetric real matrices](#).^[2] The set of all skew-Hermitian $n \times n$ matrices forms the $\mathfrak{u}(n)$ [Lie algebra](#), which corresponds to the $\mathfrak{so}(n, \mathbb{C})$ Lie algebra when generalized to include [linear transformations](#) of any [complex vector space](#) with a [sesquilinear norm](#).

Note that the [adjoint](#) of an operator depends on the [scalar product](#) considered on the n dimensional complex vector space. If $\langle \cdot, \cdot \rangle$ is the standard scalar product on K^n , then saying A is skew-adjoint means that for all $\mathbf{u}, \mathbf{v} \in K^n$ one has $\langle A\mathbf{u}, \mathbf{v} \rangle = -\langle \mathbf{u}, A\mathbf{v} \rangle$.

[Imaginary numbers](#) can be thought of as skew-adjoint (since they are like 1×1 matrices), whereas [real numbers](#) are self-adjoint.

Example [\[edit \]](#)

For example, the following matrix is skew-Hermitian

$$A = \begin{bmatrix} -i & +2+i \\ -2+i & 0 \end{bmatrix}$$

because

$$-A = \begin{bmatrix} i & -2-i \\ 2-i & 0 \end{bmatrix} = \begin{bmatrix} \overline{-i} & \overline{-2+i} \\ \overline{2+i} & \overline{0} \end{bmatrix} = \begin{bmatrix} \overline{-i} & \overline{2+i} \\ \overline{-2+i} & \overline{0} \end{bmatrix}^T = A^H$$

Properties [\[edit \]](#)

- The eigenvalues of a skew-Hermitian matrix are all purely imaginary (and possibly zero). Furthermore they are diagonalizable and their eigenvectors for distinct eigenvalues must be orthogonal.^[3]
- All entries on the [main diagonal](#) of a skew-Hermitian matrix have to be pure [imaginary](#); i.e., on the imaginary axis (and thus purely imaginary).^[4]
- If A and B are skew-Hermitian, then $aA + bB$ is skew-Hermitian for all [real scalars](#) a and b .^[5]
- A is skew-Hermitian *if and only if* iA (or equivalently, $-iA$) is [Hermitian](#).^[5]
- A is skew-Hermitian *if and only if* the real part $\Re(A)$ is [skew-symmetric](#) and the imaginary part $\Im(A)$ is [symmetric](#).
- If A is skew-Hermitian, then A^k is Hermitian if k is an even integer and skew-Hermitian if k is an odd integer.
- A is skew-Hermitian if and only if $\mathbf{x}^H A \mathbf{y} = -\overline{\mathbf{y}^H A \mathbf{x}}$ for all vectors $\mathbf{x}, \mathbf{y}$.
- If A is skew-Hermitian, then the [matrix exponential](#) e^A is [unitary](#).
- The space of skew-Hermitian matrices forms the [Lie algebra](#) $\mathfrak{u}(n)$ of the [Lie group](#) $U(n)$.

Decomposition into Hermitian and skew-Hermitian [\[edit \]](#)

- The sum of a square matrix and its conjugate transpose $(A + A^H)$ is Hermitian.
- The difference of a square matrix and its conjugate transpose $(A - A^H)$ is skew-Hermitian. This implies that any square matrix can be written as the sum of a Hermitian and a skew-Hermitian matrix.
- An arbitrary square matrix C can be written as the sum of a Hermitian matrix A and a skew-Hermitian matrix B :

$$C = A + B \quad \text{with} \quad A = \frac{1}{2} (C + C^H) \quad \text{and} \quad B = \frac{1}{2} (C - C^H)$$

$$\begin{bmatrix} 0 & -\overline{m_{2,1}} & -\overline{m_{3,1}} & -\overline{m_{4,1}} \\ m_{2,1} & 0 & -\overline{m_{3,2}} & -\overline{m_{4,2}} \\ m_{3,1} & m_{3,2} & 0 & -\overline{m_{4,3}} \\ m_{4,1} & m_{4,2} & m_{4,3} & 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & -\overline{m_{2,1}} & -\overline{m_{3,1}} & -\overline{m_{4,1}} \\ m_{2,1} & 0 & -\overline{m_{3,2}} & -\overline{m_{4,2}} \\ m_{3,1} & m_{3,2} & 0 & -\overline{m_{4,3}} \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 0 & m_{2,1} & m_{3,1} & m_{4,1} \\ -m_{2,1} & 0 & m_{3,2} & m_{4,2} \\ -m_{3,1} & -m_{3,2} & 0 & m_{4,3} \\ -m_{4,1} & -m_{4,2} & -m_{4,3} & 0 \end{bmatrix} \quad (1.1.2.4.1)$$

▼ Triangulær Matrix

$$\begin{array}{c}
 | \\
 | \\
 | \\
 | \\
 \left[\begin{array}{cccc}
 33 & 87 & -18 & -38 \\
 0 & -81 & 45 & -22 \\
 0 & 0 & -50 & -9 \\
 0 & 0 & 0 & -16
 \end{array} \right] : \\
 \left[\begin{array}{cccc}
 -74 & 0 & 0 & 0 \\
 -32 & -2 & 0 & 0 \\
 -72 & -76 & -93 & 0 \\
 27 & 57 & -77 & -98
 \end{array} \right] :
 \end{array}$$

Types of Triangular Matrices

There are different types of triangular matrices that we study. Given below is a list of some special types of triangular matrices:

- **Upper Triangular Matrix:** A triangular matrix is said to be an upper triangular matrix if all the elements below the main diagonal are zero.
- **Lower Triangular Matrix:** A triangular matrix is said to be a lower triangular matrix if all the elements above the main diagonal are zero.
- **Strictly Triangular Matrix:** A triangular matrix is said to be a strictly triangular matrix if all the elements of the main diagonal are zero.
- **Strictly Lower Triangular Matrix:** A lower triangular matrix is said to be a strictly lower triangular matrix if all the elements of the main diagonal are zero.
- **Strictly Upper Triangular Matrix:** An upper triangular matrix is said to be a strictly upper triangular matrix if all the elements of the main diagonal are zero.
- **Unit Triangular Matrix:** A triangular matrix is said to be a unit triangular matrix if all the elements of the main diagonal are equal to 1.
- **Unit Lower Triangular Matrix:** A lower triangular matrix is said to be a unit lower triangular matrix if all the elements of the main diagonal are equal to 1.
- **Unit Upper Triangular Matrix:** An upper triangular matrix is said to be a unit upper triangular matrix if all the elements of the main diagonal are equal to 1.

In the following sections, we will mainly explore two types of triangular matrices, namely an upper and lower triangular matrix.

▼ Invers Matrix

En invers matrice er en modsatrettet matrice, hvor diagonalen bliver vendt om og de resterende felter skifter fortegn.. Den betegnes som A^{-1}

$$Eks := \begin{bmatrix} 76 & 24 \\ -44 & 65 \end{bmatrix} :$$

$$Eks^{-1} = \left[\begin{array}{cc} \frac{65}{5996} & -\frac{6}{1499} \\ \frac{11}{1499} & \frac{19}{1499} \end{array} \right]$$

YOUR INPUT

Calculate $\begin{bmatrix} 76 & 24 \\ -44 & 65 \end{bmatrix}^{-1}$ using the Gauss-Jordan elimination.

SOLUTION

To find the inverse matrix, augment it with the identity matrix and perform row operations trying to make the identity matrix to the left. Then to the right will be the inverse matrix.

So, augment the matrix with the identity matrix:

$$\left[\begin{array}{cc|cc} 76 & 24 & 1 & 0 \\ -44 & 65 & 0 & 1 \end{array} \right]$$

Divide row 1 by 76: $R_1 = \frac{R_1}{76}$.

$$\left[\begin{array}{cc|cc} 1 & \frac{6}{19} & \frac{1}{76} & 0 \\ -44 & 65 & 0 & 1 \end{array} \right]$$

Add row 1 multiplied by 44 to row 2: $R_2 = R_2 + 44R_1$.

$$\left[\begin{array}{cc|cc} 1 & \frac{6}{19} & \frac{1}{76} & 0 \\ 0 & \frac{1499}{19} & \frac{76}{11} & 1 \end{array} \right]$$

Multiply row 2 by $\frac{19}{1499}$: $R_2 = \frac{19R_2}{1499}$.

$$\left[\begin{array}{cc|cc} 1 & \frac{6}{19} & \frac{1}{76} & 0 \\ 0 & 1 & \frac{76}{1499} & \frac{19}{1499} \end{array} \right]$$

Subtract row 2 multiplied by $\frac{6}{19}$ from row 1: $R_1 = R_1 - \frac{6R_2}{19}$.

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{65}{5996} & -\frac{6}{1499} \\ 0 & 1 & \frac{76}{1499} & \frac{19}{1499} \end{array} \right]$$

We are done. On the left is the identity matrix. On the right is the inverse matrix.

ANSWER

ANSWER

The inverse matrix is $\begin{bmatrix} \frac{65}{5996} & -\frac{6}{1499} \\ \frac{11}{1499} & \frac{19}{1499} \end{bmatrix} \approx$

$$\begin{bmatrix} 0.010840560373582 & -0.00400266844563 \\ 0.007338225483656 & 0.012675116744496 \end{bmatrix} \cdot \mathbf{A}$$

Definition

Here is the definition:

The inverse of A is A^{-1} only when:

$$AA^{-1} = A^{-1}A = \mathbf{I}$$

Sometimes there is no inverse at all.

(Note: writing AA^{-1} means A times A^{-1})

2x2 Matrix

OK, how do we calculate the inverse?

Well, for a 2x2 matrix the inverse is:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

In other words: **swap** the positions of a and d , put **negatives** in front of b and c and **divide** everything by $ad-bc$.

Note: $ad-bc$ is called the determinant.

Let us try an example:

$$\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}^{-1} = \frac{1}{4 \times 6 - 7 \times 2} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0,6 & -0,7 \\ -0,2 & 0,4 \end{bmatrix}$$

How do we know this is the right answer?

Remember it must be true that: $AA^{-1} = \mathbf{I}$

So, let us check to see what happens when we multiply the matrix by its inverse

$$\begin{aligned} \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 0,6 & -0,7 \\ -0,2 & 0,4 \end{bmatrix} &= \begin{bmatrix} 4 \times 0,6 + 7 \times -0,2 & 4 \times -0,7 + 7 \times 0,4 \\ 2 \times 0,6 + 6 \times -0,2 & 2 \times -0,7 + 6 \times 0,4 \end{bmatrix} \\ &= \begin{bmatrix} 2,4 - 1,4 & -2,8 + 2,8 \\ 1,2 - 1,2 & -1,4 + 2,4 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

And, hey!, we end up with the Identity Matrix!
So it must be right.

It should **also** be true that: $A^{-1}A = \mathbf{I}$

Why don't you have a go at multiplying these? See if you also get the Identity Matrix:

$$\begin{bmatrix} 0,6 & -0,7 \\ -0,2 & 0,4 \end{bmatrix} \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} & \\ & \end{bmatrix}$$

Why Do We Need an Inverse?

Because with matrices we **don't divide**! Seriously, there is no concept of dividing a matrix.

But we can **multiply by an inverse**, which achieves the same thing.

Imagine we can't divide by numbers ...

... and someone asks "How do I share 10 apples with 2 people?"

But we can take the **reciprocal** of 2 (which is 0,5), so we answer:

$$10 \times 0,5 = 5$$

They get 5 apples each.

The same thing can be done with matrices:

Say we want to find matrix X , and we know matrix A and B :

$$XA = B$$

It would be nice to divide both sides by A (to get $X=B/A$), but remember **divide**.

But what if we multiply both sides by A^{-1} ?

$$XAA^{-1} = BA^{-1}$$

And we know that $AA^{-1} = I$, so:

$$XI = BA^{-1}$$

We can remove I (for the same reason we can remove "1" from $1x = ab$ numbers):

$$X = BA^{-1}$$

And we have our answer (assuming we can calculate A^{-1})

In that example we were very careful to get the multiplications correct, because matrices the order of multiplication matters. AB is almost never equal to BA .

▼ Unitær Matrix

Unitary Matrix

Unitary Matrix is a square matrix of complex numbers. The product of the conjugate transpose of a unitary matrix, with the unitary matrix, gives an identity matrix. From this, we can also understand that a unitary matrix is a nonsingular matrix, and is invertible.

Let us learn more about the properties, and examples of the unitary matrix.

What is Unitary Matrix?

A **unitary matrix** is a square matrix of complex numbers, whose inverse is equal to its conjugate transpose. Alternatively, the product of the unitary matrix and the conjugate transpose of a unitary matrix is equal to the **identity matrix**. i.e., if U is a unitary matrix and U^H is its complex transpose (which is sometimes denoted as U^*) then one /both of the following conditions is satisfied.

- $U^H = U^{-1}$
- $U^H U = U U^H = I$

where I is the identity matrix whose order is the same as U .

Unitary Matrix



$$U^H = U^{-1}$$

OR

$$U^H \cdot U = I$$

I = Identity Matrix

Also, a unitary matrix is a nonsingular matrix. Or the determinant of a unitary matrix is not equal to zero. The columns and rows of a unitary matrix are orthonormal.

matrix are orthonormal.

Properties of Unitary Matrix

The properties of a unitary matrix are as follows.

- The unitary matrix is a [non-singular matrix](#).
- The unitary matrix is an invertible matrix
- The product of two unitary matrices is a unitary matrix.
- The inverse of a unitary matrix is another unitary matrix.
- A matrix is unitary, if and only if its transpose is unitary.
- A matrix is unitary if its rows are orthonormal, and the columns are orthonormal.
- The unitary matrices can also be non-square matrices but have orthonormal columns and rows.

Note: The sum or difference of two unitary matrices doesn't need to be a unitary matrix. For example, if A is a unitary matrix, then $A - A = O$ ([null matrix](#)), which is NOT unitary.

Terms Related to Unitary Matrix

The following terms related to matrices are helpful for a better understanding of this concept of unitary matrix.

- **Non-Singular Matrix:** The [determinant](#) of a non singular matrix is a non zero value. For a square matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the condition of it being a non singular matrix is $|A| = ad - bc \neq 0$.
- **Invertible Matrix:** The matrix whose inverse matrix can be computed, is called an invertible matrix. The [inverse of a matrix](#) A is $A^{-1} = \text{Adj } A / |A|$.
- **Conjugate Matrix:** The conjugate matrix of a given matrix is obtained by replacing the corresponding elements of the given matrix, with their [complex conjugates](#).
- **Transpose Matrix:** The transpose of a matrix A is represented as A^T , and the transpose of a matrix is obtained by changing the rows into columns and columns into rows for a given matrix.
- **Orthogonal Matrix:** If the product of a matrix and its transpose is an identity matrix, then it is called an [orthogonal matrix](#). $A \cdot A^T = I$.
- **Hermitian Matrix:** A hermitian matrix is a square matrix, which is equal to its conjugate transpose matrix. The non-diagonal elements of a hermitian matrix are all complex numbers. $A = \bar{A}^T$.

▮ Related Topics:

The following topics are helpful for a better understanding of the non-unitary matrix.

- [Unit Matrix](#)
- [Symmetric Matrix](#)
- [Diagonal Matrix](#)
- [Skew Symmetric Matrix](#)

▼ Hankel Matrix

En hankel matrice er en specialmatrice som vi nok aldrig kommer til at møde igen, men den ser pæn ud. Den skal opfylde at hver diagonal linje har samme værdier, med start i venstre hjørne af matricen:

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 & 8 \end{bmatrix} :$$

$$\begin{bmatrix} a & b & c & d & e \\ b & c & d & e & f \\ c & d & e & f & g \\ d & e & f & g & h \\ e & f & g & h & i \end{bmatrix} .$$

Properties [\[edit \]](#)

- Any Hankel matrix is [symmetric](#).
- Let J_n be the $n \times n$ [exchange matrix](#). If H is an $m \times n$ Hankel matrix, then $H = TJ_n$ where T is
 - If T is [real](#) symmetric, then $H = TJ_n$ will have the same [eigenvalues](#) as T up to sign.^[1]
- The [Hilbert matrix](#) is an example of a Hankel matrix.
- The [determinant](#) of a Hankel matrix is called a [catalecticant](#).

▼ Toeplitz Matrix

En toeplitz matrice minder meget om hankel matricen, i dette tilfælde er det bare langs den normale diagonal man sætter værdierne:

$$\begin{bmatrix} a & b & c & d & e \\ f & a & b & c & d \\ g & f & a & b & c \\ h & g & f & a & b \\ i & h & g & f & a \end{bmatrix}$$

$$\begin{bmatrix} \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & x_1 & \cdots & x_{n-2} & x_{n-1} & x_n \\ 0 & \cdots & 0 & 0 & 0 & x_1 & \cdots & x_{n-2} & x_{n-1} \end{bmatrix}$$

This approach can be extended to compute [autocorrelation](#), [cross-correlation](#), [moving average](#) etc.

▼ Selvadjungeret Matrix

Har vi slet ikke haft noget om (tror jeg) og det virker lidt spacy ifølge wiki (men alt math er spacy på wiki) til gengæld viser denne lidt om hvordan man regner med adjointed matrices:
<https://www.emathhelp.net/en/calculators/linear-algebra/adjoint-matrix-calculator/?i=%5B%5B76%2C24%5D%2C%5B-44%2C65%5D%5D>

YOUR INPUT

Find the adjoint matrix of $\begin{bmatrix} 76 & 24 \\ -44 & 65 \end{bmatrix}$.

SOLUTION

The classical adjoint, adjugate, or adjunct of a square matrix is the transpose of its cofactor matrix.

The cofactor matrix is $\begin{bmatrix} 65 & 44 \\ -24 & 76 \end{bmatrix}$ (for steps, see [cofactor matrix calculator](#)).

The transpose of the cofactor matrix is $\begin{bmatrix} 65 & -24 \\ 44 & 76 \end{bmatrix}$ (for steps, see [matrix transpose calculator](#)).

ANSWER

The adjoint matrix is $\begin{bmatrix} 65 & -24 \\ 44 & 76 \end{bmatrix}$ **A**.

YOUR INPUT

Find the cofactor matrix of $\begin{bmatrix} 76 & 24 \\ -44 & 65 \end{bmatrix}$.

SOLUTION

The cofactor matrix consists of all cofactors of the given matrix, which are calculated according to the formula $C_{ij} = (-1)^{i+j} M_{ij}$, where M_{ij} is the [minor](#), i.e. the determinant of the submatrix formed by deleting row i and column j from the given matrix.

Calculate all cofactors:

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 65 \end{vmatrix} = 65 \text{ (for steps, see [determinant calculator](#)).$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} -44 \end{vmatrix} = 44 \text{ (for steps, see [determinant calculator](#)).$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 24 \end{vmatrix} = -24 \text{ (for steps, see [determinant calculator](#)).$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 76 \end{vmatrix} = 76 \text{ (for steps, see [determinant calculator](#)).$$

Thus, the cofactor matrix is $\begin{bmatrix} 65 & 44 \\ -24 & 76 \end{bmatrix}$.

ANSWER

The cofactor matrix is $\begin{bmatrix} 65 & 44 \\ -24 & 76 \end{bmatrix}$ **A**.

YOUR INPUT

Calculate $\begin{vmatrix} 65 \end{vmatrix}$.

SOLUTION

The determinant of a 1x1 matrix is $\begin{vmatrix} a \end{vmatrix} = a$.

$$\begin{vmatrix} 65 \end{vmatrix} = 65$$

ANSWER

$$\begin{vmatrix} 65 \end{vmatrix} = 65 \text{ A}$$

Og så lidt fra wiki:

In [mathematics](#), an [element](#) of a [*-algebra](#) is called **self-adjoint** if it is the same as its adjoint (i.e. $a = a^*$).

Definition [\[edit\]](#)

Let $\mathcal{A}$ be a $*$ -algebra. An element $a \in \mathcal{A}$ is called self-adjoint if $a = a^*$.^[1]

The [set](#) of self-adjoint elements is referred to as $\mathcal{A}_{sa}$.

A [subset](#) $\mathcal{B} \subseteq \mathcal{A}$ that is [closed](#) under the [involution](#) $*$, i.e. $\mathcal{B} = \mathcal{B}^*$, is called self-adjoint.^[2]

A special case from particular importance is the case where $\mathcal{A}$ is a [complete normed \$*\$ -algebra](#), that satisfies the [C*-condition](#), which is called a [C*-algebra](#).

Especially in the older literature on $*$ -algebras and C*-algebras, such elements are often called hermitian. $H(\mathcal{A})$ for the set of self-adjoint elements are also sometimes used, even in the more recent literature.

Examples [\[edit\]](#)

- Each [positive element](#) of a C*-algebra is self-adjoint.^[3]
- For each element a of a $*$ -algebra, the elements aa^* and a^*a are self-adjoint, since $*$ is an [involutive](#).
- For each element a of a $*$ -algebra, the [real and imaginary parts](#) $\operatorname{Re}(a) = \frac{1}{2}(a + a^*)$ and $\operatorname{Im}(a) = \frac{1}{2i}(a - a^*)$ are self-adjoint, and the [imaginary unit](#).^[1]
- If $a \in \mathcal{A}_N$ is a [normal element](#) of a C*-algebra $\mathcal{A}$, then for every [real-valued function](#) f , which is [continuous](#) on the [spectrum](#) of a , the [functional calculus](#) defines a self-adjoint element $f(a)$.^[5]

Criteria [\[edit\]](#)

Let $\mathcal{A}$ be a $*$ -algebra. Then:

- Let $a \in \mathcal{A}$, then a^*a is self-adjoint, since $(a^*a)^* = a^*(a^*)^* = a^*a$. A similar calculation yields that aa^* is self-adjoint.
- Let $a = a_1a_2$ be the [product](#) of two self-adjoint elements $a_1, a_2 \in \mathcal{A}_{sa}$. Then a is self-adjoint if a_1 and a_2 commute, i.e. $(a_1a_2)^* = a_2^*a_1^* = a_2a_1$ always holds.^[1]
- If $\mathcal{A}$ is a C*-algebra, then a normal element $a \in \mathcal{A}_N$ is self-adjoint if and only if its spectrum is real, i.e. $\sigma(a) \subseteq \mathbb{R}$.

Properties [\[edit\]](#)

In $*$ -algebras [\[edit\]](#)

Let $\mathcal{A}$ be a $*$ -algebra. Then:

- Each element $a \in \mathcal{A}$ can be uniquely decomposed into real and imaginary parts, i.e. there are unique elements $a_1, a_2 \in \mathcal{A}_{sa}$ such that $a = a_1 + ia_2$ holds. Where $a_1 = \frac{1}{2}(a + a^*)$ and $a_2 = \frac{1}{2i}(a - a^*)$.^[1]
- The set of self-adjoint elements $\mathcal{A}_{sa}$ is a [real linear subspace](#) of $\mathcal{A}$. From the previous property, it follows that $\mathcal{A}$ is the complexification of $\mathcal{A}_{sa}$, i.e. $\mathcal{A} = \mathcal{A}_{sa} \oplus i\mathcal{A}_{sa}$.^[7]

- If $a \in \mathcal{A}_{sa}$ is self-adjoint, then a is normal.^[1]
- The $*$ -algebra $\mathcal{A}$ is called a hermitian $*$ -algebra if every self-adjoint element $a \in \mathcal{A}_{sa}$ has a real spectrum.

In C^* -algebras [edit]

Let $\mathcal{A}$ be a C^* -algebra and $a \in \mathcal{A}_{sa}$. Then:

- For the spectrum $\|a\| \in \sigma(a)$ or $-\|a\| \in \sigma(a)$ holds, since $\sigma(a)$ is real and $r(a) = \|a\|$ holds for $a \in \mathcal{A}_{sa}$.
- According to the continuous functional calculus, there exist uniquely determined positive elements a_+ and a_- such that $a = a_+ - a_-$ and $a_+ a_- = a_- a_+ = 0$. For the norm, $\|a\| = \max(\|a_+\|, \|a_-\|)$ holds.^[10] The elements a_+ and a_- are called the positive and negative parts. In addition, $|a| = a_+ + a_-$ holds for the absolute value defined for every element $a \in \mathcal{A}$.
- For every $a \in \mathcal{A}_+$ and odd $n \in \mathbb{N}$, there exists a uniquely determined $b \in \mathcal{A}_+$ that satisfies $b^n = a$. This is proved using the continuous functional calculus.^[12]

▼ Skævdjungenet Matrix

Nu blir det så spacy der ikke engang er en wiki-side specifikt til det:

Definition of skew-adjoint operator

Asked 2 years, 9 months ago Modified 2 years, 9 months ago Viewed 3k times



3

Let $n \in \mathbb{N}$. It is customary to call an $n \times n$ matrix A *self-adjoint* iff the (complex conjugate) transpose of A is equal to A , and to call A *skew-adjoint* iff the (complex conjugate of the) transpose of A is equal to $-A$.



Now let's go to the infinite-dimensional case: Let H be a Hilbert space over $\mathbb{C}$ with the inner product $\langle \cdot, \cdot \rangle$ and let $T : D(T) \rightarrow H$ be a linear operator, where $D(T) \subset H$ is dense.



It is customary to call T *self-adjoint* iff

1. $\langle Tx, y \rangle = \langle x, Ty \rangle$ for all $x, y \in D(T)$; and
2. $\{y \in H : \text{The operator } T_y : D(T) \rightarrow \mathbb{C}, x \mapsto \langle Tx, y \rangle \text{ is bounded}\} = D(T)$.

My question. Is there an analogous notion for skew-adjoint?

My idea. An operator T is called *skew-adjoint* iff iT is self-adjoint.

functional-analysis

operator-theory

self-adjoint-operators

skew-symmetric-matrices

Share Cite Follow

asked Aug 14, 2021 at 1



Maximilian Jani

13.7k 2 23

- 3 Your idea is exactly the common definition. Equivalently, it is an operator with $T^* = -T$, where one has to take proper care of the domains (as you did in the definition of self-adjoint operator). – [MaoWao](#) Aug 14, 2021 at 21:21

Add a comment

Skew-Hermitian matrix

[Article](#) [Talk](#)

From Wikipedia, the free encyclopedia

In [linear algebra](#), a [square matrix](#) with [complex](#) entries is said to be **skew-Hermitian** or **anti-Hermitian** if the conjugate transpose of the original matrix.^[1] That is, the matrix *A* is skew-Hermitian if it satisfies the relation

$$A \text{ skew-Hermitian} \iff A^H = -A$$

where *A*^H denotes the conjugate transpose of the matrix *A*. In component form, this means that

$$A \text{ skew-Hermitian} \iff a_{ij} = -\overline{a_{ji}}$$

for all indices *i* and *j*, where *a*_{*ij*} is the element in the *i*-th row and *j*-th column of *A*, and the overline denotes the complex conjugate.

Skew-Hermitian matrices can be understood as the complex versions of real [skew-symmetric matrices](#), which have real entries. ^[2] The set of all skew-Hermitian *n* × *n* matrices forms the *u*(*n*) [Lie algebra](#), which corresponds to the set of all *u*(*n*) [linear transformations](#) of any [complex vector space](#) with a [sesquilinear norm](#).

Note that the [adjoint](#) of an operator depends on the [scalar product](#) considered on the *n* dimensional complex vector space. If *A* is the adjoint of *A* with respect to the standard scalar product on *K*^{*n*}, then saying *A* is skew-[adjoint](#) means that for all **u**, **v** ∈ *K*^{*n*} one has (*A***u** | **v**) = −(**u** | *A***v**).

[Imaginary numbers](#) can be thought of as skew-[adjoint](#) (since they are like 1 × 1 matrices), whereas [real numbers](#) are [self-adjoint](#).

Example [\[edit \]](#)

For example, the following matrix is skew-Hermitian

$$A = \begin{bmatrix} -i & +2+i \\ -2+i & 0 \end{bmatrix}$$

because

$$-A = \begin{bmatrix} i & -2-i \\ 2-i & 0 \end{bmatrix} = \begin{bmatrix} \overline{-i} & \overline{-2+i} \\ \overline{2+i} & \overline{0} \end{bmatrix} = \begin{bmatrix} \overline{-i} & \overline{2+i} \\ \overline{-2+i} & \overline{0} \end{bmatrix}^T = A^H$$

Properties [\[edit \]](#)

- The eigenvalues of a skew-Hermitian matrix are all purely imaginary (and possibly zero). Furthermore, the matrix is diagonalizable and their eigenvectors for distinct eigenvalues must be orthogonal.^[3]
- All entries on the [main diagonal](#) of a skew-Hermitian matrix have to be pure [imaginary](#); i.e., on the imaginary axis of the complex plane.

purely imaginary).^[4]

- If A and B are skew-Hermitian, then $aA + bB$ is skew-Hermitian for all real scalars a and b .^[5]
- A is skew-Hermitian if and only if iA (or equivalently, $-iA$) is Hermitian.^[5]
- A is skew-Hermitian if and only if the real part $\Re(A)$ is skew-symmetric and the imaginary part $\Im(A)$ is symmetric.
- If A is skew-Hermitian, then A^k is Hermitian if k is an even integer and skew-Hermitian if k is an odd integer.
- A is skew-Hermitian if and only if $\mathbf{x}^H A \mathbf{y} = -\overline{\mathbf{y}^H A \mathbf{x}}$ for all vectors $\mathbf{x}, \mathbf{y}$.
- If A is skew-Hermitian, then the matrix exponential e^A is unitary.
- The space of skew-Hermitian matrices forms the Lie algebra $\mathfrak{u}(n)$ of the Lie group $U(n)$.

Isometrisk Matrix

Er ikke engang beskrevet i kreyszig.

Chat siger dog:

En isometrisk matrix er en kvadratisk matrix A , der bevarer længden af vektorer under transformation. Det betyder, at for enhver vektor $\mathbf{x}$ gælder det, at:

$$\|A\mathbf{x}\| = \|\mathbf{x}\|$$

hvor $\|\cdot\|$ betegner euklidisk norm. Algebraisk betyder dette, at en isometrisk matrix opfylder følgende egenskab:

$$A^T A = I$$

hvor A^T er transponeringen af A , og I er enhedsmatricen. Hvis A er en kompleks matrix, gælder den isometriske egenskab:

$$A^* A = I$$

hvor A^* er den Hermiteske konjugerede (det vil sige den transponerede og kompleks konjugeret) af A .

Egenskaber ved isometriske matricer

1. **Ortogonale rækker og søjler:** I en isometrisk matrix er rækkerne og søjlerne ortogonale enhedsvektorer.
2. **Bevarelse af indre produkt:** For enhver vektor $\mathbf{x}$ og $\mathbf{y}$, gælder det, at:
$$(A\mathbf{x}) \cdot (A\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$$
3. **Egenverdier:** Egenverdierne af en isometrisk matrix har en absolut værdi på 1.
4. **Invers:** En isometrisk matrix A er invertibel, og dens inverse er dens transponerede (for reelle matricer) eller dens Hermiteske konjugerede (for komplekse matricer):

$$A^{-1} = A^T \quad \text{eller} \quad A^{-1} = A^*$$

Og så med et træls eksempel (chat fucker lige nu)

Eksempel

En simpel type isometrisk matrix er en rotationsmatrix i to dimensioner:

```
\cos \theta & -\sin \theta \\
\sin \theta & \cos \theta \\
\end{pmatrix}
```

For denne matrix gælder det, at:

```
\[ A^T = \begin{pmatrix}
\cos \theta & \sin \theta \\
-\sin \theta & \cos \theta
\end{pmatrix}
```

og

```
\[ A^T A = \begin{pmatrix}
\cos \theta & \sin \theta \\
-\sin \theta & \cos \theta
\end{pmatrix} \begin{pmatrix}
\cos \theta & -\sin \theta \\
\sin \theta & \cos \theta
\end{pmatrix} = \begin{pmatrix}
\cos^2 \theta + \sin^2 \theta & 0 \\
0 & \cos^2 \theta + \sin^2 \theta
\end{pmatrix} = \begin{pmatrix}
1 & 0 \\
0 & 1
\end{pmatrix} = I
```

Sammenfatning

En isometrisk matrix bevarer længden af vektorer, som den transformerer. Den opfylder $\|Ax\| = \|x\|$ for reelle matricer eller $\|A^*x\| = \|x\|$ for komplekse matricer. Dette betyder, at en isometrisk matrix har ortogonale rækker og søjler og egenverdier med absolut værdi 1, og den bevarer vektorernes indre produkt.

▼ Stokastisk Matrix

Hører nok snarere til i ASTA kurset, så det bliver bare en wiki-snippet:

In mathematics, a **stochastic matrix** is a [square matrix](#) used to describe the transitions of a [Markov chain](#) [number](#) representing a [probability](#).^{[1][2]: 9–11} It is also called a **probability matrix**, **transition matrix**, *substochastic matrix*. A stochastic matrix was first developed by [Andrey Markov](#) at the beginning of the 20th century, and has found applications in many fields, including [probability theory](#), statistics, [mathematical finance](#) and [linear algebra](#), as well as [computer science](#). There are several different definitions and types of stochastic matrices.^{[2]: 9–11}

A **right stochastic matrix** is a square matrix of nonnegative real numbers, with each row summing to 1.

A **left stochastic matrix** is a square matrix of nonnegative real numbers, with each column summing to 1.

A *doubly stochastic matrix* is a square matrix of nonnegative real numbers with each row and column summing to 1.

In the same vein, one may define a *stochastic vector* (also called *probability vector*) as a [vector](#) whose elements sum to 1. Thus, each row of a right stochastic matrix (or column of a left stochastic matrix) is a stochastic vector. The standard language mathematics literature is to use [row vectors](#) of probabilities and right stochastic matrices rather than column vectors and left stochastic matrices; this article follows that convention.^{[2]: 1–8} In addition, a **substochastic matrix** is a real square matrix with nonnegative entries such that the sum of each row is at most 1.

Definition and properties [\[edit \]](#)

A stochastic matrix describes a [Markov chain](#) X_t over a [finite state space](#) S with [cardinality](#) α .

If the [probability](#) of moving from i to j in one time step is $\Pr(j|i) = P_{i,j}$, the stochastic matrix P is given by element, e.g.,

$$P = \begin{bmatrix} P_{1,1} & P_{1,2} & \cdots & P_{1,j} & \cdots & P_{1,\alpha} \\ P_{2,1} & P_{2,2} & \cdots & P_{2,j} & \cdots & P_{2,\alpha} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ P_{i,1} & P_{i,2} & \cdots & P_{i,j} & \cdots & P_{i,\alpha} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ P_{\alpha,1} & P_{\alpha,2} & \cdots & P_{\alpha,j} & \cdots & P_{\alpha,\alpha} \end{bmatrix}.$$

Since the total of transition probability from a state i to all other states must be 1,

$$\sum_{j=1}^{\alpha} P_{i,j} = 1;$$

thus this matrix is a right stochastic matrix.[\[2\]: 1–8](#)

The above elementwise sum across each row i of P may be more concisely written as $P\mathbf{1} = \mathbf{1}$, where $\mathbf{1}$ is Using this, it can be seen that the product of two right stochastic matrices P' and P'' is also right stochastic. The k -th power P^k of a right stochastic matrix P is also right stochastic. The probability of transitioning from element of the square of P :

$$(P^2)_{i,j}.$$

In general, the probability transition of going from any state to another state in a finite Markov chain given

An initial probability distribution of states, specifying where the system might be initially and with what pro

A *stationary* [probability vector](#) π is defined as a distribution, written as a row vector, that does not change is, it is defined as a probability distribution on the set $\{1, \dots, n\}$ which is also a row [eigenvector](#) of the pr

$$\pi P = \pi.$$

It can be shown that the [spectral radius](#) of any stochastic matrix is one. By the [Gershgorin circle theorem](#) have absolute values less than or equal to one. Additionally, every right stochastic matrix has an "obvious" eigenvalue 1: the vector $\mathbf{1}$ used above, whose coordinates are all equal to 1. As left and right eigenvalues stochastic matrix has, at least, a row [eigenvector](#) associated to the [eigenvalue](#) 1 and the largest absolute [Brouwer Fixed Point Theorem](#) (applied to the compact convex set of all probability distributions of the finite eigenvector which is also a stationary probability vector.

On the other hand, the [Perron–Frobenius theorem](#) also ensures that every [irreducible](#) stochastic matrix has absolute value of an eigenvalue is always 1. However, this theorem cannot be applied directly to such ma

In general, there may be several such vectors. However, for a matrix with strictly positive entries (or, more generally, an irreducible and aperiodic stochastic matrix), this vector is unique and can be computed by observing that for any i we have the following:

$$\lim_{k \rightarrow \infty} (P^k)_{i,j} = \pi_j,$$

where π_j is the j -th element of the row vector π . Among other things, this says that the long-term probability of being in state j is π_j , regardless of the initial state i . That both of these computations give the same stationary vector is a form of an [ergodic theorem](#). In the context of [dissipative dynamical systems](#), the system evolves, over time, to a [stationary state](#).

Intuitively, a stochastic matrix represents a Markov chain; the application of the stochastic matrix to a probability vector represents one step of the Markov chain, moving the mass of the original distribution while preserving its total mass. If this process is applied repeatedly, the distribution converges to the stationary vector. [2]: 55–59

Example: Cat and Mouse [edit]

Suppose there is a timer and a row of five adjacent boxes. At time zero, a cat is in the first box, and a mouse is in the fifth box. Both jump to a random **adjacent** box when the timer advances. For example, if the cat is in the second box, the probability that *the cat will be in the first box and the mouse in the fifth after the timer advances* is one fourth. If the cat is in the second box, the probability that *the cat will be in box two and the mouse will be in box four after the timer advances* is one fourth. The game ends when the cat and mouse are in the same box, at which time the game ends. Let the **random variable** K be the time the mouse stays in the first box.

The **Markov chain** that represents this game contains the following five states specified by the combination of the cat's and mouse's positions. A naive enumeration of states would list 25 states, many are impossible either because the mouse can never move past the cat (which would mean the mouse occupied the cat's box and survived to move past it), or because the sum of the cat's and mouse's positions is not equal to 6. In addition, the 3 possible states that lead to the mouse's death are combined into one:

- State 1: (1,3)
- State 2: (1,5)
- State 3: (2,4)
- State 4: (3,5)
- State 5: game over: (2,2), (3,3) & (4,4).

We use a stochastic matrix, P (below), to represent the **transition probabilities** of this system (rows and columns are the possible states listed above, with the pre-transition state as the row and post-transition state as the column). The 1st row – it is impossible for the system to stay in this state, so $P_{11} = 0$; the system also cannot transition to state 2 – it stayed in the same box – so $P_{12} = 0$, and by a similar argument for the mouse, $P_{14} = 0$. Transitions to state 5 are possible, so $P_{13}, P_{15} \neq 0$.

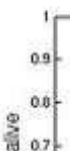
$$P = \begin{bmatrix} 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1 & 0 & 0 \\ 1/4 & 1/4 & 0 & 1/4 & 1/4 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Long-term averages [edit]

No matter what the initial state, the cat will eventually catch the mouse (with probability 1) and a stationary distribution exists in the long-term limit.^{[2]: 55–59} To compute the long-term average or expected value of a stochastic variable Y , for each state j , we compute $Y_{j,k} \cdot P(S = S_j, t = t_k)$. Survival can be treated as a binary variable with $Y = 1$ for a surviving state and $Y = 0$ for a state where the mouse has perished. States with $Y = 0$ do not contribute to the long-term average.

Phase-type representation [edit]

As State 5 is an absorbing state, the distribution of time to absorption is **discrete phase-type distributed**. Suppose the system starts in state 2, represented by the vector $[0, 1, 0, 0, 0]$. The states where the mouse has perished don't contribute to the survival average so state five can be ignored. The initial state and transition matrix can be reduced to,



$$\tau = [0, 1, 0, 0], \quad T = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 \\ \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} \\ 0 & 0 & \frac{1}{2} & 0 \end{bmatrix},$$

and

$$(I - T)^{-1} \mathbf{1} = \begin{bmatrix} 2.75 \\ 4.5 \\ 3.5 \\ 2.75 \end{bmatrix},$$

where I is the [identity matrix](#), and $\mathbf{1}$ represents a column matrix of all ones that acts as a sum over states

Since each state is occupied for one step of time the expected time of the mouse's survival is just the [sum](#) of surviving states and steps in time,

$$E[K] = \tau (I + T + T^2 + \dots) \mathbf{1} = \tau (I - T)^{-1} \mathbf{1} = 4.5.$$

Higher order moments are given by

$$E[K(K-1) \dots (K-n+1)] = n! \tau (I - T)^{-n} T^{n-1} \mathbf{1}.$$

Shit hvor er der mange matrix typer vi ikke har hørt om

▼ Normal Matrix



The survival at least th

A matrix A is said to be a **Normal matrix** if the pre and post matrix multiplication of the conjugate transpose of A with the matrix A is commutative. In other words, normal matrices are those matrices whose **matrix multiplication** with its own conjugate transpose is commutative.

Normal Matrix is a generalisation of Unitary, Hermitian, skew-Hermitian as well as symmetric and skew-symmetric matrices.

A normal matrix is also unitarily diagonalizable, which means to diagonalize a normal matrix the required modal matrix is a unitary matrix. Let us define a normal matrix more formally in mathematical terms.

Definition of Normal Matrix

A square matrix A of order $n \times n$ of complex numbers, that is $A \in \mathbb{C}^{n \times n}$ is said to be a normal matrix if $A^H A = A A^H$ where A^H is the conjugate transpose of A .

The matrix $A \in \mathbb{C}^{n \times n}$ is a Normal Matrix $\Leftrightarrow A^H A = A A^H$

The matrix A is also said to be a Normal matrix if there exists a Unitary matrix U such that U acts as a modal matrix which diagonalizes A .

The matrix $A \in \mathbb{C}^{n \times n}$ is a Normal Matrix $\Leftrightarrow D = U A U^{-1}$

where D is a diagonal matrix of A .

Examples of Normal Matrix

Let us take an example of a complex matrix of order 2×2 ,

$$\begin{bmatrix} 4i & -1+i \\ 1-i & 4i \end{bmatrix}$$

So according to the definition of normal matrix let us find the conjugate transpose of the given matrix.

Let

$$A = \begin{bmatrix} 4i & -1+i \\ 1-i & 4i \end{bmatrix}$$

Then, $A^H = (\text{conjugate of } A)^T =$

$$\begin{aligned} & \begin{bmatrix} -4i & -1-i \\ 1+i & -4i \end{bmatrix}^T \\ &= \begin{bmatrix} -4i & 1+i \\ -1-i & -4i \end{bmatrix} \end{aligned}$$

Then $A^H A =$

$$\begin{aligned} & \begin{bmatrix} -4i & 1+i \\ -1-i & -4i \end{bmatrix} \begin{bmatrix} 4i & -1+i \\ 1-i & 4i \end{bmatrix} \\ &= \begin{bmatrix} 18 & 8i \\ -8i & 18 \end{bmatrix} \end{aligned}$$

And $AA^H =$

$$\begin{aligned} & \begin{bmatrix} 4i & -1+i \\ 1-i & 4i \end{bmatrix} \begin{bmatrix} -4i & 1+i \\ -1-i & -4i \end{bmatrix} \\ &= \begin{bmatrix} 18 & 8i \\ -8i & 18 \end{bmatrix} \end{aligned}$$

Hence $A^H A = AA^H$, that is a normal matrix.

Let us take another example of a real matrix and check whether it satisfies the condition of a normal matrix.

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

,

For a real matrix, the conjugate transpose will be the same as the transpose of matrix.

Therefore, $A^H = A^T =$

$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

Now, $A^T A =$

$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

And $AA^T =$

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Thus, $A^T A = A A^T$, A is normal.

With this above example we can also say that orthogonal matrices are also normal matrix.

Properties of Normal Matrix

There are a few interesting properties of a Normal matrix are listed below which are useful in deriving many more results in Linear Algebra.

- If A is a normal matrix then it is diagonalizable by a unitary matrix.
- A Hermitian matrix is a normal matrix.

Let A be a Hermitian matrix then $A^H = A$, then

$$\text{Now, } A^H A - A A^H = A A - A A = A^2 - A^2 = 0$$

$$\Rightarrow A^H A - A A^H = 0 \Rightarrow A^H A = A A^H$$

$\Rightarrow A$ is normal

- A skew-Hermitian matrix is a normal matrix.
- A unitary matrix is normal.

If U is a unitary matrix, then $U^H U = U U^H = I$, hence normal.

- A symmetric and a skew-symmetric matrix both are normal matrices.
- A normal matrix need not be a Hermitian, skew-Hermitian, Unitary or symmetric matrix.
- An orthogonal matrix is also a normal matrix.
- If A is normal then, $A A^H$ is a Hermitian matrix.
- If A is normal, then there exists a set of orthonormal eigenvectors of A in $\mathbb{C}^n$.
- A normal matrix is unitary if and only if its eigenvalues lie on the unit circle in complex plane.

▼ Enheds Matrix - Se identitesmatrice

Det samme som identitesmatricen

▼ Skalar Matrix

En skalar matrice er kort sagt en identitetsmatrice, som er ganget med en konstant, så værdien er alt andet end 1 diagonalt:

The scalar matrix is a square matrix having a constant value for all the elements of the principal diagonal, and the other elements of the matrix are zero. The scalar matrix is obtained by the product of the identity matrix with a numeric constant value.

Let us learn more about the definition of the scalar matrix, terms related to the scalar matrix, arithmetic operations of scalar matrix, and the examples of scalar matrix.

What Is a Scalar Matrix?

The scalar matrix is a square matrix having a constant value as every element of its principal **diagonal**, and all other elements are equal to zero. The scalar matrix is a square matrix having an equal number of rows and columns.

$$A = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$$

Here in the above matrix the principal diagonal elements are all equal to the same numeric value of 'a', and all other elements of the matrix are equal to zero.

The scalar matrix is derived from an **identity matrix**, where the product of the identity matrix with a constant value, gives the scalar matrix. Here the constant 'k' is multiplied with an identity matrix to obtain the scalar matrix, and the **order of the matrix** is 3×3 .

Scalar Matrix Formula



$$\text{Constant} \times \text{Identity Matrix} = \text{Scalar Matrix}$$

$$k \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{bmatrix}$$



Terms Related to Scalar Matrix

The following are some of the important terms, which are helpful in a better understanding of the concept of scalar matrix.

Identity Matrix

The identity matrix is a square matrix and is a multiplicative identity for matrices. The identity matrix contains 1 as its diagonal element and all other elements are equal to zero. Identity matrix has numerous applications in the [multiplication of matrices](#), and in finding the inverse of a matrix, The identity matrix on multiplying with a constant value results in a scalar matrix.

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Diagonal Matrix

The diagonal matrix is also a square matrix, having elements of different value across the principal diagonal and all other elements are equal to zero. Further, if the diagonal elements of the diagonal matrix are all made equal, then it is called a scalar matrix.

$$D = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

Principal Diagonal

The elements from the first element of the first row, to the last element of the last row, if connected with a straight line, all the elements falling on this imaginary straight line in the matrix, represent the principal diagonal. In an identity matrix, the principal diagonal elements are all equal to 1, and in a scalar matrix, all the principal diagonal elements are equal to a constant value.

Constant

This is a simple numeric value, which can be an integer, rational number, decimal number, or root value. The identity matrix is multiplied by a constant value to obtain the scalar matrix. A matrix multiplied by a constant value, multiplies with each of the elements of the matrix.

Matrix Operations Using Scalar Matrix

The matrix operations involving the scalar matrix are almost the same as the **arithmetic operations** of any other type of matrices. The addition and subtraction across a scalar matrix and any other matrix are the same as that across any two other matrices. But the multiplication of a scalar matrix with another matrix can be observed in the following few steps.

$$\text{Scalar Matrix } A = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha \end{bmatrix}, \text{ Matrix } B = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}.$$

$$A \times B = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha \end{bmatrix} \times \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$= \alpha \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$= \alpha \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$A \times B = \begin{bmatrix} \alpha a & \alpha b & \alpha c \\ \alpha d & \alpha e & \alpha f \\ \alpha g & \alpha h & \alpha i \end{bmatrix}$$

$$A \times B = \alpha B$$

Thus the multiplication of a scalar matrix with any other matrix is equal to the multiplication of the constant element of the scalar matrix with all the elements of the other matrix.

▼ Singular Matrix

Heller ikke noget vi har mødt før:

Singular Matrix

We determine whether a matrix is a singular matrix or a non-singular matrix depending on its determinant. The determinant of a matrix 'A' is denoted by ' $\det A$ ' or ' $|A|$ '. If the determinant of a matrix is 0, then it is said to be a singular matrix. Why do we need to have a specific name for the matrices with determinant 0? Let us see.

Let us learn more about the singular matrix along with its definition, formula, properties, and examples.

What is a Singular Matrix?

A **singular matrix** is a square matrix if its **determinant** is 0. i.e., a square matrix A is singular if and only if $\det A = 0$. We know that the inverse of a matrix A is found using the formula $A^{-1} = (\text{adj } A) / (\det A)$. Here $\det A$ (the determinant of A) is in the denominator. We are aware that a **fraction** is NOT defined if its **denominator** is 0. Hence A^{-1} is NOT defined when $\det A = 0$. i.e., the inverse of a singular matrix is NOT defined. i.e., there does not exist any matrix B such that $AB = BA = I$ (where I is the **identity matrix**).

From the above explanation, a square matrix 'A' is said to be singular if

- **$\det A = 0$ (which is also written as $|A| = 0$) (or)**
- **A^{-1} is NOT defined (i.e., A is **non-invertible**).**

Singular Matrix



A square matrix 'A' is singular if



$$\det A = 0$$

(or equivalently)





A^{-1} is not defined

Identifying a Singular Matrix

To determine/identify whether a given matrix is singular we need to check for two conditions:

- check whether A is a **square matrix**.
- check whether $\det A = 0$.

Here are few examples to find whether a given matrix is singular.

- $A = \begin{bmatrix} 3 & 6 \\ 2 & 4 \end{bmatrix}$ is a singular matrix because

it is a square matrix (of order 2×2) and
 $\det A$ (or) $|A| = 3 \times 4 - 6 \times 2 = 12 - 12 = 0$.

- $A = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 2 \\ 3 & 2 & -1 \end{bmatrix}$ is a singular matrix because

it is a square matrix (of order 3×3) and
as $\det A$ (or) $|A| = 0$ (as the first two rows are equal).

Properties of Singular Matrix

Here are some singular matrix properties based upon its definition.

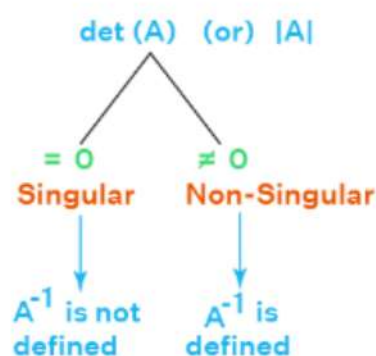
- Every singular matrix is a square matrix.
- The determinant of a singular matrix is 0.
- The **inverse** of a singular matrix is NOT defined and hence it is non-invertible.
- By **properties of determinants**, in a matrix,
 - * if any two rows or any two columns are identical, then its determinant is 0 and hence it is a singular matrix.
 - * if all the elements of a row or column are zeros, then its determinant is 0 and hence it is a singular matrix.
 - * if one of the rows (columns) is a scalar multiple of the other row (column) then the determinant is 0 and hence it is a singular matrix.
- A **null matrix** of any order is a singular matrix.

- The **rank** of a singular matrix is definitely less than the order of the matrix. For example, the rank of a 3x3 matrix is less than 3.
- All rows and columns of a singular matrix are NOT linearly independent.

Singular Matrix and Non-Singular Matrix

A non-singular matrix, as its name suggests, is a matrix that is NOT singular. Thus, the **determinant** of a non-singular matrix is a nonzero number. i.e., a square matrix 'A' is said to be a non singular matrix if and only if $\det A \neq 0$. Then it is obvious that A^{-1} is defined. i.e., a non-singular matrix always has a multiplicative inverse. Thus, we can summarize the differences between the singular matrix and non-singular matrix as follows:

Singular and Non - Singular Matrices



Singular Matrix	Non Singular Matrix
A matrix 'A' is singular if $\det (A) = 0$.	A matrix 'A' is nonsingular if $\det (A) \neq 0$.
If 'A' is singular then A^{-1} is NOT defined.	If 'A' is nonsingular then A^{-1} is defined.
Rank of A < Order of A.	Rank of A = Order of A.
Some rows and columns are linearly dependent.	All rows and columns are linearly independent.
If 'A' is singular then the system of simultaneous equations $AX = B$ has either no solution or has	If 'A' is non singular then the system of simultaneous equations $AX = B$ has a unique solution.

infinitely many solutions.

Example: $\begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix}$ is singular as

$$\begin{vmatrix} 3 & 6 \\ 1 & 2 \end{vmatrix} = 3 \times 2 - 1 \times 6 = 6 - 6 = 0.$$

solution.

Example: $\begin{bmatrix} 3 & 2 \\ 1 & -2 \end{bmatrix}$ is non-

singular as

$$\begin{vmatrix} 3 & 2 \\ 1 & -2 \end{vmatrix} = 3 \times -2 - 1 \times 2 = -6 - 2 \\ = -8 \neq 0.$$

Theorem to Generate Singular Matrices

There is one important theorem on singular matrix that can actually be used to generate a singular matrix and the theorem says: "The product of two matrices $A = [A]_{n \times k}$ and $B = [B]_{k \times n}$ (where $n > k$) is a matrix AB of order $n \times n$ and is always singular". By this theorem:

- The product AB of two matrices A of order $n \times 1$ and B of order $1 \times n$ is singular always.
- The product AB of two matrices A of order $n \times 2$ and B of order $2 \times n$ is also singular, etc.

Using this theorem, one can generate a singular matrix by multiplying two randomly generated matrices of orders $n \times k$ and $k \times n$ where $n > k$.

- - - - -

▼ Komplement - mangler forståelse

Got no clue

▼ Konisk Snit

I lineær algebra refererer et konisk snit til de kurver, der dannes ved at skære en dobbeltkeglet plan. Disse kurver er kendt som de klassiske koniske snit og omfatter ellipse, parabler og hyperbole. De kan beskrives ved kvadratiske ligninger i to variable (typisk x og y).

Generel Form af Koniske Snit

En generel kvadratisk ligning i to variable, der beskriver et konisk snit, har formen:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

hvor A, B, C, D, E og F er konstanter.

Typer af Koniske Snit

Koniske snit klassificeres baseret på værdien af diskriminanten $\Delta = B^2 - 4AC$:

1. **Ellipse** (inklusive cirkel): $\Delta < 0$
2. **Parabole**: $\Delta = 0$
3. **Hyperbole**: $\Delta > 0$

Matricer og Koniske Snit

I lineær algebra kan kvadratisk ligning skrives ved hjælp af matricer og vektorer. Ligningen $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ kan omformes til matrixform:

$$\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{p}^T \mathbf{x} + F = 0$$

hvor

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} A & \frac{B}{2} \\ \frac{B}{2} & C \end{pmatrix}, \quad \mathbf{p} = \begin{pmatrix} D \\ E \end{pmatrix}$$

Dette er en kvadratisk form, hvor:

$$\mathbf{x}^T \mathbf{Q} \mathbf{x} = Ax^2 + Bxy + Cy^2$$

$$\mathbf{p}^T \mathbf{x} = Dx + Ey$$

Anvendelse af Lineær Algebra på Koniske Snit

Ved at anvende lineær algebra kan vi analysere og klassificere koniske snit ved at studere matrix $\mathbf{Q}$:

1. **Egenverdier og egenvektorer** af $\mathbf{Q}$ kan bruges til at bestemme konisk snitets form og orientering.
2. **Diagonalisering** af $\mathbf{Q}$ kan bruges til at rotere og skalere koordinatsystemet for at bringe ligningen til en standardform, hvilket gør det lettere at identificere typen af konisk snit.

Eksempel

Lad os betragte en specifik kvadratisk ligning og omskrive den til matrixform:

$$3x^2 - 4xy + y^2 + 6x - 7y + 2 = 0$$

Vi kan skrive denne ligning som:

$$\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{p}^T \mathbf{x} + F = 0$$

hvor

$$\mathbf{Q} = \begin{pmatrix} 3 & -2 \\ -2 & 1 \end{pmatrix}, \quad \mathbf{p} = \begin{pmatrix} 6 \\ -7 \end{pmatrix}, \quad F = 2$$

Ved at analysere matrixen $\mathbf{Q}$ og dens egenverdier kan vi bestemme, hvilken type konisk snit ligning repræsenterer. Egenverdierne af $\mathbf{Q}$ fortæller os om formen (ellipse, hyperbole eller parabole) og orienteringen af konisk snit.

Denne fremgangsmåde viser, hvordan lineær algebra kan bruges til at studere geometriske objekter som koniske snit gennem matrixrepræsentationer og egenverdianalyse.

▼ Similaritetsformation

Similaritetsformation er et begreb inden for lineær algebra, der handler om forholdet mellem lineære transformationer eller matricer. To kvadratiske matricer A og B siges at være similar, hvis der eksisterer en invertibel matrix P , sådan at:

$$B = P^{-1}AP$$

Hvis A og B er similar, har de samme egenværdier, og deres egenværdier er i samme multiplisitet. Dette betyder, at de to matricer beskriver den samme lineære transformation i forskellige koordinatsystemer. Den invertible matrix P repræsenterer blot en skift af basis i vektorrummet.

At finde en similaritetsomdannelse kan være nyttigt, fordi det kan forenkle analyse af en given transformation eller en matrix. For eksempel kan en matrix A være vanskelig at arbejde med, hvis vi kan finde en matrix B , der er similar til A , men i en mere bekvem form (f.eks. en diagonalmatrix), kan det gøre det lettere at beregne egenværdier, eksponentier, og andre egenskaber ved transformationen eller matricen.

▼ Indre Produkt

Det indre produkt af to matricer refererer normalt til en operation, der kombinerer elementerne i to matricer for at producere en skalar værdi. Dette er forskelligt fra det mere almindelige "indre produkt" af to vektorer, hvor resultatet er en skalar.

For to matricer A og B med samme dimensioner $m \times n$, er det indre produkt defineret som elementvise multiplikation efterfulgt af summering af produkterne:

$$\langle A, B \rangle = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ij}$$

Hvor A_{ij} og B_{ij} er elementerne på plads (i, j) i henholdsvis matricerne A og B .

Dette resulterende indre produkt er en skalarværdi.

Det er vigtigt at bemærke, at "indre produkt" af matricer normalt bruges i specifikke kontekster, såsom i nogle typer af beregninger inden for lineær algebra eller maskinlæring. Det er ikke en sædvanlig operation sammenlignet med indre produkt af vektorer.

▼ *Augmented matrix*

En augmented matrice er en matrice hvor du har et sæt af ligninger, med tilhørende løsninger. Eksempelvis:

$$\begin{bmatrix} -81 & -98 & -76 & -4 & 29 \\ -38 & -77 & -72 & 27 & 44 \\ -18 & 57 & -2 & 8 & 92 \\ 87 & 27 & -32 & 69 & -31 \\ 33 & -93 & -74 & 99 & 67 \end{bmatrix} = \begin{bmatrix} -16 \\ -9 \\ -50 \\ -22 \\ 45 \end{bmatrix}$$

hvor den augmented matrice er med begge dele sat sammen, altså:

$$\begin{bmatrix} -81 & -98 & -76 & -4 & 29 & -16 \\ -38 & -77 & -72 & 27 & 44 & -9 \\ -18 & 57 & -2 & 8 & 92 & -50 \\ 87 & 27 & -32 & 69 & -31 & -22 \\ 33 & -93 & -74 & 99 & 67 & 45 \end{bmatrix}$$

Og det er gudhjælpemig vigtigt at vide det, fordi det skal bruges til eksamen og bliver ikke forklaret før

▼ *Coeffecient matrix*

Koeffecientmatricen er så en matrice skabt fra et ligningssystem, men UDEN løsninger (se augmented matrice).

Fra eksemplet i augmented matrice, vil det altså kun være det rødmarkerede, som er koefficientmatricen

$$\begin{bmatrix} -81 & -98 & -76 & -4 & 29 \\ -38 & -77 & -72 & 27 & 44 \\ -18 & 57 & -2 & 8 & 92 \\ 87 & 27 & -32 & 69 & -31 \\ 33 & -93 & -74 & 99 & 67 \end{bmatrix} = \begin{bmatrix} -16 \\ -9 \\ -50 \\ -22 \\ 45 \end{bmatrix}$$

Når du regner nulrum og rank er det også KUN koeffecientmatricen man benytter.

▼ Undervisning fra Gilberto

▼ *Discrete time signals - basic sequences/intro til de helt små ting*

Discrete tidssignaler kan beskrives som en sekvens af tal/værdier som spænder

$-\infty < n < \infty$ med n som en integer - det er meget vigtigt vi arbejder i heltal, da vores tidsstep

SKAL forekomme med en ensartet samplingshastighed, og tidssteppet er ikke nødvendigvis en sekundangiver, den angiver bare at nu er vi ved step nummer $n \cdot T$, hvor T er en arbitrær størrelse.

Sekvenser beskrives ved:

$$x = \{x[n]\}$$

- Discrete signals can be represented as a sequence of numbers

$$x = \{x[n]\} \quad -\infty < n < \infty$$

where n is an integer.

- In case such sequences arise from periodic sampling of an analog signal:

$$x[n] = x_a[nT_s] \quad -\infty < n < \infty$$

where T_s is the *sampling interval* and $f_s = 1/T_s$ is the *sampling frequency*.

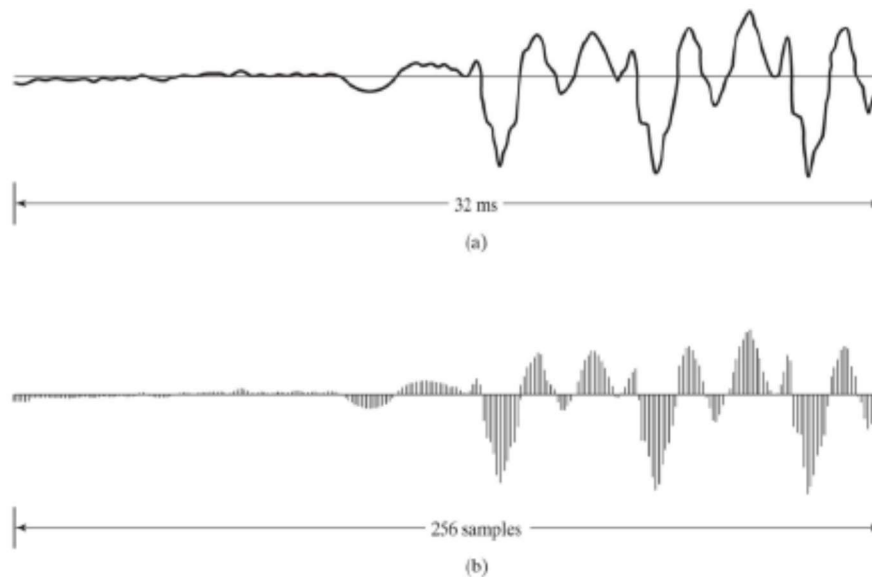


Fig 2 (13) (a) Segment of a continuous-time speech signal $x_o(t)$.

(b) Sequence of samples $x[n] = x_o(nT_s)$ obtained from the signal in part (a) with $T_s = 125 \mu s$.

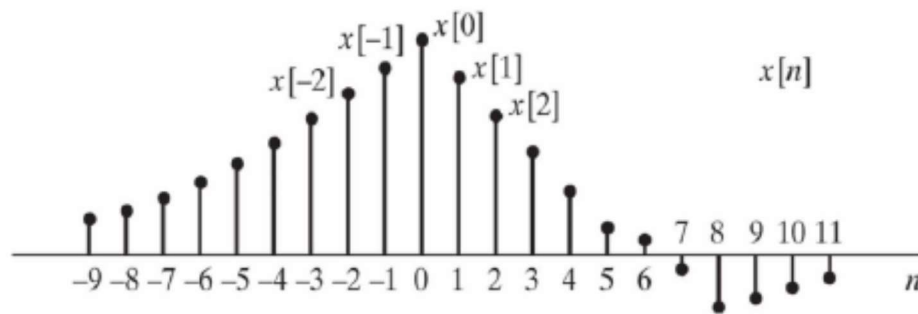


Fig 1 (13) Graphic representation of a discrete-time signal.

Når vi så forskyder en sekvens ($x[n]$) så kaldes den for $y[n]$.

Dette er særligt vigtigt at forstå, da det er en manipulation af sekvensens tidsforskydelse, men IKKE VÆRDIERNE I SEKVENSEN.

Den forskudte version af $x[n]$ er givet ved:

$$y[n] = x[n - n_0]$$

Hvor n_0 ligesom n skal være en integer.

Når det kommer til unit sample, så skal man bruge formen:

$$\delta[n] = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$$

som skal læses som en true/false statement, som i dette tilfælde betyder at udtrykket af sandt når vi aflæser lige præcis i impulsen (ved tid 0)

Se også:

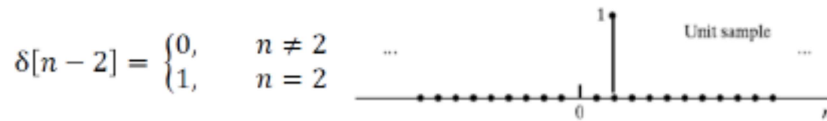
Mathematically, a unit sample sequence can be defined as:

$$x[n] = \delta[n - n_0] = \begin{cases} 1 & n = n_0 \\ 0 & n \neq n_0 \end{cases}$$

Where:

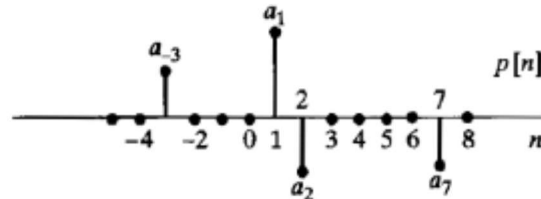
- $x[n]$ represents the value of the sequence at index n .
- $\delta[n]$ is the discrete-time impulse function or delta function.
- n_0 is the index at which the impulse occurs. At this index, $x[n]$ has the value of 1. For all other indices, $x[n]$ is zero.

Et eksempel på en forsinket unit sample sekvens er:



Et godt eksempel på hvad der kan beskrive en sekvens er dette, nakket fra slide 10 kursusgang 8:

- An arbitrary sequence can be represented as a sum of scaled, delayed, impulses.



$$p[n] = a_{-3}\delta[n+3] + a_1\delta[n-1] + a_2\delta[n-2] + a_7\delta[n-7].$$

- More generally, any sequence can be expressed as:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k].$$

I dette eksempel er hele sekvensen navngivet $p[n]$, og hver impuls er beskrevet ved $n+x$. Det virker counterintuitivt at impulser på den "negative" side navngives ved $+x$, men det giver god mening, hvis man ser variabelen x i $[n+x]$ som værende den værdi, man skal tillægge/fratrække, for at nå 0. Altså skal der ligges noget til ved impulser efter tid 0 og trækkes noget fra ved impulser efter tid 0.

Grunden til jeg skriver EFTER TID 0 er, at de negative tider ligger ude i fremtiden, altså må den ligge +X tidskonstanter ude i fremtiden

Et andet godt eksempel kommer fra slide 11:

- The unit step sequence is defined as

$$u[n] = \begin{cases} 1, & n \geq 0, \\ 0, & n < 0. \end{cases}$$



- The unit step is related to the impulse by

$$u[n] = \sum_{k=-\infty}^n \delta[k]; \quad \text{or} \quad u[n] = \sum_{k=0}^{\infty} \delta[n-k].$$

- Conversely, the impulse sequence can be expressed as the first backward difference of the unit step sequence:

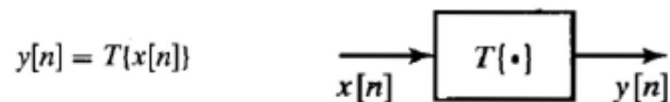
$$\delta[n] = u[n] - u[n-1].$$

Hvor vi kan se i true/false gaflen, at vi har unit steps ved alle n større end eller lig med 0, hvilket også tydeligt kan ses i grafen.

Yderligere kan vi ved definitionerne markeret med rød se, at impuls sekvensen kan beskrives med summen af alle impulser fra tidernes morgen og indtil den sidste impuls. Eller også kan den beskrives ved summen af alle impulser fra første impuls og indtil uendeligheden. - HÆNG MIG IKKE OP PÅ DENNE DEFINITION, det er bare sådan jeg lige læser udtrykkene.

Definition af diskrete tidssystemer

Et diskret tidssystem er en operator som skaber overgangen fra en input sekvens til output sekvensen. Eksempelvis:



- Examples of operators:

- Delay $y[n] = x[n - n_{\text{delay}}] \quad -\infty < n < \infty$

- Moving average $y[n] = \frac{1}{M1 + M2 + 1} \sum_{k=-M1}^{M2} x[n - k]$

- FIR filter $y[n] = \frac{1}{\sum_{k=0}^M b_k} \sum_{k=0}^M b_k \cdot x[n - k]$

▼ Linear Systems

Et lineært system er et system hvor den samme T (diskret tidssystem operator) ganges på alle led.

Altså er det noget hvor forskydninger og lignende er ens på alle unitsteps:

- The class of linear system is defined by the principle of superposition.

$$\begin{array}{lll} T\{x_1[n] + x_2[n]\} & = T\{x_1[n]\} + T\{x_2[n]\} & = y_1[n] + y_2[n] \\ T\{a \cdot x[n]\} & = a \cdot T\{x[n]\} & = a \cdot y[n] \end{array}$$



$$T\{a \cdot x_1[n] + b \cdot x_2[n]\} = a \cdot T\{x_1[n]\} + b \cdot T\{x_2[n]\} = a \cdot y_1[n] + b \cdot y_2[n]$$

- The accumulator system

$$y[n] = \sum_{k=-\infty}^n x[k] \quad \longrightarrow \quad \text{Linear system}$$

- Consider the following

$$w[n] = \log_{10}(|x[n]|). \quad \longrightarrow \quad \text{Non-Linear system}$$

Det nederste eksempel er **IKKE** lineært, da logaritmer ikke er lineære

Stability

Key note er at et stabilt system kun har 1 output for hvert 1 input. Det betyder at output forbliver begrænset for ethvert begrænset input. Det betyder at hvis inputsekvensen har en endelig maksimal værdi, vil outputsekvensen også have en endelig maksimal værdi.

Et ustabilt system har til gengæld mere end 1 output for hvert 1 input.

- A system is stable if and only if every bounded input sequence produces a bounded output sequence.

$$|x[n]| \leq B_x < \infty, \quad \text{for all } n. \quad \longrightarrow \quad |y[n]| \leq B_y < \infty, \quad \text{for all } n.$$

- Examples

$$y[n] = \sum_{k=-\infty}^n u[k] \quad \longrightarrow \quad \text{Not stable}$$

$$y[n] = x[n - n_d], \quad \longrightarrow \quad \text{Stable}$$

$$y[n] = \frac{1}{M_1 + M_2 + 1} \sum_{k=-M_1}^{M_2} x[n - k] \quad \longrightarrow \quad \text{Stable}$$

What about $y[n] = \log_{10}(|x[n]|)$?

Det første eksempel er ustabil fordi vi ikke har nogen reel begrænsning på summen (endelig maksimal værdi) da vi går mod negativ uendelig

Så er der også noget med at lave overføringsfunktioner og på den måde finde poler, og ud fra dem bedømme om et system er stabilt eller ustabil, men det hører sig vist til IIDS i stedet for.

▼ *Chat besvarelse*

Stabiliteten af et diskret tidssystem handler om systemets opførsel over tid, især om hvordan systemets output reagerer på input og initiale betingelser. Lad os definere, hvad det betyder for et system at være stabilt eller ustabilt.

Stabilitet af Diskrete Tidsystemer

1. Stabilt System:

- Et diskret tidssystem er stabilt, hvis dets output forbliver begrænset for ethvert begrænset input. Dette betyder, at selvom inputsekvensen har en endelig maksimal værdi, vil outputsekvensen også have en endelig maksimal værdi.
- Matematisk set, hvis inputsekvensen $x[n]$ er begrænset, dvs. der eksisterer en konstant M_x sådan at $|x[n]| \leq M_x$ for alle n , så er systemet stabilt, hvis der eksisterer en konstant M_y sådan at $|y[n]| \leq M_y$ for alle n , hvor $y[n]$ er outputsekvensen.

2. Asymptotisk Stabilt System:

- Et system er asymptotisk stabilt, hvis dets naturlige respons (dvs. responsen når input er nul) aftager til nul som tiden går mod uendelig.
- Dette indebærer, at alle poler af systemets overføringsfunktion $H(z)$ skal ligge inden for enhedscirklen i z -planen (dvs. $|z| < 1$ for alle poler).

3. Marginalt Stabilt System:

- Et system er marginalt stabilt, hvis alle poler af systemets overføringsfunktion ligger på enhedscirklen $|z| = 1$, men ingen poler har flere gentagelser (altså ingen poler på enhedscirklen har en multiplicitetsorden større end 1).



Ustabilt System

- Et diskret tidssystem er ustabilt, hvis det ikke er stabilt. Det betyder, at systemets output kan blive ubegrænset for et begrænset input.
- Matematisk, hvis inputsekvensen $x[n]$ er begrænset, og outputsekvensen $y[n]$ ikke er begrænset, er systemet ustabilt.
- I termer af poler af systemets overføringsfunktion $H(z)$, er systemet ustabilt, hvis en eller flere poler ligger uden for enhedscirklen i z -planen (dvs. $|z| > 1$ for mindst én pol).

Eksempler og Analyse

Eksempel på Stabilt System

Overføringsfunktion:

$$H(z) = \frac{z}{z-0.5}$$

- Polen ligger ved $z = 0.5$, som er inden for enhedscirklen ($|0.5| < 1$).
- Derfor er systemet stabilt.

Eksempel på Ustabilt System

Overføringsfunktion:

$$H(z) = \frac{z}{z-1.5}$$

- Polen ligger ved $z = 1.5$, som er uden for enhedscirklen ($|1.5| > 1$).
- Derfor er systemet ustabilt.

Eksempel på Marginalt Stabilt System

Overføringsfunktion:

$$H(z) = \frac{z}{z-1}$$

- Polen ligger ved $z = 1$, som er på enhedscirklen ($|1| = 1$), og har en multiplicitetsorden på 1.

- Derfor er systemet marginalt stabilt.

Sammenfatning

- **Stabilt System:** Output er begrænset for et begrænset input; alle poler ligger inden for enhedscirklen.
- **Asymptotisk Stabilt System:** Output går mod nul over tid; alle poler ligger inden for enhedscirklen.
- **Marginalt Stabilt System:** Output forbliver begrænset og oscillationer kan eksistere; poler ligger på enhedscirklen med multiplicitetsorden 1.
- **Ustabilt System:** Output kan blive ubegrænset for et begrænset input; mindst én pol ligger uden for enhedscirklen.

Stabilitet er afgørende for systemdesign og analyse, da et stabilt system sikrer pålidelig og forudsigelig opførsel over tid.

▼ Causality

Causality betyder i store træk at systemet vi kigger på ikke må være afhængig af fremtiden. Vi må benytte inputs fra tidernes morgen og så til nu, eller nu, eller nu, men ikke inputs der opstår om 1 sekund, da de principielt er lige så ukendte som inputs der opstår om 1000 år. Tid er som udgangspunkt causale, vores liv er causale, da alt indtil nu har været afhængig af alt det foregående.

I eksemplet herunder er det forskellen at i "forward difference system" vil vi kigge på en impuls ved $x[n+1]$ som betyder at vi vil se på en impuls i fremtiden, og det kan vi sjovt nok ikke, medmindre vi accepterer at systemet ikke er causalt.

- A system is causal if, for every choice of n_0 , the output sequence at the index $n=n_0$ depends only on the input sequence values for $n \leq n_0$.

- Forward difference system

$$y[n] = x[n + 1] - x[n]. \quad \Rightarrow \quad \text{Non-causal}$$

- Backward difference system

$$y[n] = x[n] - x[n - 1], \quad \Rightarrow \quad \text{Causal}$$

In [control theory](#), a **causal system** (also known as a [physical](#) or **nonanticipative system**) is a [system](#) where inputs but not future inputs—i.e., the output $y(t_0)$ depends only on the input $x(t)$ for values of $t \leq t_0$.

The idea that the output of a function at any time depends only on past and present values of input is defined **causality**. A system that has some dependence on input values from the future (in addition to possible dependence on past values) is termed a non-causal or [acausal system](#), and a system that depends *solely* on future input values is an [anticausal system](#). A system that depends *solely* on future *and present* input values is defined an anticausal system as one that depends solely on future *and present* input values or, more simply, on *present* input values. ^[1]

Classically, [nature](#) or physical reality has been considered to be a causal system. Physics involving [special relativity](#) has led to careful definitions of causality, as described elaborately in [Causality \(physics\)](#).

The causality of systems also plays an important role in [digital signal processing](#), where [filters](#) are constructed by altering a non-causal formulation to remove the lack of causality so that it is realizable. For more information, see [Causality in digital signal processing](#).

For a causal system, the [impulse response](#) of the system must use only the present and past values of the input. This requirement is a necessary and sufficient condition for a system to be causal, regardless of linearity. Note that this definition applies to both discrete and continuous cases. By this definition of requiring no future input values, systems must be causal to process signals in real time.

▼ Time Invariance - mangler forståelse

Et tids invariant system er et system hvor systemet ikke skifter karakter uanset hvorhenne i tiden det bliver placeret.

- A time invariant system is a system for which a time delay/shift of the input sequence causes a corresponding shift in the output sequence.

$$x_1[n] = x[n - n_0] \quad \Rightarrow \quad y_1[n] = y[n - n_0].$$

- The accumulator is a time invariant system.
- Compressor is a non-time invariant system

$$y[n] = x[Mn], \quad -\infty < n < \infty,$$

▼ *Nakket fra bogen - indeholder eksempler på compressor og accumulator side 45 i pdf'en*

A time-invariant system (often referred to equivalently as a shift-invariant system) is a system for which a time shift or delay of the input sequence causes a corresponding shift in the output sequence. Specifically, suppose that a system transforms the input sequence with values $x[n]$ into the output sequence with values $y[n]$. Then the system is said to be time invariant if, for all n_0 , the input sequence with values $x_1[n] = x[n - n_0]$ produces the output sequence with values $y_1[n] = y[n - n_0]$.

As in the case of linearity, proving that a system is time invariant requires a general proof making no specific assumptions about the input signals. All of the systems in Examples 2.3–2.7 are time invariant. The style of proof for time invariance is illustrated in Examples 2.8 and 2.9.

Example 2.8 The Accumulator as a Time-Invariant System

Consider the accumulator from Example 2.6. We define $x_1[n] = x[n - n_0]$. To show time invariance, we solve for both $y[n - n_0]$ and $y_1[n]$ and compare them to see whether they are equal. First,

$$y[n - n_0] = \sum_{k=-\infty}^{n-n_0} x[k]. \quad (2.37)$$

Next, we find

$$y_1[n] = \sum_{k=-\infty}^n x_1[k] \quad (2.38)$$

$$= \sum_{k=-\infty}^n x[k - n_0]. \quad (2.39)$$

Substituting the change of variables $k_1 = k - n_0$ into the summation gives

$$y_1[n] = \sum_{k_1=-\infty}^{n-n_0} x[k_1] = y[n - n_0]. \quad (2.40)$$

Thus, the accumulator is a time-invariant system.

The following example illustrates a system that is not time invariant.

Example 2.9 The Compressor System

The system defined by the relation

$$y[n] = x[Mn], \quad -\infty < n < \infty, \quad (2.41)$$

with M a positive integer, is called a *compressor*. Specifically, it discards $(M - 1)$ samples out of M ; i.e., it creates the output sequence by selecting every M th sample. This system is not time invariant. We can show that it is not by considering the response $y_1[n]$ to the input $x_1[n] = x[n - n_0]$. In order for the system to be time invariant, the output of the system when the input is $x_1[n]$ must be equal to $y[n - n_0]$. The output $y_1[n]$ that results from the input $x_1[n]$ can be directly computed from Eq. (2.41) to be

$$y_1[n] = x_1[Mn] = x[Mn - n_0]. \quad (2.42)$$

2 Discrete-Time Systems

21

Delaying the output $y[n]$ by n_0 samples yields

$$y[n - n_0] = x[M(n - n_0)]. \quad (2.43)$$

Comparing these two outputs, we see that $y[n - n_0]$ is not equal to $y_1[n]$ for all M and n_0 , and therefore, the system is not time invariant.

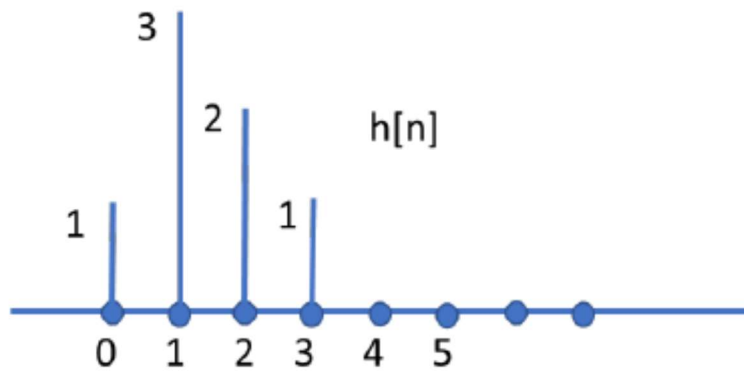
It is also possible to prove that a system is not time invariant by finding a single counterexample that violates the time-invariance property. For instance, a counterexample for the compressor is the case when $M = 2$, $x[n] = \delta[n]$, and $x_1[n] = \delta[n - 1]$. For this choice of inputs and M , $y[n] = \delta[n]$, but $y_1[n] = 0$; thus, it is clear that $y_1[n] \neq y[n - 1]$ for this system.

Impulse Response

Impulsresponsen er givet ved:

$$h[n]$$

Men for at gøre det mere håndterbart så kig på følgende graf:

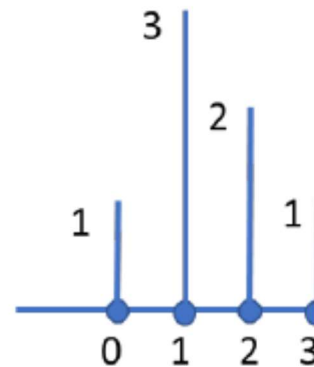
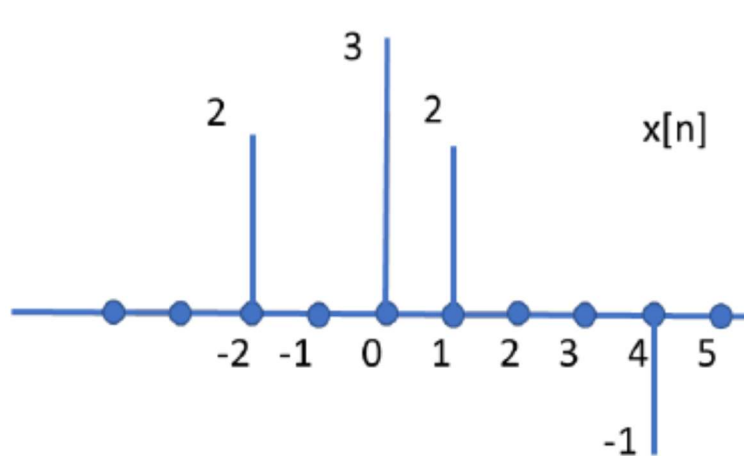


For lige at ligge nogen ord til hvad ovenstående betyder, så kan $h[n]$ beskrives med grundformlen: $h[n] = \text{amplituden Eller Værdien På Pladsen} \cdot \delta[n - \text{positionen Langs X Aksen}]$ og positionen langs x-aksen bliver selvfølgelig positiv, når man er på venstre side af 0 da $- = +$. Nu defineres impulsresponsen $h[n]$ ved:

$$h[n] = 1 \delta[n] + 3 \delta[n - 1] + 2 \delta[n - 2] + 1 \delta[n - 3] :$$

▼ Eksempelopgave fra kursusgang 8 opgave 3

3. Calculate (and sketch) the outcome of the linear time invariant system with given the input $x[n]$ (both displayed below)



For at kunne regne impulsresponsens indvirkning på det lineære tidssystem, skal de begge først skrives op:

$$x[n] = 2 \delta[n + 2] + 3 \delta[n] + 2 \delta[n - 1] - 1 \delta[n - 4] :$$

For lige at ligge nogen ord til hvad ovenstående betyder, så er $x[n]$ skrevet efter grundformlen:

$x[n] = \text{amplituden Eller Værdien På Pladsen} \cdot \delta[n - \text{positionen Langs X Aksen}]$ og positionen langs x-aksen bliver selvfølgelig positiv, når man er på venstre side af 0 da $- = +$. Nu defineres impulsresponsen $h[n]$ på samme vis:

$$h[n] = 1 \delta[n] + 3 \delta[n - 1] + 2 \delta[n - 2] + 1 \delta[n - 3] :$$

Det resulterende tids invariante system er givet ved convolution:

$$y[n] = x[k] \cdot h[n - k] :$$

Hvor k er den første relative tidsunit fra $x[n]$ (i dette tilfælde -2) og n er tiden.

Et samlet tidssystem vil derfor være:

$$y[n] = x[-2] \cdot h[n+2] + x[0] \cdot h[n] + x[1] \cdot h[n-1] + x[4] \cdot h[n-4] :$$

nu skal alle $x[\text{position}]$ leddene udskiftes med værdien på den givne position - jeg har ikke fanget helt præcist hvorfor, men tror det skal læses som om at vi så ganger impuls-responset med den givne deltaværdi og på den måde kan skabe et færdigt signal givet ved delta gange impulsresponsen. altså:

$$y[n] = 2 \cdot h[n+2] + 3 \cdot h[n] + 2 \cdot h[n-1] - 1 \cdot h[n-4] :$$

Nu skal vi så gange impulsresponsen med de givne deltaværdier, og for overskuelighedens skyld bliver det lige gjort for hver impuls:

$$x[-2] = 2 \cdot (1 \delta[n+2] + 3 \delta[n-1+2] + 2 \delta[n-2+2] + 1 \delta[n-3+2]) :$$

bemærk at der er indsat "+2" ved hvert led. Det er for at impulsresponsen matcher med det fundne fra 2 ligninger før.

HVIS MAPLE SKAL KUNNE REGNE FOR DIG **SKAL** DU HUSKE AT BRUGE ALMINDELIGE PARANTESER OG IKKE SKARPE!

$$x(-2) = 2 \cdot (1 \delta(n+2) + 3 \delta(n-1+2) + 2 \delta(n-2+2) + 1 \delta(n-3+2)) = \\ x(-2) = 2 \delta(n+2) + 6 \delta(n+1) + 4 \delta(n) + 2 \delta(n-1)$$

For at denne note matcher vores tavleudregninger, skal det siges at det der sker efter 2^* er det øverste på tavlen til højre, altså:

$$h[n+2] := 1 \delta(n+2) + 3 \delta(n-1+2) + 2 \delta(n-2+2) + 1 \delta(n-3+2) = \\ \delta(n+2) + 3 \delta(n+1) + 2 \delta(n) + \delta(n-1)$$

$$\text{således at } 2 \cdot h[n+2] = 2 \delta(n+2) + 6 \delta(n+1) + 4 \delta(n) + 2 \delta(n-1)$$

Derfor er det måske i virkeligheden nemmere at definere hver $h[n]$ værdi ligesom vi gjorde på tavlen:

$$h[n] := 1 \delta(n) + 3 \delta(n-1) + 2 \delta(n-2) + 1 \delta(n-3) = \\ \delta(n) + 3 \delta(n-1) + 2 \delta(n-2) + \delta(n-3)$$

$$h[n-1] := 1 \delta(n-1) + 3 \delta(n-1-1) + 2 \delta(n-2-1) + 1 \delta(n-3-1) = \\ \delta(n-1) + 3 \delta(n-2) + 2 \delta(n-3) + \delta(n-4)$$

$$h[n-4] := 1 \delta(n-4) + 3 \delta(n-1-4) + 2 \delta(n-2-4) + 1 \delta(n-3-4) = \\ \delta(n-4) + 3 \delta(n-5) + 2 \delta(n-6) + \delta(n-7)$$

Alle disse kan derfor nu indsættes i $y[n] = 2 \cdot h[n+2] + 3 \cdot h[n] + 2 \cdot h[n-1] - 1 \cdot h[n-4] :$

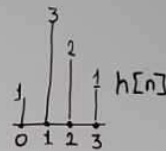
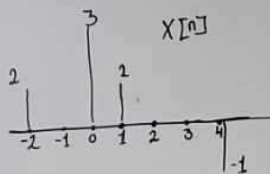
$$y[n] = 2 \cdot h[n+2] + 3 \cdot h[n] + 2 \cdot h[n-1] - 1 \cdot h[n-4] =$$

$$y_n = 2 \delta(n+2) + 6 \delta(n+1) + 7 \delta(n) + 13 \delta(n-1) + 12 \delta(n-2) + 7 \delta(n-3) + \delta(n-4) - 3 \delta(n-5) - 2 \delta(n-6) - \delta(n-7)$$

Hvor resultatet skal læses som om at tallet foran delta er amplituden og positionen stadig er fundet ved x i $[n+-x]$.

Ud fra ovenstående kan man så skabe en visuel repræsentation af signalet med impulsresponsen sat ind:

opg 3



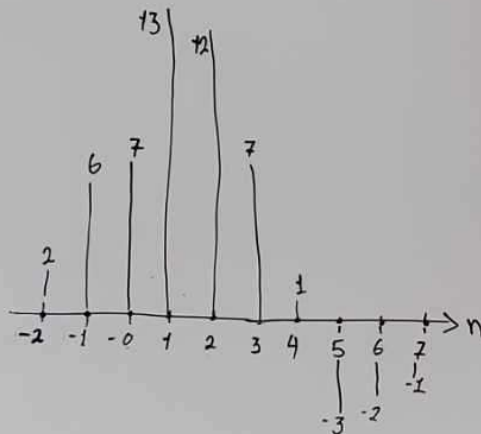
$$y[n] = \sum_{k=-\infty}^{+\infty} x[k] h[n-k] \quad k = -2, 0, 1, 4$$

$$x[n] = 2\delta[n+2] + 3\delta[n] + 2\delta[n-1] - 1\delta[n-4]$$

$$h[n] = 1\delta[n] + 3\delta[n-1] + 2\delta[n-2] + 1\delta[n-3]$$

$$y[n] = x[-2]h[n+2] + x[0]h[n] + x[1]h[n-1] + x[4]h[n-4]$$

$$= 2h[n+2] + 3h[n] + 2h[n-1] - 1h[n-4]$$



opg 3)

$$h[n+2] = 1\delta[n+2]$$

$$h[n-1] = 1\delta[n-1]$$

$$h[n-4] = 1\delta[n-4]$$

$$y[n] = 2 \cdot (1\delta[n+2] + 3\delta[n] + 2\delta[n-1] - 1\delta[n-4])$$

$$= 2(1\delta[n+2] + 3\delta[n] + 2\delta[n-1] - 1\delta[n-4])$$

$$= 2\delta[n+2] + 6\delta[n] + 4\delta[n-1] - 2\delta[n-4]$$

$$+ 2\delta[n-4]$$

$$= 2\delta[n+2] + 6\delta[n] + 4\delta[n-1] - 2\delta[n-4]$$

amplitude pos

Convolution

Convolution er når vi ganger et impulsrespons på vores diskrete tidssystem. Convolution er commutativ og tvinger sig ind foran plus ved paranteser:

- Convolution operation is commutative

$$x[n] * h[n] = h[n] * x[n].$$

- Convolution operation distributes over addition

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n].$$

For at eksemplificere lidt, benyttes nedenstående figur hvor følgende formler kan stilles op (læs impulsrespons først hvis noget af det her er nyt):

vi har givet et system $x[n] := 2\delta(n+2) + 3\delta(n) - 1\delta(n-3)$: - der bruges alm paranteser for at maple kan læse det ordentligt

og en impulsrespons $h[n] := 3\delta(n) + 2\delta(n-1) + 1\delta(n-2)$:

I figurække 2 kigges der kun på impulsen på position -2 (+2 i formlerne). Hvis impulsresponsen sættes ind vha convolution gøres det ved at gange delta (amplituden) med impulsresponsen hvor n tillægges positionen for impulsen:

$$y_{-2} := 2 \cdot h_{-2}[n] = 6\delta(n+2) + 4\delta(n+1) + 2\delta(n)$$

$$\text{hvor } h_{-2}[n] \text{ er givet ved: } h_{-2}[n] := 3\delta(n+2) + 2\delta(n-1+2) + 1\delta(n-2+2) = \\ 3\delta(n+2) + 2\delta(n+1) + \delta(n)$$

Det er vigtigt at notere at i graferne under er det bare vist som hjælp, og ikke baseret på nogen egentlige værdier, derfor kan de kun bruges som svag visualisering.

Men om ikke andet kan vi nu tydeligt aflæse, at vi har en impuls som falder fra -2 og frem.

proceduren gentages så for hver impuls, og til sidst har man responsen for hele sit system. Se fremgangsmetoden under impulsrespons's eksempelopgave for et komplet system

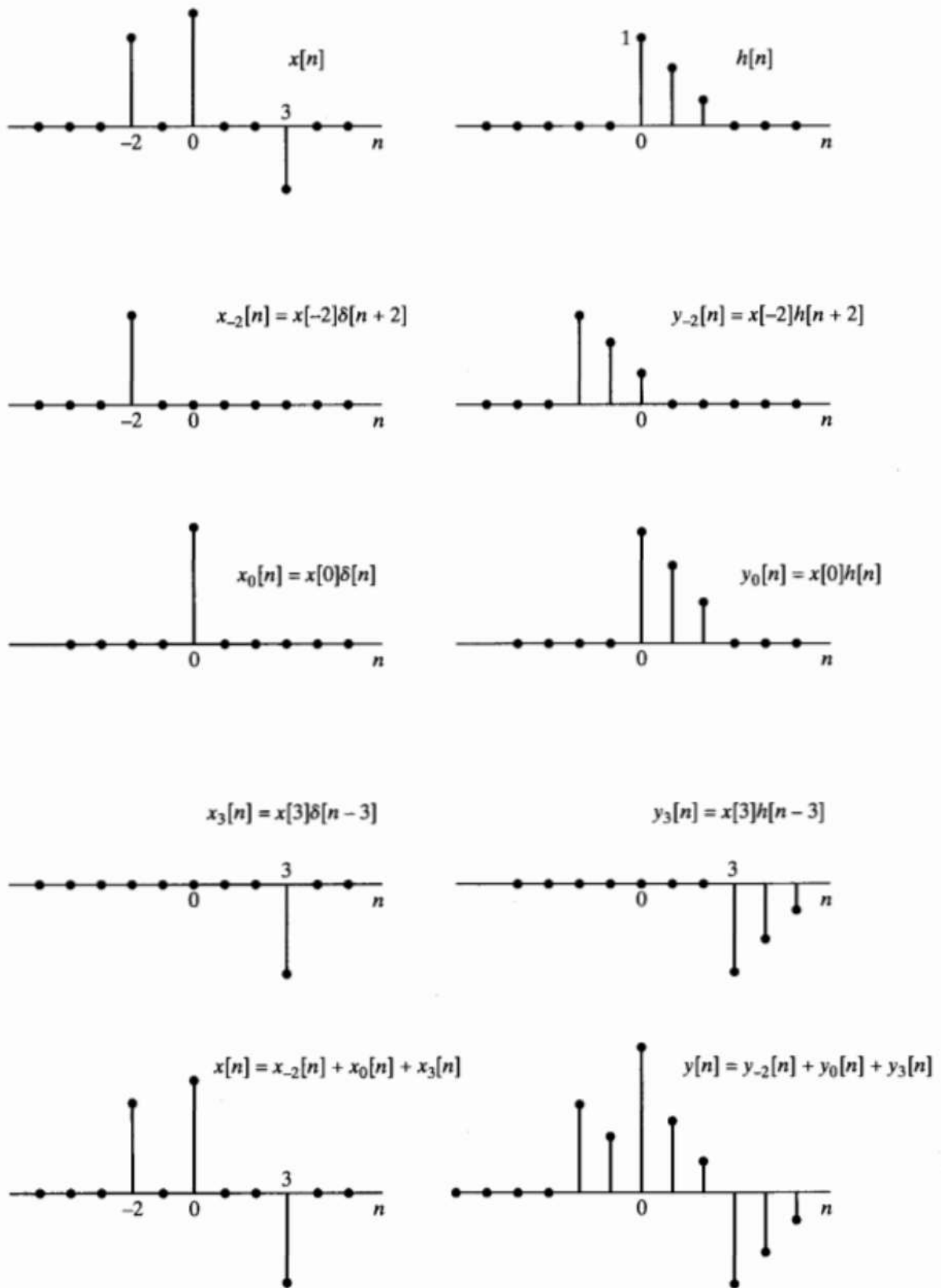


Figure 2.8 Representation of the output of a linear time-invariant system as the superposition of responses to individual samples of the input.

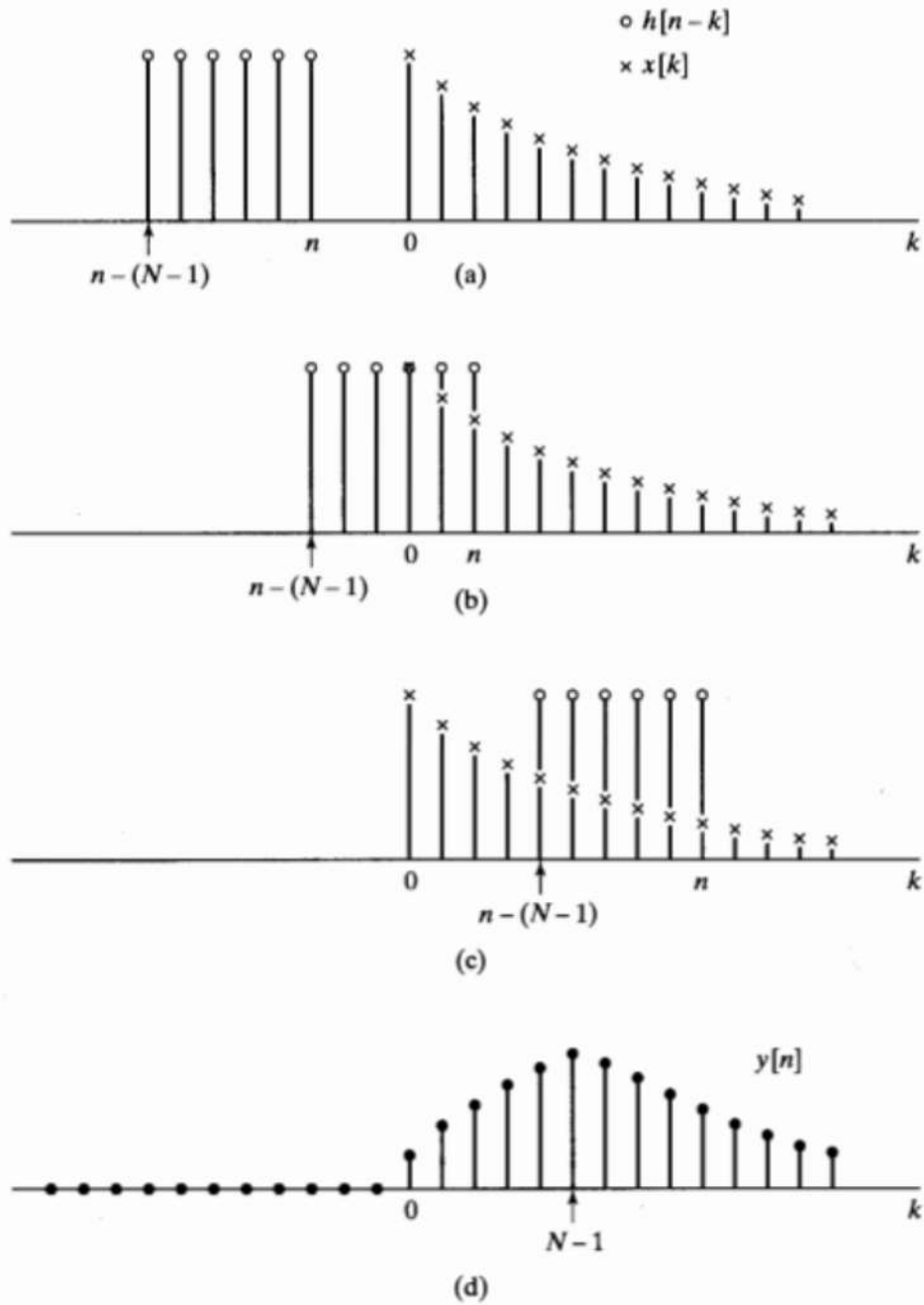


Figure 2.10 Sequence involved in computing a discrete convolution. (a)–(c) The sequences $x[k]$ and $h[n-k]$ as a function of k for different values of n . (Only nonzero samples are shown.) (d) Corresponding output sequence as a function of n .

▼ *Cascading System*

▼ *System Combination*

▼ *Fourier Transform of LTI Systems*

Fourier transformen er defineret som:

$$X(\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt :$$

og den inverse som:

$$x(t) = \int_{-\infty}^{+\infty} X(\omega)e^{j\omega t} d\omega$$

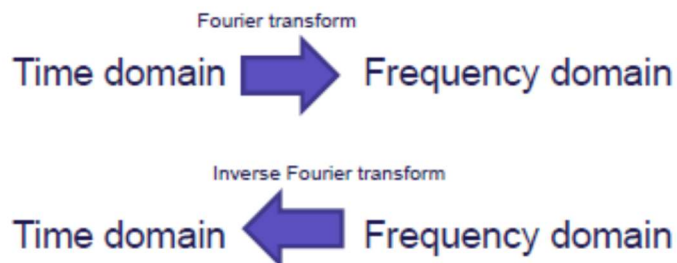
omega er vinklen mellem vektoren til et punkt z på enhedscirklen

to the Fourier transform. Note that ω is the angle between the vector to a point z on the unit circle and the real axis of the complex z-plane. If we evaluate $X(z)$ at points on the

fra bogen side 118.

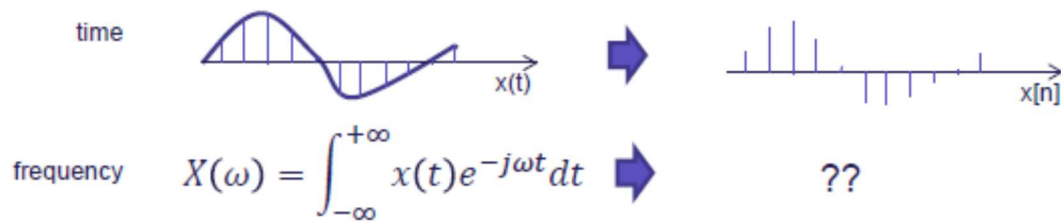
Men udover det ved jeg ikke lige hvad jeg skal sige, da vi ikke har regnet nogen opgaver med det

Physical interpretation: spectrum of a signal



- The Fourier transform returns the amplitude and phase of sinusoidal signals at the different frequencies that compose the time domain signal.
- The Inverse Fourier transform returns the time domain signal which is composed of the sinusoidal signals at different amplitude and phases of the frequency domain representation.

- What about the spectrum of a discrete signal, obtained by sampling $x(t)$?



- It can be shown that the spectrum of a discrete time signal can be calculated as

$$X(\omega) = \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n}$$

Discrete time Fourier transform (DTFT)

- The frequency response of a linear time invariant system is the Fourier transform of the impulse response,

$$H(\omega) = \sum_{n=-\infty}^{+\infty} h[n]e^{-j\omega n}$$



$$h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega})e^{j\omega n} d\omega$$

▼ *Definitions and conditions for existence*

- What is the condition of existence of the Fourier transform?

Determining the class of signals that can be represented by

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega,$$

is equivalent to considering the convergence of the infinite sum in $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

$$\begin{aligned} |X(e^{j\omega})| < \infty \quad \text{for all } \omega, & \quad \Rightarrow \quad |X(e^{j\omega})| = \left| \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \right| \\ & \leq \sum_{n=-\infty}^{\infty} |x[n]| |e^{-j\omega n}| \\ & \leq \sum_{n=-\infty}^{\infty} |x[n]| < \infty. \end{aligned}$$

If $x[n]$ is absolutely summable, then the Fourier transform exists.

▼ Z-transform

Z-transformen af en sekvens er givet ved:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n}$$

z-transform operatoren transformerer en sekvens $x[n]$ til en funktion $X[z]$, hvor z er en kontinuert kompleks variabel.

Hvis man så udfører fourier på z-transformen får man:

$$z = e^{j\omega}$$

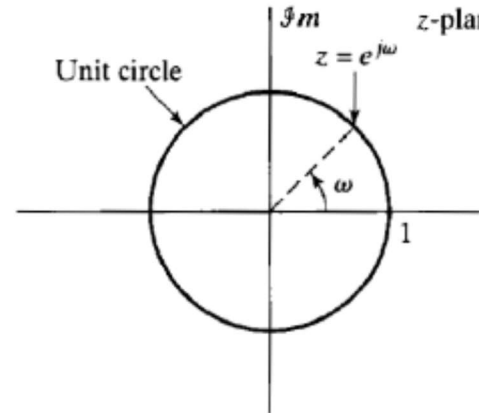
- More generally, the complex variable z can be expressed as $z = r e^{j\omega}$.

$$X(r e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] (r e^{j\omega})^{-n},$$



$$X(r e^{j\omega}) = \sum_{n=-\infty}^{\infty} (x[n] r^{-n}) e^{-j\omega n}.$$

- The z-transform evaluated in the unit circle corresponds to the Fourier transform.
- When it exists, the Fourier transform is simply $X(z)$ with $z = e^{j\omega}$



▼ Z-transform cookbook/fremgangsmetode

Først opskrives sekvensen:

$$x[n] := a^n \cdot u(n) :$$

Den indsættes så i z-transformen:

$$X(z) = \sum_{n=-\infty}^{+\infty} x(n) \cdot z^{-n} \rightarrow X(z) = \sum_{n=-\infty}^{+\infty} a^n \cdot u[n] \cdot z^{-n}$$

Derefter kigger vi på hvad $u[n]$ er:

$$u[n] = \begin{cases} 1, n \geq 0 \\ 0, n < 0 \end{cases}$$

Og får derfor at $u[n]=1$.

Det indsættes så i formelen:

$$\sum_{n=-\infty}^{+\infty} a^n \cdot 1 \text{ eller } 0 \cdot z^{-n} :$$

Derfor står der reelt set nu:

$$\sum_{n=-\infty}^{+\infty} a^n \cdot z^{-n} :$$

Nu sættes $n=-\infty$ til 0 i stedet, for at skabe en realistisk analyse.

$$\sum_{n=0}^{+\infty} a^n \cdot z^{-n} :$$

Så sættes a og z leddene sammen til:

$$\sum_{n=0}^{+\infty} (a \cdot z^{-1})^n :$$

og nu kommer tricket som jeg først har indset nu:

vi udregner hvad a^n bliver til for hver n fra 0 til uendelig, og skal derefter indse at når common

ratio står som $a \cdot z^{-1}$, så skal formelen $\frac{FT}{1 - CR}$ benyttes, hvor FT står for first tome (første

resultat i rækken) og CR står for common ratio. I dette tilfælde prøver vi lige at regne lidt

forskellige steps ud (med forskellige n'er) og $(a \cdot z^{-1})^n = \left(\frac{1}{3z}\right)^n$:

$$\begin{aligned} \left(\frac{1}{3z}\right)^0 &= 1 \left(\frac{1}{3z}\right)^1 = \frac{1}{3z} \left(\frac{1}{3z}\right)^2 = \frac{1}{9z^2} \left(\frac{1}{3z}\right)^3 = \frac{1}{27z^3} \left(\frac{1}{3z}\right)^4 = \frac{1}{81z^4} \left(\frac{1}{3z}\right)^5 = \frac{1}{243z^5} \\ \left(\frac{1}{3z}\right)^6 &= \frac{1}{729z^6} \left(\frac{1}{3z}\right)^7 = \frac{1}{2187z^7} \left(\frac{1}{3z}\right)^8 = \frac{1}{6561z^8} \left(\frac{1}{3z}\right)^9 \end{aligned}$$

Nu da det der tydeliggjort at CR aldrig vil komme over 1, er det tydeligt at vi overholder

følgende: $|CR| < 1$ som er kravet for at vi kan bruge $\frac{FT}{1 - CR}$

Derfor kan vi nu indsætte FT (givet ved $\left(\frac{1}{3z}\right)^0 = 1$) ind på sin plads, og CR på sin (husk at sætte CR i $\wedge -1$) og derved få:

$$\frac{1}{1 - \left(\frac{1}{3z}\right)^{-1}} = \frac{1}{1 - 3z}$$

Og vupti, så var z-transformen fundet. Nu kan ROC så findes ved at kigge i tabellen

TABLE 3.1 SOME COMMON z-TRANSFORM PAIRS

Sequence	Transform	ROC
1. $\delta[n]$	1	All z
2. $u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
3. $-u[-n - 1]$	$\frac{1}{1 - z^{-1}}$	$ z < 1$
4. $\delta[n - m]$	z^{-m}	All z except 0 (if $m > 0$) or ∞ (if $m < 0$)
5. $a^n u[n]$	$\frac{1}{1 - az^{-1}}$	$ z > a $
6. $-a^n u[-n - 1]$	$\frac{1}{1 - az^{-1}}$	$ z < a $
7. $na^n u[n]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
8. $-na^n u[-n - 1]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
9. $[\cos \omega_0 n] u[n]$	$\frac{1 - [\cos \omega_0] z^{-1}}{1 - [2 \cos \omega_0] z^{-1} + z^{-2}}$	$ z > 1$
10. $[\sin \omega_0 n] u[n]$	$\frac{[\sin \omega_0] z^{-1}}{1 - [2 \cos \omega_0] z^{-1} + z^{-2}}$	$ z > 1$
11. $[r^n \cos \omega_0 n] u[n]$	$\frac{1 - [r \cos \omega_0] z^{-1}}{1 - [2r \cos \omega_0] z^{-1} + r^2 z^{-2}}$	$ z > r$
12. $[r^n \sin \omega_0 n] u[n]$	$\frac{[r \sin \omega_0] z^{-1}}{1 - [2r \cos \omega_0] z^{-1} + r^2 z^{-2}}$	$ z > r$
13. $\begin{cases} a^n, & 0 \leq n \leq N - 1, \\ 0, & \text{otherwise} \end{cases}$	$\frac{1 - a^N z^{-N}}{1 - az^{-1}}$	$ z > 0$

TABLE 3.2 SOME z-TRANSFORM PROPERTIES

Section Reference	Sequence	Transform	ROC
	$x[n]$	$X(z)$	R_x
	$x_1[n]$	$X_1(z)$	R_{x_1}
	$x_2[n]$	$X_2(z)$	R_{x_2}
3.4.1	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	Contains $R_{x_1} \cap R_{x_2}$
3.4.2	$x[n - n_0]$	$z^{-n_0} X(z)$	R_x , except for the possible addition or deletion of the origin or ∞
3.4.3	$z_0^n x[n]$	$X(z/z_0)$	$ z_0 R_x$
3.4.4	$nx[n]$	$-z \frac{dX(z)}{dz}$	R_x , except for the possible addition or deletion of the origin or ∞
3.4.5	$x^*[n]$	$X^*(z^*)$	R_x
	$\mathcal{Re}\{x[n]\}$	$\frac{1}{2}[X(z) + X^*(z^*)]$	Contains R_x
	$\mathcal{Im}\{x[n]\}$	$\frac{1}{2j}[X(z) - X^*(z^*)]$	Contains R_x
3.4.6	$x^*[-n]$	$X^*(1/z^*)$	$1/R_x$
3.4.7	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	Contains $R_{x_1} \cap R_{x_2}$
3.4.8	Initial-value theorem: $x[n] = 0, \quad n < 0 \quad \lim_{z \rightarrow \infty} X(z) = x[0]$		

Z Transform Table

	$f[n]$	$F(z).$	Region of Convergence
1.	$\delta[n]$	1	
2	$\delta[n - m]$	z^{-m}	
3	$a^n u_0[n]$	$\frac{z}{z - a}$	$ z > a$
4	$u_0[n]$	$\frac{z}{z - 1}$	$ z > 1$
5	$(e^{-anT_s})u_0[n]$	$\frac{z}{z - e^{-aT_s}}$	$ e^{-aT_s} z^{-1} < 1$
6	$(\cos naT_s)u_0[n]$	$\frac{z^2 - z \cos aT_s}{z^2 - 2z \cos aT_s + 1}$	$ z > 1$
7	$(\sin naT_s)u_0[n]$	$\frac{z \sin aT_s}{z^2 - 2z \cos aT_s + 1}$	$ z > 1$
8	$(a^n \cos naT_s)u_0[n]$	$\frac{z^2 - az \cos aT_s}{z^2 - 2az \cos aT_s + a^2}$	$ z > 1$
9	$(a^n \sin naT_s)u_0[n]$	$\frac{az \sin aT_s}{z^2 - 2az \cos aT_s + a^2}$	$ z > 1$
10	$u_0[n] - u_0[n - m]$	$\frac{z^m - 1}{z^{m-1}(z - 1)}$	
11	$nu_0[n]$	$\frac{z}{(z - 1)^2}$	
12	$n^2 u_0[n]$	$\frac{z(z + 1)}{(z - 1)^3}$	

13	$[n + 1]u_0[n]$	$\frac{z^2}{(z - 1)^2}$
----	-----------------	-------------------------

14	$a^n n u_0[n]$	$\frac{az}{(z - a)^2}$
----	----------------	------------------------

15	$a^n n^2 u_0[n]$	$\frac{az(z + a)}{(z - a)^3}$
----	------------------	-------------------------------

16	$a^n n[n + 1]u_0[n]$	$\frac{2az^2}{(z - a)^3}$
----	----------------------	---------------------------

PROPERTY 1: The ROC is a ring or disk in the z -plane centered at the origin; i.e., $0 \leq r_R < |z| < r_L \leq \infty$.

PROPERTY 2: The Fourier transform of $x[n]$ converges absolutely if and only if the ROC of the z -transform of $x[n]$ includes the unit circle.

PROPERTY 3: The ROC cannot contain any poles.

PROPERTY 4: If $x[n]$ is a *finite-duration sequence*, i.e., a sequence that is zero except in a finite interval $-\infty < N_1 \leq n \leq N_2 < \infty$, then the ROC is the entire z -plane, except possibly $z = 0$ or $z = \infty$.

PROPERTY 5: If $x[n]$ is a *right-sided sequence*, i.e., a sequence that is zero for $n < N_1 < \infty$, the ROC extends outward from the *outermost* (i.e., largest magnitude) finite pole in $X(z)$ to (and possibly including) $z = \infty$.

PROPERTY 6: If $x[n]$ is a *left-sided sequence*, i.e., a sequence that is zero for $n > N_2 > -\infty$, the ROC extends inward from the *innermost* (smallest magnitude) nonzero pole in $X(z)$ to (and possibly including) $z = 0$.

PROPERTY 7: A *two-sided sequence* is an infinite-duration sequence that is neither right sided nor left sided. If $x[n]$ is a two-sided sequence, the ROC will consist of a ring in the z -plane, bounded on the interior and exterior by a pole and, consistent with property 3, not containing any poles.

PROPERTY 8: The ROC must be a connected region.

Region of Convergence (ROC)

TABLE 3.1 SOME COMMON z-TRANSFORM PAIRS

Sequence	Transform	ROC
1. $\delta[n]$	1	All z
2. $u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
3. $-u[-n - 1]$	$\frac{1}{1 - z^{-1}}$	$ z < 1$
4. $\delta[n - m]$	z^{-m}	All z except 0 (if $m > 0$) or ∞ (if $m < 0$)
5. $a^n u[n]$	$\frac{1}{1 - az^{-1}}$	$ z > a $
6. $-a^n u[-n - 1]$	$\frac{1}{1 - az^{-1}}$	$ z < a $
7. $na^n u[n]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
8. $-na^n u[-n - 1]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
9. $[\cos \omega_0 n] u[n]$	$\frac{1 - [\cos \omega_0] z^{-1}}{1 - [2 \cos \omega_0] z^{-1} + z^{-2}}$	$ z > 1$
10. $[\sin \omega_0 n] u[n]$	$\frac{[\sin \omega_0] z^{-1}}{1 - [2 \cos \omega_0] z^{-1} + z^{-2}}$	$ z > 1$
11. $[r^n \cos \omega_0 n] u[n]$	$\frac{1 - [r \cos \omega_0] z^{-1}}{1 - [2r \cos \omega_0] z^{-1} + r^2 z^{-2}}$	$ z > r$
12. $[r^n \sin \omega_0 n] u[n]$	$\frac{[r \sin \omega_0] z^{-1}}{1 - [2r \cos \omega_0] z^{-1} + r^2 z^{-2}}$	$ z > r$
13. $\begin{cases} a^n, & 0 \leq n \leq N - 1, \\ 0, & \text{otherwise} \end{cases}$	$\frac{1 - a^N z^{-N}}{1 - az^{-1}}$	$ z > 0$

Right, Leftsided and Finite Duration Sequences

Rightsided sequence

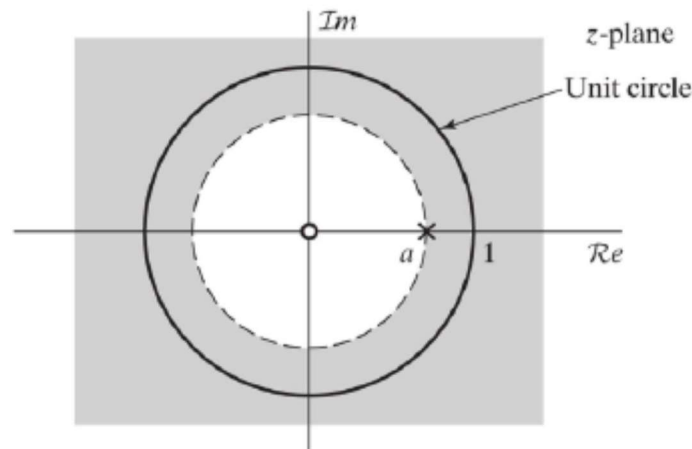
$$X(z) = \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n.$$

For convergence of $X(z)$, we require that

$$\sum_{n=0}^{\infty} |az^{-1}|^n < \infty.$$

Thus, the ROC is the range of values of z for which $|az^{-1}| < 1$ or, equivalently, $|z| > |a|$. Inside the ROC, the infinite series converges to

$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad |z| > |a|. \quad (12)$$



$$x[n] = \left(\frac{1}{2}\right)^n \cdot u[n]$$

$$a = |a| = \left(\frac{1}{2}\right)$$

- For $|a| > 1$, the ROC does not contain the unit circle $\rightarrow$ Fourier transform does not exist

Leftsided sequence

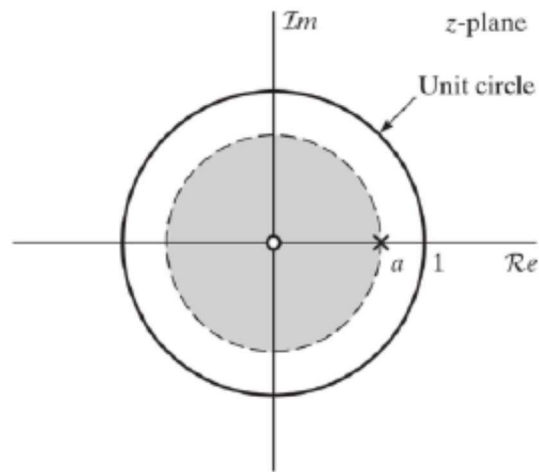
$$x[n] = -a^n u[-n-1] = \begin{cases} -a^n & n \leq -1 \\ 0 & n > -1. \end{cases}$$

Since the sequence is nonzero only for $n \leq -1$, this is a *left-sided* sequence. The z -transform in this case is

$$\begin{aligned} X(z) &= - \sum_{n=-\infty}^{\infty} a^n u[-n-1] z^{-n} = - \sum_{n=-\infty}^{-1} a^n z^{-n} \\ &= - \sum_{n=1}^{\infty} a^{-n} z^n = 1 - \sum_{n=0}^{\infty} (a^{-1}z)^n. \end{aligned} \quad (15)$$

If $|a^{-1}z| < 1$ or, equivalently, $|z| < |a|$, the last sum in Eq. (15) converges, and using again the formula for the sum of terms in a geometric series,

$$X(z) = 1 - \frac{1}{1 - a^{-1}z} = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad |z| < |a|. \quad (16)$$



$$x[n] = -\left(\frac{3}{2}\right)^n \cdot u[-n - 1]$$

$$a = \left(\frac{3}{2}\right)$$

- For $|a| < 1$, the sequence grows exponentially and therefore the Fourier transform does not exist.

Finite length sequence:

z-transform of $x[n] = \begin{cases} a^n & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$

$$X(z) = \sum_{n=0}^{N-1} a^n z^{-n} = \sum_{n=0}^{N-1} (az^{-1})^n \quad \text{finite length sequence}$$

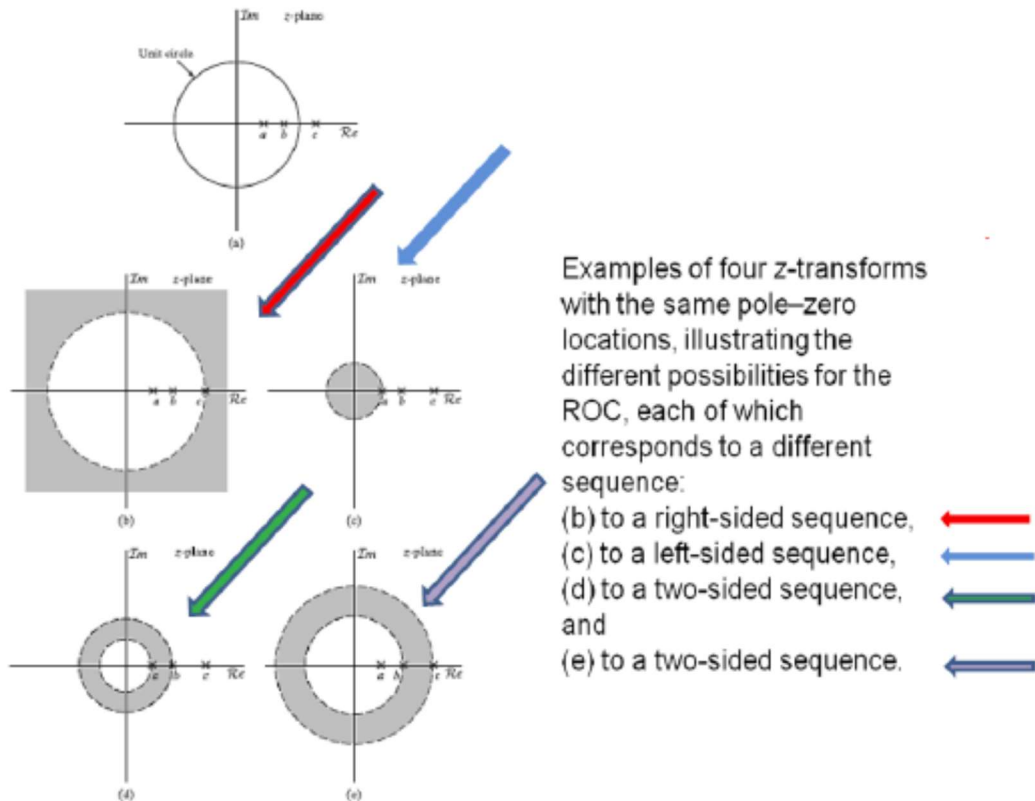
$$\begin{aligned} \left| \sum_{n=0}^{N-1} a^n z^{-n} \right. &= \sum_{n=0}^{\infty} (az^{-1})^n - \sum_{n=N}^{\infty} (az^{-1})^n \\ &= \sum_{n=0}^{\infty} (az^{-1})^n - (az^{-1})^N \sum_{n=0}^{\infty} (az^{-1})^n \\ &= \sum_{n=0}^{\infty} (az^{-1})^n (1 - (az^{-1})^N) = \\ &= \frac{1 - (az^{-1})^N}{1 - az^{-1}} \\ &= \frac{1 - \frac{a^N}{z^N}}{1 - \frac{a}{z}} = \frac{1}{z^{N-1}} \frac{z^N - a^N}{z - a} \end{aligned}$$

1 pole at $z=a \rightarrow$ it is canceled by one of the zeros
 $N-1$ order pole at 0

N zeros (roots of numerator polynomial)
 $z^k = a e^{j(2\pi k/N)}$

In case az^{-1} is finite, i.e. $|a| < \infty$ and $z \neq 0$,
 $\sum_{n=0}^{N-1} |az^{-1}|^n < \infty \rightarrow$ ROC include the entire
 - plane, with the exception of $z=0$.

ROC Analysis



Inverse Z-transform

Inspection method cookbook

a) $X(z) = (1 + 3z)(1 + 2z^{-1})(1 - z^{-1})$

Først benyttes inspektionsformlen, og vi kan ikke se noget åbenlyst, så derfor reduceres formelen mest muligt:

$$x(z) = (1 + 3z) \cdot (1 + 2z^{-1}) \cdot (1 - z^{-1}) :$$

men for at gøre det mere overskueligt, gør vi det et skridt af gangen:

$$x(z) = (1 + 3z) \cdot (1 + 2z^{-1}) \rightarrow (1 \cdot 1 + 1 \cdot 2z^{-1} + 3z \cdot 1 + 3z \cdot 2z^{-1})$$

$$x(z) = (1 \cdot 1 + 1 \cdot 2z^{-1} + 3z \cdot 1 + 3z \cdot 2z^{-1}) = 7 + \frac{2}{z} + 3z - \text{har ikke kunnet finde nogen}$$

nemmere måde end denne at få maple til at regne det for os

Det kan vi så sætte sammen med det sidste led:

$$x(z) = \left(7 + \frac{2}{z} + 3z \right) \cdot \left(1 - \frac{1}{z} \right) \rightarrow \left(7 \cdot 1 + \frac{2}{z} \cdot 1 + 3z \cdot 1 + 7 \cdot (-z^{-1}) + \frac{2}{z} \cdot (-z^{-1}) + 3z \cdot (-z^{-1}) \right)$$

$$\left(7 \cdot 1 + \frac{2}{z} \cdot 1 + 3z \cdot 1 + 7 \cdot (-z^{-1}) + \frac{2}{z} \cdot (-z^{-1}) + 3z \cdot (-z^{-1}) \right) = 4 - \frac{5}{z} - \frac{2}{z^2} + 3z$$

Nu kan vi så få det skrevet om på en mere z-transformeret formel:

$$X(z) = 4 + 3z - 5z^{-1} - 2z^{-2}$$

Den benytter vi så inspektionsmetoden og Dittes noter til at omskrive så vi får 4 led:

$$4\delta[n] + 3\delta[n+1] - 5\delta[n-1] - 2\delta[n-2]$$

Når der ikke er noget "z" på den transformerede, så omskrives der bare til $x\delta[n]$, og derved fåes der: $4 = 4\delta[n]$

Når der så er et z uden nogen potens skal der ligges +1 til n, da der reelt set står z^1 ved z, og derved fåes der: $3z = 3\delta[n+1]$

Når der så er et z^{-1} på, skal der et -1 til n, så man derved får: $5z^{-1} = 5\delta[n-1]$

Ligeledes skal 1-tallet bare erstattes med potensen, når man har højere potenser, så man får:

$$2z^{-2} = 2\delta[n-2]$$

Dittes note forklarer vildere eksempler bedre end jeg kan med skrift, husk derfor:

$$2 \cdot \frac{z^{-1}}{1 - 1/z}$$

$$2 \cdot 1^{n-1} [n-1]$$

$$\frac{z^3}{1 - \frac{4}{7} z^{-1}}$$

$$\frac{4n+3}{7} u [n+3]$$

▼ Partial fraction cookbook

Se opgave 10.b for at se det hele i brug

Denne gang skal vi gøre brug af partial fraction expansion i stedet (fra slide 9-10-11). Derfor skal vi starte med at finde graden af N og M (denominator og nominator), som findes ved graden af det højeste led. Altså antallet af z'er over brøkstregen (M) og under brøkstregen (N). Hvis m så er mindre end N, er alle poler 1. ordens, og vi kan derfor bruge formlen:

$$X(z) = \sum_{k=1}^N \frac{A_k}{1 - d_k \cdot z^{-1}}$$

hvor A_k er koeffecienter fundet ved: $A_k = (1 - d_k \cdot z^{-1}) \cdot X(z)$ hvor $z = d_k$

d_k er så givet ved at opstille følgende gaffelligning med to ubekendte:

$$\begin{cases} -(d_1 + d_2) = \\ d_1 d_2 = \end{cases}$$

▼ 2 ligninger med 2 ubekendte løses nemmest ved:

$$1 - (d_1 + d_2) \cdot z^{-1} + d_1 d_2 \cdot z^{-2} = \begin{cases} -(d_1 + d_2) = \frac{9}{4} \\ d_1 d_2 = \frac{1}{2} \end{cases}$$

$$\begin{aligned} -\left(\frac{\frac{1}{2}}{d_2} + d_2\right) &= \frac{9}{4} \xrightarrow{\text{solve}} \{d_2 = -2\}, \left\{d_2 = -\frac{1}{4}\right\} \\ d_1 \cdot (-2) &= \frac{1}{2} \xrightarrow{\text{solve}} \left\{d_1 = -\frac{1}{4}\right\} \end{aligned}$$

Substitution

Substitutionsmetoden går ud på at isolere en af de ubekendte i en af ligningerne. Derpå kan man sætte dette udtryk ind i den anden ligning, og så har man kun en ubekendt tilbage. Lad os prøve at bruge metoden på ligningerne ovenfor. Vi tager udgangspunkt i den første ligning og isolerer y .

$$\begin{aligned}3x + 4y &= 35 \Leftrightarrow \\4y &= 35 - 3x \Leftrightarrow \\y &= \frac{35-3x}{4}.\end{aligned}$$

Nu kan vi sætte dette udtryk for y ind i den anden ligning

$$\begin{aligned}4x + 2y &= 20 \Leftrightarrow \\4x + 2 \cdot \left(\frac{35-3x}{4}\right) &= 20.\end{aligned}$$

Nu er der kun én ubekendt tilbage (x), og vi løser ligningen på almindelig vis

$$\begin{aligned}4x + 2 \cdot \left(\frac{35-3x}{4}\right) &= 20 \Leftrightarrow \\4x + \frac{70-6x}{4} &= 20 \Leftrightarrow \\4x + \frac{70}{4} - \frac{6x}{4} &= 20 \Leftrightarrow\end{aligned}$$

$$\begin{aligned}\frac{5}{2}x + \frac{35}{2} &= 20 \Leftrightarrow \\ \frac{5}{2}x &= 20 - \frac{35}{2} \Leftrightarrow \\ \frac{5}{2}x &= \frac{5}{2} \Leftrightarrow \\ x &= 1.\end{aligned}$$

Nu ved vi, at x skal være 1. Dvs at prisen på en slikkepind er 1 kr. Men vi mangler stadig at finde prisen på en slikpose. Heldigvis fandt vi jo tidligere frem til et udtryk for y , hvor vi nu kan indsætte vores værdi for x

$$\begin{aligned}y &= \frac{35-3x}{4} \Leftrightarrow \\ y &= \frac{35-3 \cdot 1}{4} \Leftrightarrow \\ y &= \frac{32}{4} \Leftrightarrow \\ y &= 8.\end{aligned}$$

Altså er prisen på en slikpose 8 kr. Vi kan prøve at sætte $x = 1$ og $y = 8$ i vores oprindelige ligninger for at tjekke, om det er det rigtige svar

$$3x + 4y = 3 \cdot 1 + 4 \cdot 8 = 3 + 32 = 35$$

$$4x + 2y = 4 \cdot 1 + 2 \cdot 8 = 4 + 16 = 20,$$

og det er det heldigvis.

Husk at dette kun skal gøres når $M < N!!!$

Når man så har fundet d_{-1} og d_{-2} , indsættes de i hver sin respektive udgave af A_{-k} :

$$A_1 = (1 - d_1 \cdot z^{-1}) \cdot X(z) \text{ hvor } z = d_{-1}$$

$$A_2 = (1 - d_2 \cdot z^{-1}) \cdot X(z) \text{ hvor } z = d_{-2}$$

Næste skridt er at du tager det øverste led af $X(z)$ og indsætter værdien af z deri, og dividerer derefter med sådan som vi egentlig har regnet A_{-k} , altså $(1 - d_k \cdot z^{-1})$ og så skal det være den anden k -værdi:

$$\text{i dette tilfælde er øverste led/det over brøkstregen } 1 - \frac{1}{2}z^{-1}$$

som med z indsat bliver:

$$z := -\frac{1}{4} :$$

$$1 - \frac{1}{2}z^{-1} = 3$$

og

$$(1 - d_2 \cdot z^{-1}) = -1$$

hvor $d_2 = -1/2$ - præcist hvorfor det skal være d_2 har jeg ikke luret endnu

De to ting divideres derefter med hinanden:

$$\frac{\left(1 - \frac{1}{2}z^{-1}\right)}{(1 - d_2 \cdot z^{-1})} = -3$$

Og hvis vi så gentager successen med A_2 får vi:

$$z := -\frac{1}{2} :$$

$$1 - \frac{1}{2}z^{-1} = 2$$

og

$$(1 - d_1 \cdot z^{-1}) = \frac{1}{2}$$

hvor $d_1 = 1/4$

De to ting divideres derefter med hinanden:

$$\frac{\left(1 - \frac{1}{2}z^{-1}\right)}{(1 - d_1 \cdot z^{-1})} = 4$$

Altså er A_1 og A_2 givet ved:

$$z := -\frac{1}{4} :$$

$$A_1 := \frac{\left(1 - \frac{1}{2}z^{-1}\right)}{(1 - d_2 \cdot z^{-1})} = -3$$

$$z := -\frac{1}{2} :$$

$$A_2 := \frac{\left(1 - \frac{1}{2}z^{-1}\right)}{(1 - d_1 \cdot z^{-1})} = 4$$

Derfor kan de nu indsættes over hver deres del i formlen:

$$\frac{A_1}{1 - \left(-\frac{1}{4}\right)z^{-1}} + \frac{A_2}{1 - \left(-\frac{1}{2}\right)z^{-1}}$$

Og så bruger vi dittes guide til at få den skrevet om til $x[n]$:

$$\frac{2z^{-1}}{1 - 1z^{-1}}$$

$$2 \cdot 1^{n-1} [n-1]$$

$$\frac{z^3}{1 - \frac{4}{7}z^{-1}}$$

$$\frac{4n+3}{7} u[n+3]$$

▼ Partial Fraction Expansion

Regler for Partialbrøksopdeling

1. Graden af Tæller og Nævner:

- Hvis graden af tælleren (M) er mindre end graden af nævneren (N), kan du direkte anvende partialbrøksopdeling.
- Hvis graden af tælleren (M) er lig med eller større end graden af nævneren (N), skal du først udføre polynomiumdivision for at få graden af tælleren til at være mindre end graden af nævneren. Derefter kan du anvende partialbrøksopdeling på resten.

2. Faktoriser Nævneren:

- Faktoriser nævneren i irreducible polynomier (lineære eller kvadratiske faktorer). Nævneren kan skrives som et produkt af polynomier i formen $(z - a)^k$ eller $(az^2 + bz + c)^k$.

3. Opskriv Partialbrøker:

- For hver lineær faktor $(z - a)$ i nævneren, skal du inkludere en brøk af formen $\frac{A}{z-a}$.
- For hver gentaget lineær faktor $(z - a)^k$, skal du inkludere brøker af formen $\frac{A_1}{z-a} + \frac{A_2}{(z-a)^2} + \dots + \frac{A_k}{(z-a)^k}$.
- For hver irreducible kvadratisk faktor $(az^2 + bz + c)$ i nævneren, skal du inkludere en brøk af formen $\frac{Az+B}{az^2+bz+c}$.
- For hver gentaget irreducible kvadratisk faktor $(az^2 + bz + c)^k$, skal du inkludere brøker af formen $\frac{A_1z+B_1}{az^2+bz+c} + \frac{A_2z+B_2}{(az^2+bz+c)^2} + \dots + \frac{A_kz+B_k}{(az^2+bz+c)^k}$.

4. Bestem Koefficienter:

- Løs systemet af ligninger for at finde koefficienterne $A, B, \dots$

Dette kan gøres ved at udligne brøkenes nævner efter en fællesnævner er fundet, og derefter sammenligne koefficienterne for hver potens af z i tælleren.

Eksempel på Partialbrøksopdeling

Lad os tage et eksempel for at illustrere processen.

$$\text{Givet } X(z) = \frac{1 - \frac{1}{2z}}{1 - \frac{1}{4z}}:$$

1. Omskriv Brøken:

$$X(z) = \frac{1 - \frac{1}{2z}}{1 - \frac{1}{4z}} = \frac{z - \frac{1}{2}}{z - \frac{1}{4}}$$

2. Polynomiumdivision:

Da graden af tælleren og nævneren er den samme, udfører vi polynomiumdivision:

$$\begin{aligned} \frac{z - \frac{1}{2}}{z - \frac{1}{4}} &= 1 + \frac{(z - \frac{1}{2}) - (z - \frac{1}{4})}{z - \frac{1}{4}} = 1 + \frac{-\frac{1}{4}}{z - \frac{1}{4}} \\ &= 1 - \frac{1}{4(z - \frac{1}{4})} \end{aligned}$$

3. Opskriv Partialbrøker:

$$X(z) = 1 - \frac{1}{4} \cdot \frac{1}{z - \frac{1}{4}}$$

4. Bestem Koefficienter:

Dette trin er allerede udført i polynomiumdivisionen. Vi har fundet koefficienten for den enkelte brøkdelen.

5. Inverse z-transform:

Vi finder den inverse z-transform af hver del:

$$\mathcal{Z}^{-1}\{1\} = u[n]$$

$$\mathcal{Z}^{-1}\left\{\frac{1}{z - \frac{1}{4}}\right\} = \left(\frac{1}{4}\right)^n u[n]$$

$$\mathcal{Z}^{-1}\left\{\frac{-1}{4(z - \frac{1}{4})}\right\} = -\frac{1}{4} \left(\frac{1}{4}\right)^n u[n]$$

Så den samlede inverse z-transform er:

$$x[n] = u[n] - \left(\frac{1}{4}\right)^{n+1} u[n]$$

Dette er en grundlæggende gennemgang af reglerne og processen for partialbrøksopdeling.

▼ *Power Series Expansion*

▼ *Transform Analysis of LTI Systems*

▼ *Linear Constant Coefficient Difference Equations*

▼ *Stability and Causality*

Stabiliteten er givet ved om ROC indeholder enhedscirklen. En ROC på $0 < z < 2$ vil f.eks. være stabil, hvorimod en ROC på $1.5 < z < 2$ IKKE vil være stabil

causality er givet ved om unit stepsne har $u[n-1]$ eller $u[n+1]$. Så længe der ikke er nogen $+1$ er systemet causalt, da det så ikke er afhængige af fremtiden.

▼ *Inverse Systems*

▼ *FIR and IIR Systems*

▼ **Gaffel-regler**

Grundreglen for gaffeldelen er:

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

Ud fra den skal man kigge på hvad $u[n]$ er i den givne opgave:

I opgaven fra eksamenssættet er $u[n]$ leddet $u[n-1]$ hvilket giver en gaffelfunktion som hedder:

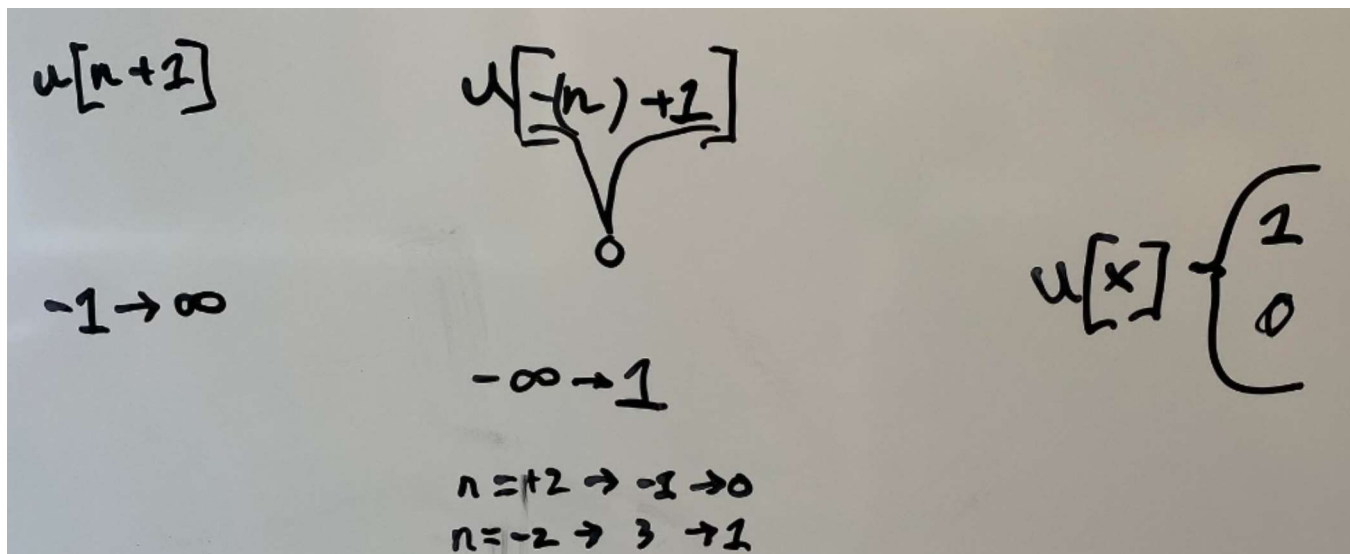
$$u[n-1] = \begin{cases} 1 & n \geq 1 \\ 0 & n < 1 \end{cases}$$

Da for at $u[n]$ bliver til 0 eller derover skal n være mindst 1 i dette eksempel. Yderligere skal det bemærkes at når der står $+n$ har vi at gøre med right hand sequence.

Et andet eksempel er der fra den anden del af eksamen hvor $u[n]$ ledder er $u[-n+1]$ som giver denne gaffelfunktion:

$$u[-n+1] = \begin{cases} 1 & n \leq 1 \\ 0 & n > 1 \end{cases}$$

Da for at $u[n]$ leddet bliver til 0 eller derover, skal n maks være 1 i dette eksempel. Yderligere skal det bemærkes at når der står $-n$ har vi at gøre med left hand sequence.



▼ Long division eksempel (gilberto)

I dette eksempel starter vi ud med en z-transformeret som:

$$x(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

med polerne 1 og 1/2

Næste step er at få skrevet om til følgende formel:

$$x(z) = B_0 + \frac{\quad}{(1 - z^{-1})\left(1 - \frac{1}{2}z^{-1}\right)}$$

Så nu skal B_0 bestemmes. Det gøres ved at stille de to polynomier op med det mindste led forrest, altså som:

$$z^{-2} + 2z^{-1} + 1$$

og

$$\frac{1}{2}z^{-2} - \frac{3}{2}z^{-1} + 1$$

Så deles første led med hinanden så man får:

$$B_0 = \frac{z^{-2}}{\frac{1}{2}z^{-2}} = 2 \text{ (step 1)}$$

Dernæst ganges B_0 med nederste del af brøken:

$$2 \cdot \left(\frac{1}{2}z^{-2} - \frac{3}{2}z^{-1} + 1 \right) = z^{-2} - 3z^{-1} + 2 \text{ (step 2)}$$

Den del skal vi så nu trække fra øverste del af brøken:

$$1 + 2z^{-1} + z^{-2} - (z^{-2} - 3z^{-1} + 2) = -1 + \frac{5}{z}$$

HUSK AT MAPLE IKKE VIL HAVE $^{-1}$ OG I STEDET LAVER DET TIL EN BRØK

Nu står der derfor:

$$5^{-1} - 1$$

Dette led skal indsættes for at finde koeffecienter til partialbrøkopspaltningen, så der nu står:

$$x(z) = 2 + \frac{5^{-1} - 1}{(1 - z^{-1}) \cdot \left(1 - \frac{1}{2}z^{-1}\right)}$$

Som partialbrøkopspaltet bliver til:

$$x(z) = 2 + \frac{A_1}{1 - z^{-1}} + \frac{A_2}{1 - \frac{1}{2}z^{-1}}$$

A__1 og A__2 findes derefter ved:

$$A_1 := (1 - z^{-1}) \cdot \frac{5z^{-1} - 1}{(1 - z^{-1}) \cdot \left(1 - \frac{1}{2}z^{-1}\right)} \text{ for } z := 1 :$$

$$A_1 := \frac{5z^{-1} - 1}{1 - \frac{1}{2}z^{-1}} = 8$$

Processen gentages for A__2:

$$A_2 := \left(1 - \frac{1}{2}z^{-1}\right) \cdot \frac{5z^{-1} - 1}{(1 - z^{-1}) \cdot \left(1 - \frac{1}{2}z^{-1}\right)} \text{ for } z := \frac{1}{2} :$$

$$A_2 := \frac{5z^{-1} - 1}{1 - z^{-1}} = -9$$

Med A__1 og A__2 kendt kan vi nu færdigskrive x(z):

$$x(z) = 2 + \frac{8}{1 - z^{-1}} + \frac{-9}{1 - \frac{1}{2}z^{-1}}$$

Næste step er at tage den inverse z-transform på x(z):

$$x(n) = 2\delta[n] + 8 \cdot 1^n u[n] - 9 \cdot \left(\frac{1}{2}\right)^n u[n]$$

Og så er showet ovre

Handwritten notes showing the inverse z-transform of the partial fractions. The first fraction, $\frac{2}{1 - z^{-1}}$, is transformed to $2 \cdot 1^{n-1} u[n-1]$. The second fraction, $\frac{-9}{1 - \frac{1}{2}z^{-1}}$, is transformed to $-\frac{9}{7} \cdot \left(\frac{1}{2}\right)^{n+3} u[n+3]$. There are red scribbles above and below the first result.

$$X(z) = \frac{z + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} \rightarrow \text{poles: } 2, \frac{1}{2}$$

$$X(z) = \frac{5z^2 - 1}{(z - 2)(z - \frac{1}{2})} \rightarrow X(z) = 2 + \frac{A_1}{z - 2} + \frac{A_2}{z - \frac{1}{2}}$$

$$A_k = (1 - d_k \cdot z^{-1}) \cdot X(z)$$

$$A_1 = \left[\left(\frac{5z^2 - 1}{(z - \frac{1}{2})(z - 2)} \right) \cdot (z - \frac{1}{2}) \right]_{z=2} = 8$$

$$A_2 = \left[-11 \cdot (z - \frac{1}{2}) \right]_{z=\frac{1}{2}} = -9$$

$$X(z) = 2 + \frac{8}{z - 2} + \frac{-9}{z - \frac{1}{2}}$$

$$z = 2 \delta[n]$$

Find B_0 :

$$\frac{1}{2}z^{-2} - \frac{3}{2}z^{-1} + 1 \quad \frac{z}{z^{-2} + 2z^{-1} + 1}$$

$$B_0 = 2$$

Step 1: $\frac{z^{-2}}{\frac{1}{2}z^{-2}} = 2$

Step 2: $2 \cdot (\frac{1}{2}z^{-2} - \frac{3}{2}z^{-1} + 1) = z^{-2} - 3z^{-1} + 2$

Step 3: $\frac{z^{-2} + 2z^{-2} + 2}{-(z^{-2} - 3z^{-1} + 2)}$

$$\frac{0 + 5z^{-2} - 1}{-1}$$

▼ Formler fra kursusgang 1

▼ Formler fra kursusgang 2

▼ Formler fra kursusgang 3

▼ Formler fra kursusgang 4

L

▼ **Formler fra kursusgang 5**

L

▼ **Formler fra kursusgang 6**

L

▼ **Formler fra kursusgang 7**

L

▼ **Formler fra kursusgang 8**

L

▼ **Formler fra kursusgang 9**

L

▼ **Formler fra kursusgang 10**

L

▼ **Formler fra kursusgang 11**

L

▼ **Formler fra kursusgang 12**

L